



CLC

Educational Institute

DIGITAL SAT — MATH

Theory Companion — Part 1

Algebra & Data Analysis

Lines | Systems | Inequalities | Absolute Value | Exponents | Radicals

Ratios & Proportions | Probability | Unit Conversions | Statistics | Data Displays | Scatterplots | Sampling

A shortened, high-yield version of CLC's full Math theory reference —

condensed rules for all 13 skills, each followed by 8 Hard/Very Hard worked problems broken into Formula, Solution, Tip, Trap, Process of Elimination, Shortcut, and Desmos Tip.

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Digital SAT Math — Overview

Math accounts for half of the Digital SAT score, split across Algebra, Advanced Math, Problem-Solving & Data Analysis, and Geometry & Trigonometry. This companion (Part 1) covers 13 skills from Algebra and Problem-Solving & Data Analysis — the ones built on proportional reasoning, linear relationships, and statistical thinking. Each skill is condensed to its essential rules, then drilled with 8 genuinely difficult problems built around the specific twists the real exam uses to separate students who memorized a formula from students who understand it.

Group	Skills Covered
Algebra	Lines · Systems of Linear Equations · Linear Inequalities · Absolute Value · Exponent Rules · Radicals, Rational Exponents & Scientific Notation
Problem-Solving & Data Analysis	Ratios & Proportions · Probability · Unit Conversions & Rates · Statistics — Center & Spread · Data Displays · Scatterplots · Sampling, Bias & Margin of Error

This is deliberately a SHORTENED companion: the full theory reference (with every derivation, extra example, and edge case) exists separately. The purpose here is speed and critical thinking under time pressure — recognizing which rule applies when a question is disguised, combined with another rule, or padded with irrelevant information, the way the actual Digital SAT writes Hard and Very Hard questions.

Every worked problem follows the same seven-part breakdown used across CLC's companion series: the Formula/Rule the problem hinges on, a full step-by-step Solution, a Tip (the reliable first move for this problem type), a Trap (the specific wrong turn this problem is built to tempt), a Process of Elimination table naming the concrete wrong answers a rushed student lands on and exactly which slip produces each one, a Shortcut where one genuinely exists, and a Desmos Tip showing how to confirm or accelerate the solution on the Digital SAT's built-in graphing calculator.

PART 1 — ALGEBRA

1. Lines

Lines — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Slope	Κλίση
Y-intercept	Σημείο τομής με τον άξονα y
Slope-intercept form	Μορφή κλίσης-τομής
Parallel lines	Παράλληλες ευθείες
Perpendicular lines	Κάθετες ευθείες
Point-slope form	Μορφή σημείου-κλίσης

Lines — Condensed Theory

Linear functions model any relationship with a constant rate of change. Every Lines question comes down to slope, one of four equation forms, or a small set of coordinate-geometry formulas.

Slope Formula

$$m = (y_2 - y_1) / (x_2 - x_1)$$

Positive slope rises left to right; negative slope falls. A horizontal line has slope 0; a vertical line has undefined slope.

Slope-Intercept Form

$$y = mx + b$$

m is the slope (rate of change); b is the y-intercept (value of y when x = 0).

Point-Slope Form

$$y - y_1 = m(x - x_1)$$

Use when you know the slope and one point; solve for y to convert to slope-intercept form.

Standard Form

$$Ax + By = C \rightarrow \text{slope} = -A/B, \text{ y-intercept} = C/B$$



Always convert to slope-intercept form to read off the slope or y-intercept directly.

Parallel & Perpendicular Lines

Parallel: $m_1 = m_2$. **Perpendicular:** $m_1 \times m_2 = -1$

Perpendicular slopes are negative reciprocals of each other.

Intercepts

x-intercept: set $y = 0$. **y-intercept:** set $x = 0$.

Solve the resulting one-variable equation.

Midpoint Formula

$$M = ((x_1+x_2)/2, (y_1+y_2)/2)$$

Average the x-coordinates and the y-coordinates separately.

Distance Formula

$$d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

Derived from the Pythagorean theorem; can also be found by drawing the right triangle between the two points.



Watch Out

Never count grid squares to estimate slope from a graph — the axes are often scaled differently. Always use the actual coordinate values.

The SAT deliberately writes many Lines questions so that graphing alone won't get you the exact answer — know the algebra, don't just rely on Desmos.

Lines — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Lines — Problem 1 (Very Hard)



Segment AB has endpoints A(-2, 5) and B(6, 1). Line m is the perpendicular bisector of segment AB. What is the y-intercept of line m?

Formula / Rule Used: Midpoint formula and negative reciprocal slopes (perpendicular bisector).

Solution

Step 1: Find the midpoint of segment AB: $M = ((-2+6)/2, (5+1)/2) = (4/2, 6/2) = (2, 3)$.

Step 2: Find the slope of AB: $m_{AB} = (1 - 5)/(6 - (-2)) = -4/8 = -1/2$.

Step 3: Line m is the perpendicular bisector, so its slope is the negative reciprocal of $-1/2$, which is 2.

Step 4: Line m passes through the midpoint (2, 3) with slope 2. Use point-slope form: $y - 3 = 2(x - 2)$.

Step 5: Simplify: $y - 3 = 2x - 4$, so $y = 2x - 1$.

Step 6: The y-intercept occurs where $x = 0$: $y = 2(0) - 1 = -1$.

Answer: y-intercept = -1, so line m crosses the y-axis at (0, -1).

Tip

For a perpendicular bisector problem, always compute the midpoint AND the perpendicular slope before writing the equation -- both pieces are required.

Trap

A common error is building the bisector line through one of the original endpoints (A or B) instead of the midpoint, which produces a different, incorrect y-intercept.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
y-intercept = 9	Used endpoint A(-2, 5) instead of the midpoint (2, 3) as the point for the bisector line, keeping the correct perpendicular slope of 2 but building the equation through the wrong point.
y-intercept = 4	Used the slope of segment AB itself ($-1/2$) instead of its negative reciprocal (2) when writing the bisector's equation through the midpoint.
y-intercept = 7	Made a sign error finding the slope of AB, computing $(1-5)/(6+2)$ as $4/8 = 1/2$ instead of $-4/8 = -1/2$, which flipped the perpendicular slope to -2 instead of 2.

Shortcut

Since only the y-intercept is needed, you can skip fully simplifying to standard form -- just solve the point-slope equation for y and read off the constant term directly.

Desmos Tip

Plot A(-2,5) and B(6,1), graph the segment, then graph $y = 2x - 1$ and confirm it crosses AB at a right angle exactly at the midpoint (2,3).



Line p satisfies the equation $3(2y - x) = 4(y + 6)$. Line q is parallel to line p and passes through the point (4, -1). Write the equation of line q in standard form $Ax + By = C$, where A, B, and C are integers and A is positive.

Formula / Rule Used: Simplify to slope-intercept form; parallel lines share slope; convert to standard form.

Solution

Step 1: Distribute both sides of line p's equation: $6y - 3x = 4y + 24$.

Step 2: Collect y-terms on one side: $6y - 4y = 3x + 24$, so $2y = 3x + 24$.

Step 3: Solve for y: $y = (3/2)x + 12$, so line p has slope $3/2$.

Step 4: Line q is parallel to line p, so line q also has slope $3/2$.

Step 5: Use point-slope form with (4, -1): $y - (-1) = (3/2)(x - 4)$, so $y + 1 = (3/2)(x - 4)$.

Step 6: Multiply both sides by 2 to clear the fraction: $2y + 2 = 3(x - 4) = 3x - 12$.

Step 7: Rearrange into standard form: $3x - 2y = 14$.

Answer: $3x - 2y = 14$

Tip

When an equation is wrapped in parentheses or mixes x and y on both sides, distribute and collect like terms first -- don't try to read off the slope before it's isolated.

Trap

A rushed student distributes only part of the equation or mishandles the coefficient when isolating y, landing on slope 3 instead of the correct slope $3/2$, which produces the wrong final equation for line q.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$3x - y = 13$	Mishandled isolating y from $2y = 3x + 24$, dividing only the x-term by 2 and not the constant, landing on slope 3 instead of $3/2$, then building line q's equation with that wrong slope.
$3x - 2y = 10$	Used the correct slope $3/2$ but dropped the negative sign on -1 when substituting the point, writing $y - 1 = (3/2)(x - 4)$ instead of $y - (-1) = (3/2)(x - 4)$.
$3x - 2y = 6$	Forgot to multiply the constant by 3 when distributing on the right side, writing $3(x - 4)$ as $3x - 4$ instead of $3x - 12$, which shifted the final constant term.

Shortcut

Once you know line q needs the same slope as line p, you can skip fully solving line p for y -- just read the slope directly as the ratio of coefficients from $2y = 3x + 24$ ($3/2$) without finishing the simplification.

Desmos Tip

Graph $3(2y - x) = 4(y + 6)$ directly (Desmos accepts implicit equations) alongside $3x - 2y = 14$ and confirm the two lines never intersect and that the second passes through (4, -1).

**Lines — Problem 3** (Very Hard)

Point P has coordinates $(k, 3)$, and point Q has coordinates $(-2, -3)$. The distance between P and Q is 10 units. What is the sum of all possible values of k ?

Formula / Rule Used: Distance formula, solved as a quadratic for an unknown coordinate.

Solution

Step 1: Apply the distance formula: $d = \sqrt{(k - (-2))^2 + (3 - (-3))^2}$.

Step 2: Substitute $d = 10$: $10 = \sqrt{(k + 2)^2 + 6^2}$.

Step 3: Square both sides: $100 = (k + 2)^2 + 36$.

Step 4: Isolate the squared term: $(k + 2)^2 = 64$.

Step 5: Take the square root of both sides, keeping both signs: $k + 2 = 8$ or $k + 2 = -8$.

Step 6: Solve each case: $k = 6$ or $k = -10$.

Step 7: Add the two possible values: $6 + (-10) = -4$.

Answer: -4

Tip

Whenever a distance-formula problem gives you the distance and asks for a coordinate, expect a squared equation with two valid solutions -- solve it like any other quadratic and keep both roots unless the context rules one out.

Trap

Many students take the square root of $(k+2)^2 = 64$ and keep only the positive root ($k = 6$), missing $k = -10$ entirely and reporting an incomplete answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
6	Took the square root of $(k+2)^2 = 64$ and kept only the positive root ($k+2 = 8$, so $k = 6$), discarding the negative branch entirely and reporting that single value instead of the sum of both roots.
4	Misread the x-coordinate of Q as +2 instead of -2, so assumed the two k -values were symmetric about $k = 2$ and reported their sum as $2(2) = 4$ instead of $2(-2) = -4$.
-60	Correctly found both roots $k = 6$ and $k = -10$, but multiplied them together instead of adding them, computing the product (-60) instead of the requested sum (-4) .

Shortcut

Because the problem only asks for the SUM of the possible k -values, you don't need to solve both equations individually: the two roots of $(k+2)^2 = 64$ are symmetric around $k = -2$, so their sum is always $2(-2) = -4$.

Desmos Tip



Graph the circle $(x + 2)^2 + (y + 3)^2 = 100$ centered at Q with radius 10, then find where it crosses the horizontal line $y = 3$ -- the x-coordinates of the two intersection points are the two values of k.

Lines — Problem 4 (Hard)

A bike-rental kiosk in Rotterdam charges a fixed unlocking fee plus a constant rate per minute. Nikos rents a bike for 15 minutes and pays 4.75 euros total. His friend Elena rents a bike for 40 minutes and pays 9.00 euros total. Assuming cost is a linear function of rental time, how much would a 70-minute rental cost, in euros?

Formula / Rule Used: Build a linear model $C = mt + b$ from two data points, then evaluate it.

Solution

Step 1: Translate the two rentals into coordinate points (time, cost): (15, 4.75) and (40, 9.00).

Step 2: Find the rate (slope) using these points: $m = (9.00 - 4.75)/(40 - 15) = 4.25/25 = 0.17$ euros per minute.

Step 3: Use point-slope form with (15, 4.75) to find the fixed fee (y-intercept): $4.75 = 0.17(15) + b$.

Step 4: Compute $0.17(15) = 2.55$, so $4.75 = 2.55 + b$, giving $b = 2.20$ euros.

Step 5: Write the full cost model: $C = 0.17t + 2.20$.

Step 6: Substitute $t = 70$: $C = 0.17(70) + 2.20 = 11.90 + 2.20$.

Step 7: Add: $C = 14.10$ euros.

Answer: 14.10 euros

Tip

In a linear word problem, immediately convert each described scenario into an (input, output) ordered pair -- once you have two points, it's a standard slope-and-intercept problem.

Trap

A student who only computes the per-minute rate and multiplies it by 70 (getting $0.17 \times 70 = 11.90$) forgets the fixed unlocking fee entirely and arrives at the wrong total.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
11.90 euros	Computed only the per-minute rate (0.17) and multiplied it by 70, forgetting to add the fixed unlocking fee of 2.20 euros.
15.75 euros	Rounded the per-minute rate from 0.17 to 0.2 too early, then carried that rounding error through the intercept ($b = 1.75$) and the final calculation instead of keeping the exact rate.
11.55 euros	Misread the question as asking for the cost of 70 additional minutes beyond Nikos's original 15-minute rental, substituting $t = 55$ instead of $t = 70$ into the cost model.

Shortcut



No shortcut beats setting up the two points and solving for m and b directly -- this is already the fastest reliable path for a two-point linear word problem.

Desmos Tip

Enter the two points (15, 4.75) and (40, 9.00) in a Desmos table, graph $y = 0.17x + 2.20$ through them, and read the y -value at $x = 70$ directly off the graph.

Lines — Problem 5 (Hard)

Triangle ABC has vertices A(-4, -3), B(6, -3), and C(2, 9). Side AB is horizontal, and the slope of side BC is -3. Find the perimeter of triangle ABC, in exact simplified radical form.

Formula / Rule Used: Distance formula applied to each side of a triangle to find perimeter.

Solution

Step 1: Since A and B share the same y -coordinate (-3), side AB is horizontal, so its length is simply the difference in x -coordinates: $AB = |6 - (-4)| = 10$.

Step 2: The slope of BC is given but not needed to find lengths -- apply the distance formula directly to B(6, -3) and C(2, 9): $BC = \sqrt{(2-6)^2 + (9-(-3))^2}$.

Step 3: Compute inside the square root: $(2-6)^2 = 16$ and $(9-(-3))^2 = 144$, so $BC = \sqrt{16 + 144} = \sqrt{160}$.

Step 4: Simplify $\sqrt{160} = \sqrt{16 \times 10} = 4\sqrt{10}$.

Step 5: Apply the distance formula to A(-4, -3) and C(2, 9): $AC = \sqrt{(2-(-4))^2 + (9-(-3))^2}$.

Step 6: Compute inside the square root: $(2-(-4))^2 = 36$ and $(9-(-3))^2 = 144$, so $AC = \sqrt{36 + 144} = \sqrt{180}$.

Step 7: Simplify $\sqrt{180} = \sqrt{36 \times 5} = 6\sqrt{5}$.

Step 8: Add all three sides for the perimeter: $\text{Perimeter} = 10 + 4\sqrt{10} + 6\sqrt{5}$.

Answer: $10 + 4\sqrt{10} + 6\sqrt{5}$ units (approximately 36.0 units)

Tip

For a perimeter or side-length problem, apply the distance formula to each pair of vertices directly -- don't try to use slope information, which only tells you direction, not length.

Trap

The slope of BC (-3) is a tempting piece of data, and a student might try to fold it into a length calculation (treating it as a ratio of legs), which doesn't give the actual side length and wastes time or produces a wrong value.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$2 + 4\sqrt{10} + 6\sqrt{5}$ units (approximately 28.1 units)	Misread vertex A's x -coordinate as 4 instead of -4, computing $AB = 6 - 4 = 2$ instead of the correct $AB = 6 - (-4) = 10$.
$10 + 4\sqrt{10} + 3\sqrt{2}$ units (approximately 27.9 units)	When computing AC, added the coordinate differences directly instead of squaring them first: $\sqrt{6 + 12} = \sqrt{18} = 3\sqrt{2}$, instead of correctly



	squaring to get $\sqrt{36 + 144} = \sqrt{180} = 6\sqrt{5}$.
10 + 4sqrt(10) units (approximately 22.6 units)	Computed AB and BC correctly but forgot to include side AC in the perimeter sum, stopping after only two of the three sides.

Shortcut

Recognizing that AB is horizontal (equal y-coordinates) lets you find that side by simple subtraction of x-values instead of running the full distance formula, saving a step on one of the three sides.

Desmos Tip

Plot the three vertices in Desmos, draw the triangle with the polygon feature, and use the distance measurement between each pair of points to confirm $BC = \sqrt{160}$ and $AC = \sqrt{180}$ numerically.

Lines — Problem 6 (Very Hard)

Line r is given by $kx + 3y = 12$, and line s is given by $2x - 5y = 10$. If line r is perpendicular to line s , what is the x-intercept of line r ?

Formula / Rule Used: Standard-form slope ($-A/B$), perpendicular negative reciprocal, then x-intercept.

Solution

Step 1: Find the slope of line s from standard form $Ax + By = C$, where slope = $-A/B$: slope of $s = -2/(-5) = 2/5$.

Step 2: Since line r is perpendicular to line s , its slope must be the negative reciprocal of $2/5$, which is $-5/2$.

Step 3: Find the slope of line r in terms of k from its standard form $kx + 3y = 12$: slope of $r = -k/3$.

Step 4: Set the two expressions for line r 's slope equal: $-k/3 = -5/2$.

Step 5: Solve for k : $k/3 = 5/2$, so $k = 15/2$.

Step 6: Substitute $k = 15/2$ back into line r 's equation: $(15/2)x + 3y = 12$.

Step 7: Multiply every term by 2 to clear the fraction: $15x + 6y = 24$, which simplifies (divide by 3) to $5x + 2y = 8$.

Step 8: Find the x-intercept by setting $y = 0$: $5x = 8$, so $x = 8/5$.

Answer: x-intercept = $(8/5, 0)$

Tip

When a line's equation contains an unknown coefficient and a perpendicularity or parallelism condition, first write both slopes in terms of A and B (slope = $-A/B$ for standard form), then set up one equation to solve for the unknown before touching the rest of the problem.

Trap

A student who confuses the perpendicular condition with the parallel condition sets $-k/3 = 2/5$ (matching slopes instead of using negative reciprocals), which yields the wrong value of k and a wrong final intercept.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
x-intercept = (-10, 0)	Confused the perpendicular condition with the parallel condition, setting $-k/3 = 2/5$ (matching slopes) instead of $-k/3 = -5/2$ (negative reciprocal), which gave $k = -6/5$ and a completely different line.
x-intercept = (72/5, 0)	Correctly set up $k/3 = 5/2$ but solved it by dividing $5/2$ by 3 instead of multiplying by 3, getting $k = 5/6$ instead of the correct $k = 15/2$.
x-intercept = (0, 4)	Found the correct simplified line $5x + 2y = 8$ but set $x = 0$ instead of $y = 0$, computing the y-intercept (0, 4) and mislabeling it as the x-intercept.

Shortcut

There isn't a faster route than solving for k first -- the x-intercept genuinely depends on k , so both steps (solve for k , then find the intercept) are necessary in this order.

Desmos Tip

After solving for k , graph $5x + 2y = 8$ and $2x - 5y = 10$ together in Desmos and confirm visually that the lines cross at a right angle and that the first line crosses the x-axis at $8/5 = 1.6$.

Lines — Problem 7 (Very Hard)

For what value of k does the system of equations $4x - 6y = 10$ and $kx + 9y = -20$ have no solution?

Formula / Rule Used: Standard-form slope ($-A/B$); parallel lines share slope but must have different constants for a true 'no solution' case.

Solution

Step 1: Find the slope of the first line from standard form: $4x - 6y = 10$ has slope $= -A/B = -4/(-6) = 2/3$.

Step 2: For the system to have no solution, the two lines must be parallel (equal slopes) but not identical. Find the slope of the second line in terms of k : $kx + 9y = -20$ has slope $= -k/9$.

Step 3: Set the slopes equal: $-k/9 = 2/3$.

Step 4: Solve for k : multiply both sides by 9: $-k = 6$, so $k = -6$.

Step 5: Check that the lines aren't identical when $k = -6$ (which would give infinitely many solutions instead of no solution): substitute $k = -6$, so the second line becomes $-6x + 9y = -20$.

Step 6: Scale the first line ($4x - 6y = 10$) by -1.5 to match the second line's x-coefficient: $-6x + 9y = -15$.

Step 7: Since -15 does not equal -20 , the two lines are parallel but distinct, confirming the system truly has no solution rather than infinitely many.

Step 8: Therefore $k = -6$.

Answer: $k = -6$

Tip

Whenever a system-of-equations problem asks for 'no solution', translate it as 'same slope, different intercept' -- find both lines' slopes from standard form using $-A/B$ and set them equal.

Trap



The most common error is misreading the standard-form slope formula $-A/B$ when B is already negative (here $B = -6$): students often compute $-4/6 = -2/3$ instead of correctly canceling the double negative to get $2/3$, which flips the sign of the final answer.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
k = 6	Miscomputed the slope of the first line from standard form as $-4/6 = -2/3$ (mishandling the double negative from $B = -6$), instead of the correct $2/3$, which flipped the sign of the final answer.
k = 13.5	Used the perpendicular condition (product of slopes = -1) instead of the parallel/no-solution condition (equal slopes), solving $(2/3)(-k/9) = -1$ instead of $-k/9 = 2/3$.
k = -9	Made an arithmetic slip computing $9 \times (2/3)$, getting 9 instead of 6, so $-k = 9$ was solved as $k = -9$ instead of the correct $k = -6$.

Shortcut

Skip computing each slope separately -- cross-multiply the coefficients directly (parallel condition $A_1 \times B_2 = A_2 \times B_1$): $4(9) = k(-6)$, so $36 = -6k$, giving $k = -6$ in one line.

Desmos Tip

Graph $4x - 6y = 10$ and $kx + 9y = -20$ for a few trial k -values near -6 and watch the second line rotate until it looks parallel to the first, then confirm $k = -6$ makes the lines never touch.

Lines — Problem 8 (Hard)

Line w passes through the points $(2, 5)$ and $(6, -3)$. Line w crosses the x -axis at point R and the y -axis at point S . What is the area of triangle ORS , where O is the origin?

Formula / Rule Used: Slope formula, point-slope form, x - and y -intercepts, triangle area = $(1/2)(\text{base})(\text{height})$.

Solution

Step 1: Find the slope of line w using the two given points, keeping the same point order in numerator and denominator: $m = (-3 - 5)/(6 - 2) = -8/4 = -2$.

Step 2: Use point-slope form with $(2, 5)$: $y - 5 = -2(x - 2)$.

Step 3: Simplify: $y - 5 = -2x + 4$, so $y = -2x + 9$.

Step 4: Find the y -intercept (point S) by setting $x = 0$: $y = -2(0) + 9 = 9$, so $S = (0, 9)$.

Step 5: Find the x -intercept (point R) by setting $y = 0$: $0 = -2x + 9$, so $2x = 9$, $x = 9/2 = 4.5$, so $R = (4.5, 0)$.

Step 6: Triangle ORS is a right triangle with legs OS (along the y -axis, length 9) and OR (along the x -axis, length 4.5), since the axes meet at the origin at a right angle.

Step 7: Area = $(1/2)(\text{base})(\text{height}) = (1/2)(4.5)(9) = (1/2)(40.5) = 20.25$.

Answer: Area = 20.25 square units ($81/4$).

**Tip**

When a triangle is formed by a line and both coordinate axes, find the line's x- and y-intercepts first -- those two intercepts are automatically the legs of a right triangle with the origin, so the area formula applies directly.

Trap

The most common slip is losing track of point order when computing slope from two points -- pairing the y-values in one order but the x-values in the reverse order silently flips the sign of the slope, which then shifts both intercepts and produces a completely different (and much smaller) triangle.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Area = 0.25 (1/4) square units	Mismatched the point order in the slope formula, computing $m = (-3-5)/(2-6) = 2$ instead of pairing coordinates in the same order ($m = (-3-5)/(6-2) = -2$), which flipped the sign of the slope and shrank both intercepts.
Area = 40.5 square units	Found both intercepts correctly ($R = (4.5, 0)$, $S = (0, 9)$) but forgot to multiply by $1/2$ in the triangle area formula, computing base $\times$ height instead of $(1/2)(\text{base})(\text{height})$.
Area = 22.5 square units	Correctly found the line $y = -2x + 9$, but rounded the x-intercept $9/2 = 4.5$ up to 5 instead of keeping the exact value, then computed $(1/2)(5)(9) = 22.5$ instead of $(1/2)(4.5)(9) = 20.25$.

Shortcut

You can't skip any of the three formulas here, but you can save a step by reading the y-intercept b directly from $y = mx + b$ using either point ($5 = -2(2) + b$, so $b = 9$) instead of writing out full point-slope form first.

Desmos Tip

Graph the line through (2,5) and (6,-3) in Desmos -- it will automatically display where the line crosses each axis, so you can read R and S directly and check the area with Desmos's polygon/area tool on triangle O, R, S .

2. Systems of Linear Equations

Systems of Linear Equations — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
System of equations	Σύστημα εξισώσεων
Solution (of a system)	Λύση συστήματος
Substitution method	Μέθοδος αντικατάστασης
Elimination method	Μέθοδος απαλοιφής
No solution / infinite solutions	Καμία λύση / άπειρες λύσεις

Systems of Linear Equations — Condensed Theory

A system of two linear equations is solved where their graphs intersect. The SAT tests both finding that solution and reasoning about how many solutions exist without fully solving.

Substitution

Isolate one variable in one equation, substitute it into the other, solve, then back-substitute. Best when a variable already has coefficient 1.

Elimination

Multiply one or both equations so a variable's coefficients become opposites, then add the equations to cancel that variable.

Setting Equal

$$f(x) = g(x)$$

If both equations are already solved for y, set the right-hand sides equal and solve for x directly.

Solving Directly for an Expression (shortcut)

If asked for a combined value like $2x + 3y$ rather than x and y individually, check whether adding/subtracting/scaling the two equations produces that expression directly — this is almost always faster than solving the full system.

Number of Solutions

One: $m_1 \neq m_2$. **None:** $m_1 = m_2$, $b_1 \neq b_2$. **Infinite:** $m_1 = m_2$, $b_1 = b_2$.

In standard form $Ax + By = C$, no solution means $A_1/A_2 = B_1/B_2 \neq C_1/C_2$; infinite solutions means $A_1/A_2 =$



$$B1/B2 = C1/C2.$$

⚠ Watch Out

For a 'no solution' or 'infinite solutions' question with an unknown constant, use the coefficient-ratio shortcut instead of solving the whole system — it turns the problem into a simple proportion.

Systems of Linear Equations — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Systems of Linear Equations — Problem 1 (Very Hard)

Consider the system $2x = 11 - 5y$ and $ky = 30 - 6x$, where k is a constant. For what value of k does this system have no solution?

Formula / Rule Used: No-solution condition for a linear system: $A1/A2 = B1/B2$ does not equal $C1/C2$ in standard form $Ax + By = C$.

Solution

Step 1: Rewrite the first equation in standard form: $2x + 5y = 11$.

Step 2: Rewrite the second equation in standard form: $6x + ky = 30$.

Step 3: For no solution, the coefficient ratios must match: $2/6 = 5/k$.

Step 4: Cross-multiply: $2k = 30$, so $k = 15$.

Step 5: Check that the constant ratio is genuinely different (required for no solution, not infinite solutions): $C1/C2 = 11/30$, while $A1/A2 = 2/6 = 1/3 = 10/30$.

Step 6: Since $11/30$ does not equal $10/30$, the lines are parallel but distinct at $k = 15$, confirming no solution.

Answer: $k = 15$

Tip

Before comparing ratios, rearrange every equation into $Ax + By = C$ form -- don't try to compare coefficients while variables and constants are mixed on both sides.

Trap

A student who finds $k = 15$ from the coefficient ratio alone often stops and assumes matching slopes automatically means infinitely many solutions; they must also check that the constant ratio ($11/30$) differs from the coefficient ratio ($1/3$) to confirm this is actually a no-solution case.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = -15$	Rearranged $ky = 30 - 6x$ incorrectly as $6x - ky = 30$ (sign error on the y -term) instead of $6x + ky = 30$, which flips the sign in the ratio and produces



	$k = -15$ instead of $k = 15$.
$k = 12/5$	Mispaired the coefficients when writing the ratio, using $5/6 = 2/k$ (y-coefficient of equation 1 over x-coefficient of equation 2) instead of the correct $2/6 = 5/k$, mixing up which coefficients correspond to which equation.
$k = 5/3$	Cross-multiplied the proportion $2/6 = 5/k$ backwards, computing $2 \times 5 = 6 \times k$ instead of $2 \times k = 6 \times 5$, giving $k = 5/3$ instead of 15.

Shortcut

Once both equations are in standard form, skip solving the system entirely -- just cross-multiply the x- and y-coefficient ratio ($2/6 = 5/k$) to get $k = 15$ in one line.

Desmos Tip

Graph $2x + 5y = 11$ and $6x + 15y = 30$ (solved for y) in Desmos and confirm the two lines are visibly parallel and never touch.

Systems of Linear Equations — Problem 2 (Hard)

In the system below, the two lines intersect at a point located in Quadrant I of the xy -plane. $2x + y = 9$ and $x + 3y = 16$. What is the value of $3x + 4y$?

Formula / Rule Used: Linear combination shortcut: add or scale the two equations directly to build the target expression without solving for x and y .

Solution

Step 1: Notice the target expression $3x + 4y$ can be built by adding the two equations as given: $(2x + y) + (x + 3y) = 3x + 4y$.

Step 2: Add the left sides: $2x + y + x + 3y = 3x + 4y$.

Step 3: Add the right sides: $9 + 16 = 25$.

Step 4: Therefore $3x + 4y = 25$ directly, with no need to solve for x and y individually.

Step 5: As a check, solve the system fully: from equation 1, $y = 9 - 2x$; substitute into equation 2: $x + 3(9 - 2x) = 16$, so $x + 27 - 6x = 16$, giving $-5x = -11$ and $x = 11/5$.

Step 6: Then $y = 9 - 2(11/5) = 9 - 22/5 = 23/5$, and both values are positive, matching the Quadrant I detail (which is not needed to answer the question).

Step 7: Confirm: $3(11/5) + 4(23/5) = 33/5 + 92/5 = 125/5 = 25$, matching the shortcut answer.

Answer: $3x + 4y = 25$

Tip

Before solving a system fully, check whether the target expression is already a simple sum, difference, or scaled combination of the two given equations -- if it is, add or subtract directly instead of finding x and y .

Trap



A student who sees a two-variable system and a geometric detail like 'Quadrant I' often assumes they must fully solve for x and y (landing on messy fractions like $x = 11/5$) before computing $3x + 4y$, wasting time and risking an arithmetic slip -- the quadrant location is never actually used.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-7	Subtracted the two equations instead of adding them, computing $(2x + y) - (x + 3y) = 9 - 16 = -7$, then mistakenly reported -7 as the value of $3x + 4y$ instead of recognizing only addition produces that target expression.
-19	While solving the system fully as a check, distributed $3(9 - 2x)$ incorrectly as $27 - 2x$ instead of $27 - 6x$, producing wrong values $x = 11$ and $y = -13$, and an incorrect final combination of $3(11) + 4(-13) = -19$.
26	Misled by the Quadrant I detail into assuming the coordinates must be 'nice' integers, rounded $x = 11/5$ to 2 and used $y = 9 - 2(2) = 5$, then computed $3(2) + 4(5) = 26$ instead of using the exact fractional solution.

Shortcut

Just add the two equations as given -- their sum already equals $3x + 4y$, so the answer appears in one line without ever isolating x or y .

Desmos Tip

Graph both lines in Desmos, find the intersection point $(2.2, 4.6)$, then compute $3(2.2) + 4(4.6) = 25$ directly from the coordinates as a check.

Systems of Linear Equations — Problem 3 (Very Hard)

Solve the system for the ordered pair (x, y) : $x/2 + 5y/6 = 9/2$ and $4x/3 - y = 7/3$.

Formula / Rule Used: Clear fractions by multiplying each equation by its LCD, then eliminate a variable by multiplying both resulting equations by different constants.

Solution

Step 1: Clear the first equation by multiplying every term by 6 (the LCD of 2 and 6): $3x + 5y = 27$.

Step 2: Clear the second equation by multiplying every term by 3 (the LCD of 3 and 1): $4x - 3y = 7$.

Step 3: To eliminate y , find a common multiple of 5 and 3 (which is 15): multiply the first cleared equation by 3 to get $9x + 15y = 81$.

Step 4: Multiply the second cleared equation by 5 to get $20x - 15y = 35$.

Step 5: Add the two new equations: $9x + 15y + 20x - 15y = 81 + 35$, so $29x = 116$.

Step 6: Solve for x : $x = 116/29 = 4$.

Step 7: Substitute $x = 4$ into $4x - 3y = 7$: $16 - 3y = 7$, so $-3y = -9$ and $y = 3$.

Answer: $(x, y) = (4, 3)$

Tip



Clear every fraction by multiplying each equation by its own LCD before attempting elimination -- never try to eliminate variables while denominators are still in play.

Trap

A student often tries to eliminate y by scaling only one cleared equation (since 5 and -3 share no simple single multiplier), instead of multiplying BOTH cleared equations (by 3 and by 5) to reach matching y -coefficients of 15, which leads to leftover fractions and arithmetic errors.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$(x, y) = (4, -23/3)$	After correctly finding $x = 4$ and substituting into $4x - 3y = 7$ to get $16 - 3y = 7$, subtracted in the wrong direction (computed $-3y = 7 + 16$ instead of $-3y = 7 - 16$), giving $y = -23/3$ instead of 3.
$(x, y) = (3, 4)$	Correctly computed the two numeric values 4 and 3 through elimination but swapped which value is x and which is y when writing the final ordered pair.
$(x, y) = (62/23, 87/23)$	Cleared $4x/3 - y = 7/3$ by multiplying only the x -term and the constant by 3, forgetting to also multiply the y -term by 3, producing the wrong equation $4x - y = 7$ instead of $4x - 3y = 7$, which throws off the entire solution.

Shortcut

No shortcut beats clearing fractions first -- trying to eliminate a variable while fractions remain multiplies the chance of arithmetic slips, so multiplying by the LCD immediately is itself the fastest path.

Desmos Tip

Graph $x/2 + 5y/6 = 9/2$ and $4x/3 - y = 7/3$ directly in Desmos and use the intersection point to confirm the solution $(4, 3)$.

Systems of Linear Equations — Problem 4 (Hard)

The system $5x - 2y = 16$ and $bx + 3y = -1$ has a solution in which $y = -3$. What is the value of $b - x$?

Formula / Rule Used: Substitute the known value into the equation without an unknown constant first, then use the result to solve for the unknown constant in the other equation.

Solution

Step 1: Substitute $y = -3$ into the first equation, which has no unknown constant: $5x - 2(-3) = 16$.

Step 2: Simplify: $5x + 6 = 16$, so $5x = 10$ and $x = 2$.

Step 3: Substitute $x = 2$ and $y = -3$ into the second equation: $b(2) + 3(-3) = -1$.

Step 4: Simplify: $2b - 9 = -1$, so $2b = 8$ and $b = 4$.

Step 5: Compute the requested expression: $b - x = 4 - 2 = 2$.

Answer: $b - x = 2$

**Tip**

When one equation contains an unknown constant and the other doesn't, always use the equation without the unknown constant first to pin down the known variable before touching the equation that has two unknowns.

Trap

A student who plugs $y = -3$ into the second equation first gets stuck with two unknowns (b and x) in one equation and often guesses or mishandles the algebra; working equation 1 first avoids this entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-7	Correctly found $x = 2$ but substituted $y = -3$ into the second equation with a sign flip, writing $2b + 3(3) = -1$ instead of $2b + 3(-3) = -1$, getting $b = -5$ instead of $b = 4$, then computed $-5 - 2 = -7$.
-2	Correctly found $b = 4$ and $x = 2$ but computed $x - b$ instead of $b - x$, reversing the order the question asks for.
6	Correctly found $b = 4$ and $x = 2$ but added the two values instead of subtracting, computing $b + x = 6$ instead of $b - x = 2$.

Shortcut

There's no way around the two substitutions here, but always resolving the 'clean' equation (the one without the unknown constant) first keeps you from ever juggling two unknowns at once.

Desmos Tip

After finding $x = 2$, graph $bx + 3y = -1$ as $y = (-1 - bx)/3$ and adjust b until the line passes through $(2, -3)$ to visually confirm $b = 4$.

Systems of Linear Equations — Problem 5 (Hard)

Cycladic Ferries sells standard and express tickets for its Piraeus-Naxos route. The express ticket costs 15 euros more than twice the price of a standard ticket. Last week, a travel agency bought 3 standard tickets and 2 express tickets for a total of 205 euros. What is the combined price, in euros, of one standard ticket and one express ticket?

Formula / Rule Used: Translate verbal relationships into algebraic equations, then substitute to reduce the system to one variable.

Solution

Step 1: Let s = the price of a standard ticket and e = the price of an express ticket.

Step 2: Translate 'the express ticket costs 15 euros more than twice the standard price' into $e = 2s + 15$.

Step 3: Translate the purchase into an equation: $3s + 2e = 205$.

Step 4: Substitute $e = 2s + 15$ into the second equation: $3s + 2(2s + 15) = 205$.

Step 5: Expand: $3s + 4s + 30 = 205$, so $7s + 30 = 205$.

Step 6: Solve for s : $7s = 175$, so $s = 25$.

Step 7: Find e : $e = 2(25) + 15 = 65$.

Step 8: Compute the combined price of one of each: $s + e = 25 + 65 = 90$ euros.

Answer: 90 euros (standard = 25 euros, express = 65 euros)

Tip

Translate the comparison sentence into an equation first ($e = 2s + 15$), then substitute it directly into the purchase equation instead of building two equations and eliminating -- it turns the system into a single-variable equation immediately.

Trap

A student may translate 'costs 15 euros more than twice the standard price' backwards as $s = 2e + 15$ instead of $e = 2s + 15$, swapping which ticket is expressed in terms of the other and producing incorrect prices for both tickets.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
75 euros	Translated the comparison sentence backwards as $s = 2e + 15$ instead of $e = 2s + 15$, swapping which ticket price is expressed in terms of the other, giving standard = 55 and express = 20 (combined 75) instead of the correct 90.
675/7 euros (about 96.43)	Substituted $e = 2s + 15$ into $3s + 2e = 205$ but distributed incorrectly as $3s + 4s + 15 = 205$ (forgot to double the 15), instead of the correct $3s + 4s + 30 = 205$, throwing off both prices.
65 euros	Correctly solved for both prices ($s = 25$, $e = 65$) but reported only the express ticket price instead of computing the combined total $s + e = 90$.

Shortcut

Since one equation already isolates e in terms of s , substitution is faster than elimination here -- there's no need to multiply or align coefficients at all.

**Desmos Tip**

Graph $e = 2s + 15$ and $3s + 2e = 205$ (solved for e) in Desmos; the intersection point (25, 65) gives both ticket prices directly, and their sum confirms 90.

Systems of Linear Equations — Problem 6 *(Very Hard)*

The system $6x - 9y = 3k$ and $8x - 12y = 20$ has infinitely many solutions, where k is a constant. What is the value of $k^2 - 5$?

Formula / Rule Used: Infinite-solutions ratio shortcut: $A_1/A_2 = B_1/B_2 = C_1/C_2$ for the two equations to represent the same line.

Solution

Step 1: Compare the x-coefficients: $A_1/A_2 = 6/8 = 3/4$.

Step 2: Compare the y-coefficients: $B_1/B_2 = -9/-12 = 3/4$, which matches, so the two lines already share a slope.

Step 3: For infinitely many solutions, the constant ratio must also equal $3/4$: $3k/20 = 3/4$.

Step 4: Cross-multiply: $3k$ times $4 = 3$ times 20 , so $12k = 60$.

Step 5: Solve for k : $k = 5$.

Step 6: Substitute into the requested expression: $k^2 - 5 = 5^2 - 5 = 25 - 5 = 20$.

Answer: 20

Tip

Whenever a system is described as having infinitely many solutions, immediately set up $A_1/A_2 = B_1/B_2 = C_1/C_2$ instead of trying to 'solve' the system -- there is no unique point to find, only a constant to pin down.

Trap

A student who overlooks the coefficient 3 in front of k and solves $6/8 = k/20$ instead of $6/8 = 3k/20$ gets $k = 15$ (not 5), which leads to a final answer of 220 instead of the correct 20.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
220	Set up the constant ratio as $6/8 = k/20$ instead of $6/8 = 3k/20$, forgetting the coefficient 3 in front of k , getting $k = 15$ instead of 5, then computed $15^2 - 5 = 220$.
0	Correctly found $k = 5$ but misread the requested expression $k^2 - 5$ as $(k - 5)^2$, computing $(5 - 5)^2 = 0$ instead of $5^2 - 5 = 20$.
5995/81 (about 74.0)	Cross-multiplied the proportion $3k/20 = 3/4$ backward, computing $3k \times 3 = 20 \times 4$ instead of $3k \times 4 = 20 \times 3$, giving $k = 80/9$ instead of 5, then computed $(80/9)^2 - 5$.

Shortcut



You never need to find x or y at all -- the entire problem collapses to one proportion ($3k/20 = 3/4$) once the coefficient ratios are confirmed to already match.

Desmos Tip

Substitute $k = 5$ back into $6x - 9y = 15$ and graph it alongside $8x - 12y = 20$ (solved for y) to confirm the two lines overlap exactly, verifying infinitely many solutions.

Systems of Linear Equations — Problem 7 (Very Hard)

Let m equal the value of y in the solution to the system $x + y = 15$ and $x - y = 3$. Using this value of m , for what value of n does the system $3x + 4y = 7$ and $mx + ny = -2$ have no solution?

Formula / Rule Used: Solve a preliminary system to fix an unknown coefficient, then apply the no-solution ratio condition $A1/A2 = B1/B2$, with $C1/C2$ different, to the resulting system.

Solution

Step 1: Solve the first system: add the two equations, $(x + y) + (x - y) = 15 + 3$, so $2x = 18$ and $x = 9$.

Step 2: Subtract the two equations instead to get y directly: $(x + y) - (x - y) = 15 - 3$, so $2y = 12$ and $y = 6$.

Step 3: So $m = 6$, the value of y from the first system.

Step 4: Substitute $m = 6$ into the second system, which becomes $3x + 4y = 7$ and $6x + ny = -2$.

Step 5: For this system to have no solution, the coefficient ratios must be equal: $3/6 = 4/n$.

Step 6: Simplify the left side: $3/6 = 1/2$, so $1/2 = 4/n$.

Step 7: Cross-multiply: $n = 8$.

Step 8: Confirm this is a true no-solution case (not infinitely many): the constant ratio is $7/(-2) = -3.5$, which does not equal $1/2$, so the lines are parallel but distinct.

Answer: $n = 8$

Tip

When a coefficient in a ratio-condition problem is defined as 'the value of' a specific variable from a separate system, fully solve that system first and clearly identify which variable (x or y) supplies the coefficient before setting up any ratio.

Trap

The two-stage setup tempts students to rush the first system and grab the wrong variable (x instead of y) for m , or to mismatch which coefficient belongs in the numerator versus denominator when writing the ratio $3/6 = 4/n$, since the problem never explicitly labels which equation is 'first' and which is 'second'.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$n = 12$	Used the x -value (9) from the first system as m instead of the y -value (6), then solved the ratio $3/9 = 4/n$.
$n = 2$	Correctly found $m = 6$ but inverted the coefficient ratio, comparing the second equation's x -coefficient to the first's (6/3) instead of the first's to the second's (3/6), while keeping the y -coefficients in the original order.



$n = -8$	Correctly found $m = 6$ but made a sign error setting up the ratio equation, writing $3/6 = -4/n$ instead of $3/6 = 4/n$, which flips the sign of the resulting n .
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Shortcut

The first system's equations sum directly to $2x = 18$ and subtract directly to $2y = 12$, so you can read off both x and y in one line each without full elimination setup -- useful for confirming $m = 6$ quickly.

Desmos Tip

Graph $x + y = 15$ and $x - y = 3$ to confirm the intersection $(9, 6)$, then graph $3x + 4y = 7$ and $6x + 8y = -2$ (solved for y) to visually confirm the lines are parallel and never intersect.

Systems of Linear Equations — Problem 8 (Hard)

Solve the system $3x + 4y = 7$ and $2x - 3y = 16$. What is the value of $x - y$?

Formula / Rule Used: Elimination by scaling both equations to a common coefficient, then compute the specific combination the question actually asks for.

Solution

Step 1: To eliminate y , find the LCM of 4 and 3, which is 12: multiply the first equation by 3 to get $9x + 12y = 21$.

Step 2: Multiply the second equation by 4 to get $8x - 12y = 64$.

Step 3: Add the two new equations: $9x + 12y + 8x - 12y = 21 + 64$, so $17x = 85$.

Step 4: Solve for x : $x = 85/17 = 5$.

Step 5: Substitute $x = 5$ into $2x - 3y = 16$: $10 - 3y = 16$, so $-3y = 6$ and $y = -2$.

Step 6: Compute the requested expression: $x - y = 5 - (-2) = 5 + 2 = 7$.

Answer: $x - y = 7$

Tip

Once you solve for x and y , reread exactly what expression the question wants ($x - y$, not x alone, not y alone, not $y - x$) before writing your final answer.

Trap

This problem is built so that $x = 5$ and $y = -2$ are both 'clean' numbers on their own, tempting a student to report just $x = 5$ (answering 'what is x ' instead of 'what is $x - y$ ') or to subtract in the wrong order -- either looks like a perfectly reasonable final answer if you stop reading carefully.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
5	Solved correctly for $x = 5$ but reported it directly, forgetting the question asks for $x - y$, not x alone.
-2	Solved correctly for $y = -2$ but reported it directly, mistakenly treating y as



	the requested value instead of computing $x - y$.
-7	Correctly found $x = 5$ and $y = -2$ but computed $y - x$ instead of $x - y$, subtracting in the reversed order and getting -7 instead of 7.

Shortcut

No shortcut beats solving fully here -- since the question asks for a specific combination ($x - y$) rather than a simple sum or difference of the two original equations as given, you must find x and y individually before combining them.

Desmos Tip

Graph $3x + 4y = 7$ and $2x - 3y = 16$ in Desmos, read the intersection point $(5, -2)$, then compute $5 - (-2) = 7$ directly from the coordinates.

3. Linear Inequalities

Linear Inequalities — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Inequality	Ανίσωση
Solution set	Σύνολο λύσεων
Compound inequality	Σύνθετη ανίσωση
Flip the inequality sign	Αναστροφή συμβόλου ανίσωσης
Boundary line / shaded region	Οριακή ευθεία / σκιασμένη περιοχή

Linear Inequalities — Condensed Theory

Inequalities are solved like equations, with one critical exception: multiplying or dividing by a negative number reverses the inequality sign.

The Flip Rule

Multiply/divide by a negative → reverse the sign.

Adding or subtracting (even negative numbers) never flips the sign — only multiplying or dividing by a negative does.

Compound (Three-Part) Inequalities

$a < x < b$

Apply the same operation to all three parts simultaneously; flip both signs if you multiply/divide by a negative.

Graphing in the xy-Plane

Solid line for $\leq / \geq$ (boundary included); dashed line for $< / >$ (boundary excluded). Shade above for $y > / \geq$, below for $y < / \leq$, or test a point like (0,0).

Systems of Inequalities

Graph each inequality; the solution is the overlapping shaded region. To check a point, substitute it into every inequality.

Word Problems

'at least/no less than' → $\geq$. 'at most/no more than' → $\leq$.



After solving, check whether the context requires an integer answer, and round down for a maximum or up for a minimum.

Watch Out

Real-world quantities (people, tickets, boxes) often force you to round the algebraic answer to the nearest valid integer — always re-read what's being asked (max vs. min) before rounding.

Linear Inequalities — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Linear Inequalities — Problem 1 (Hard)

The inequality $3(x - 4) - 5x \geq k - 15$ has solution set $x \leq 7$. What is the value of k ?

Formula / Rule Used: Distribute and combine like terms, then divide by a negative number (which flips the inequality) to isolate x ; reverse-engineer an unknown constant from a given solution set.

Solution

Step 1: Expand the left side: $3(x - 4) - 5x = 3x - 12 - 5x = -2x - 12$, so the inequality becomes $-2x - 12 \geq k - 15$.

Step 2: Add 15 to both sides: $-2x - 12 + 15 \geq k$, which simplifies to $-2x + 3 \geq k$, or equivalently $-2x \geq k - 3$.

Step 3: Divide both sides by -2 to isolate x . Dividing by a negative number reverses the inequality sign: $x \leq (k - 3)/(-2)$, which simplifies to $x \leq (3 - k)/2$.

Step 4: The problem states the solution set is $x \leq 7$, so set the boundary expressions equal: $(3 - k)/2 = 7$.

Step 5: Multiply both sides by 2: $3 - k = 14$.

Step 6: Solve for k : $-k = 14 - 3 = 11$, so $k = -11$.

Step 7: Check: substitute $k = -11$ into the original inequality: $3(x-4) - 5x \geq -11 - 15 = -26$. The left side simplifies to $-2x - 12$, so $-2x - 12 \geq -26$, giving $-2x \geq -14$, and dividing by -2 (flip) gives $x \leq 7$. Confirmed.

Answer: $k = -11$

Tip

When a solution set is given and you need to find a hidden constant, solve the inequality symbolically in terms of that constant first, then match the resulting boundary expression to the given boundary value.

Trap

A student who forgets to flip the inequality sign when dividing by -2 will set up the equation $(k-3)/2 = 7$ instead of $(3-k)/2 = 7$, leading to the wrong value $k = 11$ instead of the correct $k = -11$.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



k = 11	Divided $-2x \geq k - 3$ by -2 without reversing the inequality, setting up $(k-3)/2 = 7$ instead of $(3-k)/2 = 7$, giving $k = 11$ instead of -11 .
k = -3	Distributed $3(x-4)$ incorrectly as $3x - 4$ (forgetting to multiply the -4 by 3 as well), which shifts the constant term and yields $k = -3$ instead of -11 .
k = 13	Mis-copied the -12 term as $+12$ when isolating k , writing $-2x + 12 \geq k - 15$ instead of $-2x - 12 \geq k - 15$, which produces $k = 13$ instead of -11 .

Shortcut

Skip solving the general inequality entirely: plug the boundary value $x = 7$ directly into the original expression as an equality, $-2(7) - 12 = k - 15$, which gives $k = -11$ in one line.

Desmos Tip

Graph $y = -2x - 12$ and $y = -26$ (using $k = -11$) and confirm the graphs intersect at $x = 7$, matching the given boundary.

Linear Inequalities — Problem 2 (Very Hard)

Solve the compound inequality for x , and express the solution set using inequality notation: $-3 < (2 - 5x)/3 \leq 4$.

Formula / Rule Used: Compound (three-part) inequality: perform the same operation on all three parts, reversing both inequality signs and swapping the endpoint order when multiplying or dividing by a negative number.

Solution

Step 1: Multiply all three parts by 3 to clear the denominator: $-3 \times 3 < 2 - 5x \leq 4 \times 3$, giving $-9 < 2 - 5x \leq 12$.

Step 2: Subtract 2 from all three parts: $-9 - 2 < -5x \leq 12 - 2$, giving $-11 < -5x \leq 10$.

Step 3: Divide all three parts by -5 . Since -5 is negative, reverse both inequality signs and flip the order of the parts: $-11/-5 > x \geq 10/-5$.

Step 4: Simplify each side: $-11/-5 = 11/5$ and $10/-5 = -2$, so the inequality reads $11/5 > x \geq -2$.

Step 5: Rewrite in standard increasing order (smallest to largest, left to right): $-2 \leq x < 11/5$.

Step 6: Check an endpoint: at $x = -2$, $(2 - 5(-2))/3 = (2 + 10)/3 = 12/3 = 4$, which satisfies ≤ 4 with equality, confirming $x = -2$ is correctly included.

Step 7: Check the other endpoint: at $x = 11/5$, $(2 - 5(11/5))/3 = (2 - 11)/3 = -9/3 = -3$, which is NOT strictly greater than -3 (it equals -3), confirming $x = 11/5$ is correctly excluded since the original inequality was strict ($>$) on that side.

Answer: $-2 \leq x < 11/5$

Tip

Clear the fraction first by multiplying every part by the denominator, then isolate x exactly as with a normal compound inequality, remembering to flip both inequality signs and reverse the order of the endpoints whenever you multiply or divide by a negative number.

**Trap**

A student who divides by -5 but forgets to also swap which endpoint is 'less than' and which is 'greater than' will report the answer as $11/5 \leq x < -2$, which is an empty (impossible) set since $11/5$ is greater than -2 .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$11/5 \leq x < -2$	Flipped the inequality symbols correctly when dividing by -5 but forgot to also swap the left-right order of the two endpoints, producing an impossible set where the 'lower' bound is larger than the 'upper' bound.
$-2/5 \leq x < 1$	Skipped clearing the fraction (never multiplied all three parts by 3) and instead subtracted 2 and divided by -5 directly on the un-cleared expression $(2 - 5x)/3$, shifting every boundary value.
$-2 < x \leq 11/5$	Correctly flipped the inequality direction and endpoint order but attached the strict '<' symbol to the wrong endpoint, swapping which side is open and which is closed.

Shortcut

No shortcut beats clearing the fraction first -- trying to divide by 3 termwise before combining like terms just recreates the same fraction later, so multiplying through immediately is already the fastest path.

Desmos Tip

Graph $y = (2 - 5x)/3$ along with the horizontal lines $y = -3$ and $y = 4$, then read off the x -interval where the slanted line stays between the two horizontal lines (open circle at $x = 11/5$, closed circle at $x = -2$).

Linear Inequalities — Problem 3 (Hard)

A region R in the xy-plane is defined by the system of inequalities: $y \leq -2x + 9$ and $y > \frac{1}{3}x - 1$. Determine which of the points P(3, 2), Q(0, 10), or S(5, -3) lies in region R.

Formula / Rule Used: A point satisfies a system of inequalities only if it satisfies every inequality in the system simultaneously -- substitute the point's coordinates directly into each inequality.

Solution

Step 1: Test point P(3, 2) in the first inequality: $2 \leq -2(3) + 9 = -6 + 9 = 3$. Since $2 \leq 3$ is true, P satisfies the first inequality.

Step 2: Test point P(3, 2) in the second inequality: $2 > \frac{1}{3}(3) - 1 = 1 - 1 = 0$. Since $2 > 0$ is true, P satisfies the second inequality too, so P is in region R.

Step 3: Test point Q(0, 10) in the first inequality: $10 \leq -2(0) + 9 = 9$. Since $10 \leq 9$ is false, Q fails the first inequality and is NOT in region R (no need to check the second inequality).

Step 4: Test point S(5, -3) in the first inequality: $-3 \leq -2(5) + 9 = -10 + 9 = -1$. Since $-3 \leq -1$ is true, S passes the first inequality.

Step 5: Test point S(5, -3) in the second inequality: $-3 > \frac{1}{3}(5) - 1 = \frac{5}{3} - 1 = \frac{2}{3}$. Since $-3 > \frac{2}{3}$ is false, S fails the second inequality, so S is NOT in region R.

Step 6: Only point P(3, 2) satisfies both inequalities simultaneously, so P is the point that lies in region R.

Answer: P(3, 2)

Tip

Never rearrange or graph the inequalities to check a specific point -- just substitute its x- and y-coordinates directly into each inequality as given and evaluate; it's faster and avoids sign errors.

Trap

A student who only checks the first inequality for each point (since both P and S pass it) might stop early and mistakenly accept S as a valid answer without testing the strict second inequality, missing that S fails $y > \frac{1}{3}x - 1$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
S(5, -3)	Checked only the first inequality $y \leq -2x + 9$ for each point (which both P and S satisfy) and stopped there without testing the strict second inequality, incorrectly accepting S even though it fails $y > \frac{1}{3}x - 1$.
Q(0, 10)	Checked only the second inequality $y > \frac{1}{3}x - 1$ for each point -- Q passes it easily ($10 > -1$) -- and picked Q without ever testing it against the first inequality, where it actually fails ($10 \leq 9$ is false).
No point lies in R	Misapplied the first inequality's substitution for P, computing $-2(3 + 9)$ instead of $-2(3) + 9$ (distributing across the sum instead of just the x-term), making P appear to fail the first inequality, so no point seemed to satisfy both conditions.

Shortcut



Since Q obviously fails the first inequality's y-value at $x = 0$ (10 versus the boundary value 9), testing Q first eliminates it fastest, leaving only two candidates left to fully check.

Desmos Tip

Graph both boundary lines $y = -2x + 9$ (solid) and $y = (1/3)x - 1$ (dashed) with the correct shading, then plot the three points to see visually that only P falls inside the overlapping shaded region.

Linear Inequalities — Problem 4 (Hard)

A student committee in Rotterdam is chartering a bus for a class trip. The bus rental costs a flat 340 euros plus 12 euros per student who rides it. Each student pays a 25 euro activity fee to the committee. The committee also printed 80 information flyers for parents. The committee wants the total fees collected to be at least 500 euros more than the total bus rental cost. What is the minimum number of students needed for this to happen?

Formula / Rule Used: Translate 'at least' as $\geq$, set up a linear inequality in the unknown count, then round the solution to the correct integer in the direction that keeps the inequality true.

Solution

Step 1: Let n represent the number of students riding the bus.

Step 2: Total fees collected by the committee: $25n$.

Step 3: Total bus rental cost: $340 + 12n$.

Step 4: Translate 'at least 500 euros more than the cost' into an inequality: $25n \geq (340 + 12n) + 500$.

Step 5: Simplify the right side: $25n \geq 840 + 12n$.

Step 6: Subtract $12n$ from both sides: $25n - 12n \geq 840$, so $13n \geq 840$.

Step 7: Divide both sides by 13 (positive, no flip needed): $n \geq 840/13 = 64.615\ldots$

Step 8: Since n must be a whole number of students, round UP to the next integer: $n = 65$ (rounding down to 64 would give $13(64) = 832$, which is less than 840 and fails the requirement).

Step 9: Note that the flyer count (80) is never used in the calculation -- it is irrelevant to the fee-versus-cost relationship.

Answer: 65 students

Tip

In minimum-value word problems with 'at least,' always solve the inequality first, then round the decimal boundary UP to the next whole number, since rounding down would fall short of the requirement.

Trap

A student might round $64.615\ldots$ down to 64 out of habit, or try to work the irrelevant flyer count of 80 into the equation -- explicitly verifying that $n = 64$ gives $832 < 840$ (a failure) shows why 65 is required.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
64 students	Computed $n \geq 840/13 = 64.615\ldots$ but rounded down to the nearest whole number out of habit instead of up, submitting 64, which actually gives $13(64) = 832 < 840$ and fails the requirement.



27 students	Dropped the '500 euros more than' condition entirely, setting up $25n \geq 340 + 12n$ instead of $25n \geq 340 + 12n + 500$, which gives $n = 27$, far too low to satisfy the actual requirement.
23 students	Combined the $12n$ term incorrectly by adding instead of subtracting when isolating n , turning $25n - 12n \geq 840$ into $25n + 12n \geq 840$, which gives $37n \geq 840$ and $n = 23$ instead of 65.

Shortcut

Instead of expanding $(340 + 12n) + 500$ fully, immediately combine the constants $340 + 500 = 840$ mentally before writing the inequality, saving a simplification step.

Desmos Tip

Graph $y = 25x$ and $y = 840 + 12x$ and find where the first crosses above the second; any integer x at or to the right of that intersection point (rounded up) is a valid answer.

Linear Inequalities — Problem 5 (Very Hard)

For what values of k is the slope of the line through the points $(2, k)$ and $(7, -3)$ less than -2 ?

Formula / Rule Used: Slope formula $m = (y_2 - y_1)/(x_2 - x_1)$, combined with solving a linear inequality where dividing by a negative number reverses the inequality.

Solution

Step 1: Write the slope formula using the two given points: $m = (-3 - k)/(7 - 2) = (-3 - k)/5$.

Step 2: Set up the inequality using the given condition that the slope is less than -2 : $(-3 - k)/5 < -2$.

Step 3: Multiply both sides by 5 (positive, no flip needed): $-3 - k < -10$.

Step 4: Add 3 to both sides: $-k < -10 + 3 = -7$.

Step 5: Divide both sides by -1 to isolate k . Since we're dividing by a negative number, reverse the inequality sign: $k > 7$.

Step 6: Check with a test value: let $k = 8$. Slope $= (-3 - 8)/5 = -11/5 = -2.2$, and $-2.2 < -2$ is true, confirming $k > 7$ is correct.

Step 7: Check the boundary: at $k = 7$, slope $= (-3 - 7)/5 = -10/5 = -2$, which is NOT less than -2 (it's equal), confirming $k = 7$ itself must be excluded, matching the strict inequality $k > 7$.

Answer: $k > 7$

Tip

When slope is described using an inequality, first write the slope formula normally as an expression in terms of the unknown, then solve that expression as an ordinary inequality -- treat the slope formula as just another algebraic expression.

Trap

A student who mishandles the final divide-by -1 step (forgetting to flip the sign) will conclude $k < 7$ instead of the correct $k > 7$ -- exactly backwards.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$k < 7$	Divided $-k < -7$ by -1 without reversing the inequality sign, concluding $k < 7$ instead of the correct $k > 7$ -- exactly backwards.
$k < -13$	Wrote the slope formula with the y-values subtracted in the wrong order relative to the x-values, using $(k + 3)/5$ instead of $(-3 - k)/5$, which flips the sign of the whole expression and leads to $k < -13$.
$k \geq 7$	Solved the inequality correctly down to the boundary $k = 7$ but reported it as $k \geq 7$ (inclusive), forgetting the original slope condition was strict less-than, which excludes the boundary $k = 7$ itself.

Shortcut

Multiply the inequality $(-3 - k)/5 < -2$ by 5 and then by -1 in one combined move, immediately reversing to $3 + k > 10$, then solve $k > 7$ directly without a separate 'add 3' step.

Desmos Tip

Graph $y = (-3 - x)/5$ and the horizontal line $y = -2$, then observe the slanted graph lies below $y = -2$ exactly when $x > 7$.

Linear Inequalities — Problem 6 (Very Hard)

A factory's quality-control rule states that a machined rod fails inspection if its length L (in centimeters) differs from the target length of 18 cm by 3 cm or more: $|L - 18| \geq 3$. The factory runs three shifts per day and produces 200 rods per shift. For what values of L does a rod PASS inspection? Express your answer as a compound inequality.

Formula / Rule Used: A rod passes inspection exactly when it does NOT satisfy the failing condition -- negate $|L - 18| \geq 3$ to get $|L - 18| < 3$, then split the absolute value inequality into a three-part compound inequality.

Solution

Step 1: Passing is the logical opposite of failing. Since failing means $|L - 18| \geq 3$, passing means the negation: $|L - 18| < 3$.

Step 2: An absolute value inequality of the form $|\text{expression}| < c$ (with c positive) splits into a compound inequality: $-c < \text{expression} < c$. Here that gives $-3 < L - 18 < 3$.

Step 3: Add 18 to all three parts of the compound inequality: $-3 + 18 < L - 18 + 18 < 3 + 18$.

Step 4: Simplify: $15 < L < 21$.

Step 5: Sanity check with the failing boundary: at $L = 15$, $|15 - 18| = |-3| = 3$, which satisfies the ORIGINAL failing condition (≥ 3 , since it equals 3), so $L = 15$ should indeed fail and correctly be excluded from the passing range -- confirming the strict inequality ($<$ not $\leq$) is correct on both sides.

Step 6: Note that the production details (three shifts, 200 rods per shift) are not needed to answer the question about acceptable lengths.

Answer: $15 < L < 21$ (centimeters)

**Tip**

When a problem defines a 'failing' or 'rejection' condition with $\geq$ or $>$, remember that the 'passing' condition is its exact logical complement, which flips the inequality to $<$ or $\leq$ before you even start splitting the absolute value.

Trap

A student who solves $|L - 18| \geq 3$ directly (getting $L \leq 15$ OR $L \geq 21$, the FAILING range) and reports that as the answer to 'which rods pass' has answered the opposite question -- the passing range is the complement, $15 < L < 21$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$L \leq 15$ or $L \geq 21$	Solved the given failing condition $ L - 18 \geq 3$ directly and reported that FAILING range as the answer, instead of first negating it to find the PASSING condition $ L - 18 < 3$.
$15 \leq L \leq 21$	Used $\leq$ instead of $<$ when splitting $ L - 18 < 3$ into $-3 < L - 18 < 3$, incorrectly including the boundary values 15 and 21, which actually belong to the failing range since $ 15 - 18 = 3$ satisfies ≥ 3 .
$12 < L < 24$	Doubled the tolerance value (used 6 instead of 3) when splitting the absolute value inequality, apparently confusing the ± 3 boundary with a full range width, giving $12 < L < 24$ instead of $15 < L < 21$.

Shortcut

The boundary values (15 and 21) are identical whether you solve for the failing or passing condition, so you can solve whichever direction feels more natural (here, directly splitting $|L - 18| < 3$) and just make sure the final inequality symbols match what 'passing' truly requires.

Desmos Tip

Graph $y = \text{abs}(L - 18)$ and the horizontal line $y = 3$, then read off the L-interval where the V-shaped graph dips below the line -- that's the passing range $15 < L < 21$.

Linear Inequalities — Problem 7 (Hard)

For how many integer values of x do both inequalities $2x - 5 < 9$ and $-3x + 4 \leq 19$ hold true at the same time?

Formula / Rule Used: Solve each linear inequality separately, then take the intersection of the two solution sets before counting integer solutions.

Solution

Step 1: Solve the first inequality: $2x - 5 < 9$. Add 5 to both sides: $2x < 14$. Divide by 2 (positive, no flip): $x < 7$.

Step 2: Solve the second inequality: $-3x + 4 \leq 19$. Subtract 4 from both sides: $-3x \leq 15$. Divide by -3, which is negative, so reverse the inequality: $x \geq -5$.

Step 3: Combine both solution sets: x must satisfy $x < 7$ AND $x \geq -5$ at the same time, giving the intersection $-5 \leq x < 7$.

Step 4: List the integers in this range, including -5 but not including 7: -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6.

Step 5: Count the listed integers: there are 12 of them.

Step 6: Verify the endpoints: at $x = -5$, $-3(-5) + 4 = 19$, which satisfies ≤ 19 (boundary correctly included); at $x = 7$, $2(7) - 5 = 9$, which does NOT satisfy < 9 (boundary correctly excluded).

Answer: 12 integer values ($x = -5$ through $x = 6$)

Tip

When a problem asks how many integers satisfy two inequalities together, solve each one separately first, then intersect the two number lines before counting -- never try to merge the original inequalities into one expression.

Trap

A student who forgets to flip the inequality sign when dividing $-3x \leq 15$ by -3, or who mishandles which boundary is included versus excluded, will miscount the integers in the intersection even though each inequality looks routine on its own.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
13	Treated the strict inequality $2x - 5 < 9$ as if it were $\leq$, including the boundary $x = 7$, which actually fails the original strict inequality since $2(7) - 5 = 9$ is not less than 9.
11	Treated $-3x + 4 \leq 19$ as if it were strictly $<$, excluding the boundary $x = -5$, which actually satisfies the original inequality with equality since $-3(-5) + 4 = 19$.
14	Added 4 to both sides instead of subtracting it when isolating $-3x$, turning $-3x \leq 15$ into $-3x \leq 23$, which shifts the lower bound down and produces an incorrect count of 14.

Shortcut

Since both inequalities solve cleanly to integer boundaries, skip listing every integer individually -- just compute the upper bound minus the lower bound, $7 - (-5) = 12$, which directly gives the count since the lower bound is included and the upper bound is excluded.

**Desmos Tip**

Graph $y = 2x - 5$ with the line $y = 9$, and $y = -3x + 4$ with the line $y = 19$, to visually confirm the two boundary x -values -5 and 7 before counting the integers between them.

Linear Inequalities — Problem 8 (Very Hard)

For a positive constant a , the inequality $a|x - 5| \geq 12$ has solution set $x \leq -1$ or $x \geq 11$. What is the value of a ?

Formula / Rule Used: Splitting an absolute value inequality of the form $|\text{expression}| \geq c$ into two linear branches, combined with reverse-engineering an unknown coefficient from a given solution set.

Solution

Step 1: Divide both sides by a . Since a is positive, the inequality direction stays the same: $|x - 5| \geq 12/a$.

Step 2: Split the absolute value inequality into its two branches: $x - 5 \geq 12/a$, or $x - 5 \leq -12/a$.

Step 3: Solve each branch for x : $x \geq 5 + 12/a$, or $x \leq 5 - 12/a$.

Step 4: Match the first branch to the given boundary $x \geq 11$: $5 + 12/a = 11$.

Step 5: Solve for a : $12/a = 11 - 5 = 6$, so $a = 12/6 = 2$.

Step 6: Verify with the second branch using $a = 2$: $5 - 12/2 = 5 - 6 = -1$, which exactly matches the given boundary $x \leq -1$, confirming $a = 2$ works for both branches.

Step 7: Since the problem specifies a is a positive constant, $a = 2$ satisfies that condition as well.

Answer: $a = 2$

Tip

When an unknown coefficient sits inside an absolute value inequality and you're given the final solution set, split the absolute value into its two linear branches symbolically first, then match either branch's boundary to the corresponding given boundary value to solve for the unknown.

Trap

A student who solves $5 + 12/a = 11$ by adding 5 to both sides instead of subtracting it will get $12/a = 16$ instead of $12/a = 6$, producing $a = 3/4$ instead of the correct $a = 2$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$a = 3/4$	Solved $5 + 12/a = 11$ by adding 5 to both sides instead of subtracting it, getting $12/a = 16$ instead of $12/a = 6$, which gives $a = 3/4$ instead of $a = 2$.
$a = 6$	Mishandled isolating the absolute value by subtracting a from 12 instead of dividing by a , turning $ x - 5 \geq 12/a$ into $ x - 5 \geq 12 - a$, which leads to $a = 6$ instead of $a = 2$.
$a = -2$	Dropped the negative sign on the second branch (wrote $x - 5 \leq 12/a$ instead of $x - 5 \leq -12/a$), then matched $5 + 12/a$ to the boundary -1 instead of 11, producing $a = -2$, which also violates the problem's requirement that a be positive.



Shortcut

Once you split into the two branches $5 + 12/a$ and $5 - 12/a$, you only need to match ONE branch to its corresponding boundary (say $5 + 12/a = 11$) to solve for a -- the other branch is just a free check, not a second equation to solve.

Desmos Tip

Graph $y = 2 \cdot \text{abs}(x - 5)$ (using the solved value $a = 2$) together with the horizontal line $y = 12$, then confirm the graph crosses $y = 12$ exactly at $x = -1$ and $x = 11$, matching the given solution set.

4. Absolute Value

Absolute Value — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Absolute value	Απόλυτη τιμή
Distance from zero	Απόσταση από το μηδέν
Two cases (positive/negative)	Δύο περιπτώσεις (θετική/αρνητική)
Extraneous solution	Ξένη (μη αποδεκτή) λύση

Absolute Value — Condensed Theory

Absolute value is distance from zero and is always non-negative. Every absolute value equation or inequality splits into two cases.

Solving $|X| = a$

If $a \geq 0$: $X = a$ or $X = -a$. If $a < 0$: no solution.

Isolate the absolute value first, then split into the two cases.

Absolute Value Inequalities

$|X| < a \rightarrow -a < X < a$. $|X| > a \rightarrow X < -a$ or $X > a$ ($a > 0$)

'Less than' becomes a compound (between) inequality; 'greater than' becomes two separate branches.

$|A| = |B|$

$A = B$ or $A = -B$

Solve both resulting equations and check each solution in the original equation.

Extraneous Solutions

Whenever a variable appears both inside and outside the absolute value bars (or gets squared), you must check every solution in the ORIGINAL equation — one branch often fails.

Distance Interpretation

$|x - a| = \text{distance between } x \text{ and } a$

Useful for tolerance / 'within so many units of a target' word problems: $|\text{actual} - \text{target}| \leq \text{tolerance}$.

⚠ Watch Out

If isolating the absolute value ever produces $|X| = (\text{negative number})$, stop — there is no solution. Don't split into cases anyway.

Absolute Value — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Absolute Value — Problem 1 (Hard)

Solve for x : $|2x - 5| = 3x - 4$. Check for extraneous solutions and give only the valid solution(s).

Formula / Rule Used: $|X| = a$ splits into $X = a$ or $X = -a$; any solution must be checked in the original equation.

Solution

Step 1: Since the variable appears both inside and outside the absolute value bars, split into two cases.

Step 2: Case 1: $2x - 5 = 3x - 4$. Solve: $2x - 3x = -4 + 5$, so $-x = 1$, giving $x = -1$.

Step 3: Case 2: $2x - 5 = -(3x - 4) = -3x + 4$. Solve: $2x + 3x = 4 + 5$, so $5x = 9$, giving $x = 9/5$.

Step 4: Check $x = -1$ in the original equation: left side = $|2(-1) - 5| = |-7| = 7$, right side = $3(-1) - 4 = -7$. Since 7 does not equal -7, $x = -1$ is extraneous and must be rejected.

Step 5: Check $x = 9/5$: left side = $|2(9/5) - 5| = |18/5 - 25/5| = |-7/5| = 7/5$, right side = $3(9/5) - 4 = 27/5 - 20/5 = 7/5$. These match.

Step 6: The only valid solution is $x = 9/5$.

Answer: $x = 9/5$

Tip

Whenever the variable also sits outside the absolute value bars, treat both algebraic results as candidates only — substitute each back into the ORIGINAL equation before accepting it.

Trap

A rushed student stops after solving the two linear cases and reports $x = -1$ as a valid answer too, never noticing it forces the absolute value expression to equal a negative number (-7), which is impossible.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = -1$	Solved Case 1 correctly to $x = -1$ but never substituted it back into the original equation, so did not catch that it makes the right side negative $(3(-1) - 4 = -7)$, which an absolute value can never equal.
$x = 1/5$	When expanding Case 2's negative sign, wrote $-(3x - 4)$ as $-3x - 4$ instead of correctly distributing to $-3x + 4$, producing the wrong linear equation $2x - 5 = -3x - 4$ and the wrong root.
$x = -1$ and $x = 9/5$	Listed both algebraic candidates as final answers, treating the extraneous-



	solution check as optional instead of mandatory whenever the variable sits outside the bars.
--	--

Shortcut

Before checking by substitution, you can flag a candidate as suspicious instantly by testing whether the bare side ($3x-4$) is negative at that x -value, since an absolute value can never equal a negative number — this marks $x=-1$ as doubtful before any arithmetic.

Desmos Tip

Graph $y = \text{abs}(2x-5)$ and $y = 3x-4$ and look at the actual intersection point — Desmos shows only one crossing, near $x = 1.8$, confirming $x = -1$ is not a real solution.

Absolute Value — Problem 2 (Hard)

Solve for all values of x : $5 + |2x - 6| = |x + 3| + 5$.

Formula / Rule Used: $|A| = |B|$ means $A = B$ or $A = -B$.

Solution

Step 1: Subtract 5 from both sides to strip away the disguise: $|2x - 6| = |x + 3|$.

Step 2: Since $|A| = |B|$ means $A = B$ or $A = -B$, set up both cases using $A = 2x - 6$ and $B = x + 3$.

Step 3: Case 1: $2x - 6 = x + 3$. Solve: $2x - x = 3 + 6$, so $x = 9$.

Step 4: Case 2: $2x - 6 = -(x + 3) = -x - 3$. Solve: $2x + x = -3 + 6$, so $3x = 3$, giving $x = 1$.

Step 5: Verify $x = 9$: $|2(9)-6| = |12| = 12$ and $|9+3| = |12| = 12$, so both sides of the simplified equation match.

Step 6: Verify $x = 1$: $|2(1)-6| = |-4| = 4$ and $|1+3| = |4| = 4$, so both sides match here too.

Step 7: Both candidates check out, so the solutions are $x = 1$ and $x = 9$.

Answer: $x = 1$ and $x = 9$

Tip

First strip away anything added or subtracted outside the bars until the equation is purely $|A| = |B|$, then split into $A = B$ and $A = -B$.

Trap

A rushed student splits into cases before simplifying, writing ' $2x-6+5 = x+3$ ' and ' $2x-6+5 = -(x+3)$ ' by dragging the extra $+5$ into the case equations, which produces the wrong x -values entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 4$	Incorrectly folded the $+5$ into the bars, solving $2x-6+5 = x+3$ (as if $5+ 2x-6 $ meant $ 2x-6+5 $) instead of first subtracting 5 from both sides to isolate $ 2x-6 = x+3 $.
$x = -2/3$	Made the same folding-in error but from the negative-case version,



	solving $2x-6+5 = -(x+3)$ instead of the correct Case 2 equation $2x-6 = -(x+3)$.
$x = 9$	Solved only Case 1 ($2x-6 = x+3$) and never set up Case 2 ($2x-6 = -(x+3)$), so reported one solution and missed the second valid solution $x = 1$.

Shortcut

Once simplified to $|2x-6| = |x+3|$, you can solve $2x - 6 = \text{plus-or-minus}(x+3)$ as one combined step instead of writing two fully separate case equations.

Desmos Tip

Graph $y = 5 + \text{abs}(2x-6)$ and $y = \text{abs}(x+3) + 5$ and read off both intersection x-coordinates directly, which confirms $x = 1$ and $x = 9$ without any algebra.

Absolute Value — Problem 3 (Very Hard)

The inequality $5|x - 3| - 7 > k$ has solution set $x < -1$ or $x > 7$. What is the value of k ?

Formula / Rule Used: $|X| > a$ ($a > 0$) splits into $X > a$ or $X < -a$.

Solution

Step 1: Isolate the absolute value expression first: add 7 to both sides to get $5|x-3| > k + 7$.

Step 2: Divide both sides by 5: $|x - 3| > (k+7)/5$.

Step 3: An inequality of the form $|X| > a$ splits into $X > a$ or $X < -a$, so $x - 3 > (k+7)/5$ or $x - 3 < -(k+7)/5$.

Step 4: Rewrite each branch by adding 3: $x > 3 + (k+7)/5$, or $x < 3 - (k+7)/5$.

Step 5: Match the given upper boundary $x > 7$ to the first branch: $3 + (k+7)/5 = 7$, so $(k+7)/5 = 4$, giving $k + 7 = 20$, so $k = 13$.

Step 6: Check against the given lower boundary $x < -1$: $3 - (k+7)/5 = -1$, so $(k+7)/5 = 4$ again, giving the same $k = 13$.

Step 7: Both boundaries agree, so $k = 13$.

Answer: $k = 13$

Tip

When you're given the finished solution set of an absolute value inequality and asked for a hidden constant, rebuild the general $|x - \text{center}| > \text{distance}$ form and match its two boundary numbers to the given ones algebraically.

Trap

A rushed student plugs $x = 7$ or $x = -1$ directly into the original inequality as if it were an equation, but forgets to first isolate $|x-3|$ by undoing the -7 and dividing by 5, and often reports $k = 7$ by ignoring the coefficient of 5 entirely.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$k = 20$	Dropped the -7 when isolating, treating $5 x-3 -7 > k$ as $5 x-3 > k$, which



	gives $ x-3 > k/5$; matching the boundary $x=7$ then gives $k/5 = 4$, so $k = 20$ instead of 13.
$k = -3$	Correctly isolated the absolute value but forgot to divide by 5, using $3 + (k+7) = 7$ instead of $3 + (k+7)/5 = 7$, giving $k = -3$.
$k = -31/5$	When solving $(k+7)/5 = 4$, cross-multiplied backwards, writing $k+7 = 4/5$ instead of $k+7 = 4(5) = 20$, producing a wrong fractional value for k .

Shortcut

Since 3 is exactly the midpoint of the two boundary values -1 and 7, the 'radius' distance is $(7 - (-1))/2 = 4$, so $(k+7)/5 = 4$ can be written immediately without deriving both boundary equations separately.

Desmos Tip

Graph $y = 5 \cdot \text{abs}(x-3) - 7$ and the horizontal line $y = k$, then adjust k until the shaded region's edges land exactly at $x = -1$ and $x = 7$ to visually confirm $k = 13$.

Absolute Value — Problem 4 (Hard)

A quality-control station in Patras accepts a 500 mL bottle only if its actual filled volume V satisfies $|V - 500| \leq 4.5$. One bottle was rejected because its measured volume was outside the acceptable tolerance by exactly 1.8 mL on the low side. What was that bottle's measured volume?

Formula / Rule Used: Tolerance statements translate to $|\text{actual} - \text{target}| \leq \text{tolerance}$, which unfolds into a compound inequality.

Solution

Step 1: Translate the tolerance rule into an inequality: $|V - 500| \leq 4.5$.

Step 2: Rewrite this without absolute value bars as a compound inequality: $-4.5 \leq V - 500 \leq 4.5$.

Step 3: Add 500 to all three parts: $495.5 \leq V \leq 504.5$, so any volume from 495.5 mL to 504.5 mL passes inspection.

Step 4: The bottle was rejected on the low side, meaning its measured volume fell below the minimum acceptable bound of 495.5 mL.

Step 5: 'Outside the tolerance by exactly 1.8 mL' means the measured volume was 1.8 mL less than that lower bound of 495.5 mL, not 1.8 mL less than the target of 500 mL.

Step 6: Compute: $V = 495.5 - 1.8 = 493.7$ mL.

Step 7: The bottle's measured volume was 493.7 mL.

Answer: $V = 493.7$ mL

Tip

Translate a tolerance statement 'within c of a target T ' as $|\text{actual} - T| \leq c$, then convert it into a normal compound inequality before doing any arithmetic with the boundary values.

Trap



A rushed student subtracts 1.8 directly from the target 500 (getting 498.2) instead of from the lower boundary 495.5, forgetting that the bottle was already outside the allowed window before being pushed 1.8 mL further away.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
V = 498.2 mL	Subtracted 1.8 from the target 500 directly instead of from the lower tolerance bound 495.5, ignoring that the bottle was already at the edge of the accepted window before failing further.
V = 506.3 mL	Added 1.8 to the upper bound 504.5 instead of subtracting from the lower bound, misreading 'low side' as meaning the upper boundary was the one exceeded.
V = 494.2 mL	Mis-copied the tolerance value as 4 mL instead of 4.5 mL, computing the lower bound as $500 - 4 = 496$, then subtracting 1.8 to get 494.2 mL instead of the correct 493.7 mL.

Shortcut

Since the failure happened only on the low side, you can skip finding the upper bound (504.5) entirely and work with just the lower bound $500 - 4.5 = 495.5$, saving a step.

Desmos Tip

Type $\text{abs}(V-500) \leq 4.5$ into Desmos as a one-variable inequality to see the accepted interval $[495.5, 504.5]$ plotted on the number line, then confirm 493.7 falls outside it on the left.

A tutor in Athens charges 35 euros per hour and is currently reviewing the equation $|4x + 1| = x - 6$ with a student during a 50-minute session. Solve the equation for x , or determine that no real solution exists.

Formula / Rule Used: $|X|$ can never equal a negative number, so any candidate solution must make the bare side non-negative.

Solution

Step 1: Ignore the tutoring and pricing details — they don't affect the algebra. Split the equation into two cases as usual.

Step 2: Case 1: $4x + 1 = x - 6$. Solve: $4x - x = -6 - 1$, so $3x = -7$, giving $x = -7/3$.

Step 3: Case 2: $4x + 1 = -(x - 6) = -x + 6$. Solve: $4x + x = 6 - 1$, so $5x = 5$, giving $x = 1$.

Step 4: Before accepting either value, note that $|4x+1|$ is never negative, so any true solution requires the right side $x - 6$ to be greater than or equal to 0, meaning $x \geq 6$.

Step 5: Check $x = -7/3$: this does not satisfy $x \geq 6$. Directly verify: $|4(-7/3)+1| = |-28/3+3/3| = |-25/3| = 25/3$, but $x - 6 = -7/3 - 18/3 = -25/3$. Since $25/3$ does not equal $-25/3$, this is extraneous.

Step 6: Check $x = 1$: this also fails $x \geq 6$. Directly verify: $|4(1)+1| = 5$, but $x - 6 = -5$. Since 5 does not equal -5 , this is extraneous too.

Step 7: Both algebraic candidates fail the check, so the equation has no real solution.

Answer: No solution

Tip

Before trusting either algebraic case in an equation like $|\text{expr}| = (\text{linear expression})$, check whether that linear expression can even be non-negative at the candidate x -value — an absolute value can never equal a negative number.

Trap

A rushed student solves both linear cases, gets $x = -7/3$ and $x = 1$, and reports one or both as answers without ever substituting back into the original equation, missing that neither actually works.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = -7/3$	Solved Case 1 ($4x+1 = x-6$) correctly but never checked whether $x-6$ could be negative, submitting the algebraic root without noticing $ 4x+1 $ cannot equal the resulting negative value there.
$x = 1$	Solved Case 2 ($4x+1 = -(x-6)$) correctly but likewise never verified the result against the original equation, missing that it also forces the right side to be negative.
$x = -7/3$ and $x = 1$	Listed both algebraic case solutions as the final answer without performing the required non-negativity check on either one, treating extraneous-solution verification as unnecessary.

Shortcut



You can often spot 'no solution' quickly by testing only the larger candidate against the requirement $x \geq 6$ first — since even $x = 1$ already fails, both branches are immediately suspect without doing the full substitution check on each.

Desmos Tip

Graph $y = \text{abs}(4x+1)$ and $y = x - 6$ — the two curves never intersect anywhere, visually confirming there is no solution.

Absolute Value — Problem 6 (Very Hard)

Let m be the larger solution to the equation $|3m - 5| = 2m + 1$. Then determine the number of real solutions of the quadratic equation $x^2 + 4x + (m - 9) = 0$ by evaluating its discriminant.

Formula / Rule Used: Solve $|X| = a$ in two cases first; then use the discriminant $b^2 - 4ac$ to count real roots of a quadratic.

Solution

Step 1: Solve $|3m - 5| = 2m + 1$ by splitting into two cases.

Step 2: Case 1: $3m - 5 = 2m + 1$. Solve: $3m - 2m = 1 + 5$, so $m = 6$. Check: $|3(6) - 5| = |13| = 13$, and $2(6) + 1 = 13$. Valid.

Step 3: Case 2: $3m - 5 = -(2m + 1) = -2m - 1$. Solve: $3m + 2m = -1 + 5$, so $5m = 4$, giving $m = 4/5$. Check: $|3(4/5) - 5| = |12/5 - 25/5| = |-13/5| = 13/5$, and $2(4/5) + 1 = 8/5 + 5/5 = 13/5$. Also valid.

Step 4: Both $m = 6$ and $m = 4/5$ satisfy the original equation, but the problem asks for the larger one, so $m = 6$.

Step 5: Substitute $m = 6$ into the quadratic: $x^2 + 4x + (6 - 9) = 0$, which simplifies to $x^2 + 4x - 3 = 0$.

Step 6: Compute the discriminant: $b^2 - 4ac = 4^2 - 4(1)(-3) = 16 + 12 = 28$.

Step 7: Since the discriminant 28 is positive, the quadratic has two distinct real solutions.

Answer: Two distinct real solutions ($m = 6$, discriminant = 28)

Tip

When a problem chains an absolute value equation into a second calculation, fully solve and verify the absolute value part first — including checking every candidate — before touching the second concept.

Trap

A rushed student uses whichever solution of the absolute value equation they find first (often $m = 4/5$ from Case 2) instead of comparing both valid solutions and correctly selecting the larger one as the problem instructs.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Two distinct real solutions ($m = 4/5$, discriminant = 48.8)	Used the smaller root $m = 4/5$ from Case 2 instead of correctly selecting the larger solution $m = 6$ as the problem requires, propagating the wrong m into the discriminant.
Two distinct real solutions (discriminant = 4)	Substituted $m = 6$ correctly but made a sign error simplifying the constant term, writing $9 - m = 3$ instead of $m - 9 = -3$, giving the wrong quadratic $x^2 + 4x + 3 = 0$ and the wrong discriminant.

**Two distinct real solutions
(discriminant = 16)**

Forgot to square b in the discriminant formula, using $b = 4$ directly instead of $b^2 = 16$, computing $4 - 4(1)(-3) = 16$ instead of the correct $16 - 4(1)(-3) = 28$.

Shortcut

Since both cases of $|3m-5| = 2m+1$ check out here, there's no extraneous-root elimination needed — you can go straight to comparing 6 and $4/5$ and move on to the quadratic without re-verifying anything twice.

Desmos Tip

Graph $y = \text{abs}(3m-5)$ and $y = 2m+1$ to instantly read off both intersection m -values (6 and 0.8), then graph $y = x^2+4x-3$ separately to see it crosses the x -axis twice, confirming two real solutions.

Absolute Value — Problem 7 (Very Hard)

Find all real solutions to the equation $||x - 4| - 7| = 3$.

Formula / Rule Used: Nested absolute value: apply the $|X| = a$ rule to the outer bars first ($X = a$ or $X = -a$), then apply the same rule again to each resulting inner absolute value equation.

Solution

Step 1: Let $X = |x - 4| - 7$, so the equation is $|X| = 3$. This splits into two outer cases: $X = 3$ or $X = -3$.

Step 2: Outer Case A: $|x - 4| - 7 = 3$, so $|x - 4| = 10$.

Step 3: Split Outer Case A into its own two inner cases: $x - 4 = 10$, giving $x = 14$, or $x - 4 = -10$, giving $x = -6$.

Step 4: Outer Case B: $|x - 4| - 7 = -3$, so $|x - 4| = 4$.

Step 5: Split Outer Case B into its own two inner cases: $x - 4 = 4$, giving $x = 8$, or $x - 4 = -4$, giving $x = 0$.

Step 6: Since the outer equation's right side (3) is a positive constant, both outer cases are automatically valid, so no candidate needs to be discarded here.

Step 7: Verify one candidate, $x = 0$: $|0 - 4| = 4$, then $|4 - 7| = |-3| = 3$. Matches the original equation.

Step 8: The four real solutions are $x = -6, 0, 8$, and 14 .

Answer: $x = -6, 0, 8, 14$

Tip

When you see an absolute value nested inside another absolute value, work from the outside in — first split the outer $|X| = a$ into $X = a$ and $X = -a$ as two full equations, then split each of those inner absolute value equations again.

Trap

The most common failure is treating this as a single-layer absolute value problem and solving only $|x-4| - 7 = 3$, never realizing that $|x-4| - 7 = -3$ is an equally valid second outer case, which silently discards two of the four correct solutions.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



$x = -6, 14$	Solved only the outer case $ x-4 - 7 = 3$, forgetting to also set up $ x-4 - 7 = -3$, so missed the two solutions (0 and 8) that come from the second outer case.
$x = 0, 8$	Solved only the outer case $ x-4 - 7 = -3$, forgetting the other outer case $ x-4 - 7 = 3$, so missed the two solutions (-6 and 14) that come from the first outer case.
$x = 8, 14$	For both inner equations $ x-4 = 10$ and $ x-4 = 4$, took only the positive branch ($x-4=10$ and $x-4=4$) and never solved the negative branch ($x-4=-10$ and $x-4=-4$), so missed $x = -6$ and $x = 0$.

Shortcut

Once you find $|x-4| = 10$ and $|x-4| = 4$, you can write all four x-values immediately as 4 ± 10 and 4 ± 4 without redoing separate case equations for each.

Desmos Tip

Graph $y = \text{abs}(\text{abs}(x-4)-7)$ and $y = 3$, then read off the four x-coordinates where they intersect: -6, 0, 8, and 14.

Absolute Value — Problem 8 (Very Hard)

For all real numbers x , what is the minimum possible value of the expression $|2x - 10| + |2x + 4|$?

Formula / Rule Used: $|x-a| + |x-b|$ represents the total distance from x to two fixed points a and b on the number line; its minimum equals $|a-b|$, attained for any x between them. Factor out a shared coefficient before applying this rule.

Solution

Step 1: Factor the coefficient 2 out of each absolute value term: $|2x - 10| = |2(x-5)| = 2|x-5|$, and $|2x + 4| = |2(x+2)| = 2|x+2|$.

Step 2: The expression becomes $2|x-5| + 2|x+2| = 2(|x-5| + |x+2|)$.

Step 3: $|x-5| + |x+2|$ is the sum of the distances from x to the two fixed points 5 and -2 on the number line.

Step 4: This sum of distances is minimized whenever x lies anywhere between -2 and 5 (inclusive), and the minimum value equals the distance between the two points: $5 - (-2) = 7$.

Step 5: Multiply by the factored-out coefficient of 2: minimum value $= 2 \times 7 = 14$.

Step 6: Confirm with a test point inside the interval, say $x = 0$: $|2(0)-10| + |2(0)+4| = |-10| + |4| = 10 + 4 = 14$, matching the predicted minimum.

Answer: 14

Tip

When a sum of two absolute values shares a common coefficient, factor it out first, then recognize $|x-a| + |x-b|$ as the total distance from x to two fixed points — a sum that is minimized between them and equals the distance between the points.

**Trap**

A rushed student forgets to factor out the shared coefficient of 2 and reports the raw anchor-point distance $|5 - (-2)| = 7$ as the final answer, never doubling it back to account for the coefficient.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
7	Forgot to factor out and reapply the shared coefficient of 2 from each term, computing only the anchor-point distance $ 5 - (-2) = 7$ and never multiplying it back by 2.
18	Incorrectly factored $ 2x+4 $ as $2 x+4 $ instead of $2 x+2 $ (a slip in the factoring step), which shifted the second anchor point to -4 and gave $2 \times 5 - (-4) = 2 \times 9 = 18$ instead of $2 \times 7 = 14$.
6	Added the two anchor points ($5 + (-2) = 3$) instead of finding the distance between them ($5 - (-2) = 7$), then multiplied by 2 to get 6 instead of 14.

Shortcut

Once you factor out the shared coefficient, no calculus or test points are needed — the minimum of $|x-a| + |x-b|$ is always just the distance between a and b , so $2 \times 7 = 14$ can be written in one line.

Desmos Tip

Graph $y = \text{abs}(2x-10) + \text{abs}(2x+4)$ and trace the flat bottom of the V-shaped graph between $x = -2$ and $x = 5$, where y stays constant at 14, confirming the minimum.

5. Exponent Rules

Exponent Rules — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Exponent (power)	Εκθέτης (δύναμη)
Base	Βάση
Exponent rules	Ιδιότητες δυνάμεων
Negative exponent	Αρνητικός εκθέτης
Common base	Κοινή βάση

Exponent Rules — Condensed Theory

Every exponent rule follows from repeated multiplication. The SAT tests both simplifying exponential expressions and solving exponential equations.

Product & Quotient Rules

Same base: add exponents when multiplying, subtract when dividing.

Power Rules

Raising a power to a power multiplies exponents; a product or quotient raised to a power distributes to each factor.

Zero & Negative Exponents

A negative exponent means 'reciprocal,' never a negative number.

Fractional Exponents

The denominator of the exponent is the root; the numerator is the power.

Solving Exponential Equations

If (same base), then $m = n$.



If the bases differ but share a common factor, rewrite both sides with that common base first, then set exponents equal.

Solving Fractional-Exponent Equations

If $a^{\frac{1}{b}} = c^{\frac{1}{d}}$, then $a^{\frac{d}{bd}} = c^{\frac{b}{bd}}$.

Raise both sides to the reciprocal power to isolate x .



Watch Out

a^{b+c} is NEVER $a^b + a^c$ — you cannot distribute an exponent over addition or subtraction.

Negative exponents create reciprocals, not negative numbers — a common and costly misread.

Exponent Rules — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Exponent Rules — Problem 1 (Very Hard)

Simplify to a single term with positive exponents only:

Formula / Rule Used: Power of a product, product rule, negative exponents, and the zero-exponent rule together.

Solution

Step 1: Recognize that any nonzero expression raised to the power 0 equals 1, so the denominator -- it does not need to be expanded.

Step 2: Apply the power rule to the first factor: $a^{\frac{1}{2}}$.

Step 3: Apply the power rule to the second factor: $b^{\frac{1}{3}}$.

Step 4: Multiply the two results using the product rule (add exponents of like bases): $a^{\frac{1}{2}}b^{\frac{1}{3}}$.

Step 5: Divide by the denominator, which is 1, so the expression is unchanged: $a^{\frac{1}{2}}b^{\frac{1}{3}}$.

Step 6: Rewrite with positive exponents only: $a^{\frac{1}{2}}b^{\frac{1}{3}}$.

Answer: $a^{\frac{1}{2}}b^{\frac{1}{3}}$

Tip

When a factor in the expression is raised to the 0 power, simplify it to 1 immediately instead of expanding it -- it never affects the rest of the problem.

Trap

A student who doesn't recognize a^0 wastes time expanding it, or worse, mistakenly distributes the 0 as if it were a real exponent to multiply through, instead of just replacing the whole denominator with 1.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
	Correctly combined all exponents but forgot the last step of rewriting negative exponents as positive -- stopped before flipping to .
	Mis-applied the power rule to , treating as instead of , which lowers the final y exponent from 7 to 5.
-	Misinterpreted the negative exponent on the coefficient, computing as -4 instead of the reciprocal , which corrupts the numeric coefficient in the final answer.

Shortcut

Besides recognizing the zero-power factor as 1, there's no shortcut here -- the fastest path is still applying the power rule to each bracket separately before combining, exactly as shown.

Desmos Tip

Pick simple values like $x=2$, $y=1$, evaluate the original expression and the simplified answer in Desmos's calculator, and confirm they give the same number.

Exponent Rules — Problem 2 (Hard)

Solve for x:

Formula / Rule Used: Rewriting exponential expressions with a common base, then setting exponents equal.

Solution

Step 1: Rewrite all bases as powers of 3: and .

Step 2: Rewrite the left side: .

Step 3: Rewrite the right side: .

Step 4: Since both sides now have base 3, set the exponents equal: $4x - 1 = 3x + 3$.

Step 5: Solve for x: $4x - 3x = 3 + 1$, so $x = 4$.

Step 6: Check: left exponent = $4(4)-1 = 15$; right exponent = $3(4)+3 = 15$. Both sides equal , confirming $x = 4$.

Answer: $x = 4$

Tip

Convert every term to the same base first, remembering to fold any leading coefficient (like the $3\cdot$) into the exponent before comparing powers.

Trap

A student who ignores the leading factor of 3 and sets $2(2x-1)$ equal directly to $3(x+1)$ gets $4x-2=3x+3$, producing the wrong answer $x = 5$ instead of the correct $x = 4$.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$x = 5$	Ignored the leading factor of 3 outside and set $2(2x-1)$ equal directly to $3(x+1)$, giving $4x-2=3x+3$, which solves to $x=5$.
$x = 2$	Incorrectly distributed the outer exponent on , writing it as instead of (forgetting to multiply the +1 by 3 as well), then solved $4x-1=3x+1$.
$x = 1.5$	Misremembered 27 as instead of , giving the right side exponent $2x+2$ instead of $3x+3$, and solved $4x-1=2x+2$.

Shortcut

There's no faster route than converting to a common base -- but once converted, you can skip fully expanding and just compare the coefficient-adjusted exponents ($4x-1$ vs $3x+3$) in one line.

Desmos Tip

Graph and find their intersection point; the x-coordinate should read exactly 4.

Exponent Rules — Problem 3 (Hard)

A positive number k satisfies . What is the value of ?

Formula / Rule Used: Fractional exponents; solve by raising both sides to the reciprocal power.

Solution

Step 1: To undo the exponent on k , raise both sides to the reciprocal power, :

Step 2: The left side simplifies because :

Step 3: Evaluate the right side: . So $k = 64$.

Step 4: Now compute the requested value, , using $k = 64$:

Step 5: Evaluate the square root first: .

Step 6: Cube the result: .

Answer:

Tip

When you're given a fractional power of an unknown and asked for a different fractional power of it, solve for the unknown first using the reciprocal-power move, then evaluate the new expression from scratch.

Trap

A student may notice that is the reciprocal of and wrongly assume is just , the reciprocal of the given value -- but flipping an exponent does not flip the value; k must actually be solved for first.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Assumed that since is the reciprocal exponent of , the value must also be the reciprocal of 16, without actually solving for k first.



	Treated the exponent as a multiplier rather than a power throughout, computing as (instead of 64) and then repeating the same error on the final step to get .
	Correctly solved $k=64$ but only took the square root and forgot to then cube the result to finish evaluating .

Shortcut

Since $k = 64$ works out cleanly, you can track it as means , then directly, without ever writing radical notation.

Desmos Tip

Graph and find where $y = 16$ to read off $k = 64$, then evaluate directly in the calculator to confirm 512.

Exponent Rules — Problem 4 (Very Hard)

Suppose b is a positive number such that and . What is the value of ?

Formula / Rule Used: Quotient rule for exponents (dividing two equations with a common base), then the zero-exponent rule.

Solution

Step 1: Divide the second equation by the first to eliminate the unknown exponent structure: .

Step 2: Simplify the left side using the quotient rule: .

Step 3: So . Solve for b by taking the fourth root (b is positive): .

Step 4: Substitute $b = 3$ into the first equation to find n : , so $n = 2$.

Step 5: Evaluate the requested expression: .

Step 6: Apply the zero-exponent rule: .

Step 7: Check by a second route: , confirming the answer.

Answer:

Tip

When two exponential equations share the same base but different exponents, divide one equation by the other to cancel the base's unknown power and isolate a numeric power of b you can solve directly.

Trap

A student who solves by writing (dividing instead of taking the fourth root) gets a nonsensical base, instead of correctly evaluating .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
0	Misinterpreted as minus (subtracting values) rather than divided by (subtracting exponents), computing $9 - 9 = 0$ instead of .
81	Divided the two given equations in the wrong order, getting , then misread the negative exponent as if the equation were , giving instead of 3;



	carrying this incorrect base through the rest of the problem produces 81 instead of the correct 1.
9	Misread the requested expression as 2^{n-2} instead of 2^{n-2} , plugging $n=2$ directly into 2^{n-2} rather than first evaluating the exponent $n-2 = 0$.

Shortcut

You can skip solving for n entirely: $2^{n-2} = 1$ and $2^0 = 1$, so $n=2$ in two lines.

Exponent Rules — Problem 5 (Hard)

Simplify to a single fraction with positive exponents only:

Formula / Rule Used: Negative exponents and the quotient rule -- distributing a negative outer exponent across a product.

Solution

Step 1: Apply the outer exponent -1 to every factor inside the parentheses in the denominator: $\frac{1}{x^2y^4}$.

Step 2: Rewrite the full expression using this result: $\frac{x^2y^4}{x^2y^4}$.

Step 3: Apply the quotient rule to the x terms: $\frac{x^2}{x^2}$.

Step 4: Apply the quotient rule to the y terms: $\frac{y^4}{y^4}$.

Step 5: Combine the results: 1 .

Step 6: Rewrite with positive exponents: 1 .

Answer: 1

Tip

When an outer exponent applies to a product of two factors, distribute it to both factors' exponents -- including any factor that is already negative -- before simplifying further.

Trap

A student may distribute the outer -1 to (x^2y^4) but leave y^4 unchanged, reasoning it's 'already negative,' getting the wrong intermediate $\frac{x^2}{y^4}$ instead of the correct $\frac{1}{x^2y^4}$ -- this flips the final sign on y and produces the wrong answer $\frac{x^2}{y^4}$ instead of the correct 1 .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$\frac{x^2}{y^4}$	Distributed the outer -1 exponent to x^2 but left y^4 unchanged (reasoning it was 'already negative'), producing $\frac{x^2}{y^4}$ instead of $\frac{1}{x^2y^4}$, which flips the sign on y 's exponent in the final step.
$\frac{x^2}{y^2}$	Applied the quotient rule to the x terms as $\frac{x^2}{x^2}$ instead of $\frac{1}{x^2}$, mishandling the double negative and getting $\frac{x^2}{y^2}$ instead of $\frac{1}{x^2y^4}$.
$\frac{1}{x^2y^2}$	Subtracted the y exponents in the wrong order ($4-2$ instead of $2-4$) when applying the quotient rule, getting $\frac{1}{x^2y^2}$ instead of $\frac{1}{x^2y^4}$, which flips y into the



	numerator incorrectly.
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Shortcut

No shortcut beats carefully distributing the outer exponent first -- but once distributed, you can combine both quotient-rule subtractions $(-3-(-2))$ and $2-4$ on the same line instead of writing them separately.

Desmos Tip

Substitute a value like $x=2$, $y=3$ into both the original expression and the simplified answer in Desmos's calculator to confirm they evaluate to the same number.

Exponent Rules — Problem 6 (Very Hard)

A biotech lab in Rotterdam is tracking a bacterial culture that grows according to $y = a \cdot 2^t$, where t is measured in hours and a is a constant growth factor. The technician recorded that at $t=2$ hours the population was 20 organisms, and at $t=5$ hours it was 160 organisms. The lab's petri dish can hold up to 10,000 organisms before needing to be split. Assuming the growth follows this exponential model exactly, what was the population N_0 at $t=0$, when the sample was first placed in the dish?

Formula / Rule Used: Exponential growth model ; use the quotient rule to eliminate N_0 and solve the resulting equation.

Solution

Step 1: Translate the two data points into equations: $20 = a \cdot 2^2$ and $160 = a \cdot 2^5$. (The 10,000-organism dish capacity is not needed for this calculation.)

Step 2: Divide the second equation by the first to eliminate N_0 : $\frac{160}{20} = \frac{a \cdot 2^5}{a \cdot 2^2}$.

Step 3: Simplify the left side using the quotient rule: $8 = 2^3$.

Step 4: So $2^3 = 2^3$, which gives $a = 2$.

Step 5: Substitute $a = 2$ back into the first equation: $20 = 2 \cdot 2^2$, so $4N_0 = 20$.

Step 6: Solve for N_0 : $N_0 = 5$.

Step 7: Check: $20 = 2 \cdot 2^2$, which matches the given data.

Answer: $N_0 = 5$ organisms

Tip

In a growth word problem with two data points instead of a known starting value, divide the two exponential equations to cancel N_0 first -- that isolates a power of the growth factor a you can solve directly.

Trap

A rushed student divides the equations in the wrong order instead of $\frac{160}{20}$, getting $\frac{20}{160}$, which leads to $N_0 = \frac{1}{4}$ and implies the culture is shrinking -- the reverse of what the data actually shows.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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80 organisms	Divided the equations in the wrong order instead of $\cdot$, getting $\frac{1}{a}$ instead of $a=2$, then solved $\frac{1}{a}$ instead of 5.
20 organisms	Misread the question and reported the population at $t=2$ (which was given directly as 20) instead of solving for N_0 , the population at $t=0$.
0.625 organisms	After correctly finding $a=2$, substituted it into the $t=5$ equation but used the $t=2$ population value (20) instead of the $t=5$ value (160), mismatching the exponent and the data point: $\frac{1}{2} \cdot 20 = 10$.

Shortcut

Dividing the two given equations to eliminate N_0 immediately, as shown, is already the fastest route -- solving for N_0 or a individually first (e.g., using logarithms) would take longer.

Desmos Tip

Graph $y = N_0 \cdot 2^t$ in Desmos and confirm the curve passes through both (2, 20) and (5, 160) to verify the fitted model before trusting $N_0 = 5$.

Exponent Rules — Problem 7 *(Very Hard)***Simplify to a single number:**

Formula / Rule Used: Rewriting mixed bases as powers of a common base, then combining fractional and negative exponents.

Solution

Step 1: Rewrite 27 and 9 as powers of 3: $27 = 3^3$ and $9 = 3^2$.

Step 2: Rewrite the first factor: $3^3 \cdot 3^2 = 3^5$.

Step 3: Rewrite the second factor: 3^{-1} .

Step 4: Multiply the numerator's two factors using the product rule: $3^5 \cdot 3^{-1} = 3^4$.

Step 5: Rewrite the denominator: 3^3 .

Step 6: Divide the numerator by the denominator: $3^4 \div 3^3 = 3^1$.

Step 7: Check with a second route: the net exponent of 3 across the whole expression is $-2 + 3 - (-1) = 2$, so the value is $3^2 = 9$, matching.

Answer: 9

Tip

When a problem mixes different bases with fractional and negative exponents, first rewrite everything as a power of the smallest common base, then combine exponents using the rules -- never evaluate the messy fractional powers by estimation.

Trap

A student who tries to evaluate 3^{-1} and $3^{\frac{1}{2}}$ as decimals right away (rather than rewriting both as powers of 3 first) is likely to make an arithmetic slip on the negative fractional exponent and lose track of the final division by 3, often forgetting that dividing by 3 means multiplying by $\frac{1}{3}$.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
1	Correctly simplified the numerator to 3^2 but then divided by 3^2 as if dividing by 3 (instead of multiplying by 3), computing $3^2 \div 3^2 = 1$ instead of $3^2 \times 3^2 = 9$.
-9	Correctly reached 3^{-2} but wrongly assumed dividing by a negative-exponent term makes the result negative, writing -9 instead of correctly computing 3^2 .
	When combining the numerator's exponents, subtracted instead of added them ($-2-3=-5$ instead of $-2+3=1$), giving 3^{-5} , then divided by 3^2 to get 3^{-7} instead of 9.

Shortcut

Since every base reduces to 3, you can skip writing 27 and 9 explicitly and just track the net exponent of 3 in one line: $-2 + 3 - (-1) = 2$, so the answer is 3^2 .

Desmos Tip

Type the original expression directly into Desmos's calculator and confirm it evaluates to exactly 9.

Exponent Rules — Problem 8 (Hard)

Suppose $3^m = 27$ and $3^n = 9$, where m and n are positive real numbers with $m + n < 10$ (this bound is not needed for the calculation). What is the value of 3^{m-n} ?

Formula / Rule Used: Power and quotient rules applied directly to two given exponential equations sharing a base.

Solution

Step 1: Rewrite the target expression using exponent rules: 3^{m-n} . (The condition $m+n < 10$ is extra information not needed here.)

Step 2: Substitute the known value for the numerator: 3^2 , so $3^{m-n} = \frac{3^2}{3^n}$.

Step 3: Substitute the known value for the denominator: 3^2 .

Step 4: Divide: $\frac{3^2}{3^2}$.

Step 5: Simplify the fraction: 1 .

Step 6: Check: this used only the two given equations combined via the power and quotient rules -- the bound $m+n < 10$ never entered the calculation.

Answer: 1

Tip

When asked for an expression like 3^{m-n} built from two given exponential equations with the same base, rewrite it as a product or quotient of powers of the two known quantities, 3^m and 3^n , rather than trying to solve for m and n individually.

Trap



A student may try to solve for m and n individually using logarithms ($m = \log_5(3)$, $n = \log_5(12)$) and then compute $2m-n$ numerically, which is slower and invites rounding errors -- or may get distracted trying to use the irrelevant bound $m+n < 10$ to set up a system of equations that was never needed.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Misapplied the power rule, treating 5^m as 5^{2m} instead of $(5^m)^2$, computing 5^{2m-n} instead of $(5^m)^{2-n}$.
108	Misread the exponent as $2m+n$ instead of $2m-n$, multiplying 9 by 12 instead of dividing, getting $9 \times 12 = 108$.
	Set up the quotient upside down, computing 5^{m-2n} instead of 5^{2m-n} .

Shortcut

Since $2m-n$ breaks cleanly into $2m$ minus n , you can substitute directly as $5^{2m-n} = (5^m)^2 / 5^n$ in one line without ever solving for m or n individually -- this is already the fastest method.

6. Radicals, Rational Exponents & Scientific Notation

Radicals, Rational Exponents & Scientific Notation — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Radical (root)	Ρίζα (ριζικό)
Rational exponent	Ρητός εκθέτης
Simplify a radical	Απλοποίηση ρίζας
Scientific notation	Επιστημονική σημειογραφία

Radicals, Rational Exponents & Scientific Notation — Condensed Theory

Radicals and fractional exponents are two notations for the same idea, and scientific notation is exponent rules applied to place value.

Radical-Exponent Conversion

n -th root of

Converting to fractional exponents before combining or simplifying makes mixed radical expressions far easier.

Simplifying Radicals

Factor out the largest perfect square (or perfect cube, etc.) hiding inside the radicand.

Combining Radicals with Exponents

Convert every term to a fractional exponent with a common denominator, then apply the product/quotient rule, then convert back if needed.

Scientific Notation

, where $1 \leq N < 10$

When multiplying, multiply the coefficients and add the exponents; when dividing, divide the coefficients and subtract the exponents — then renormalize so the coefficient is between 1 and 10.

Watch Out

After any scientific-notation multiplication or division, check that your coefficient is still between 1 and 10 — if not, shift the decimal and adjust the exponent by 1.



Radicals, Rational Exponents & Scientific Notation — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Radicals, Rational Exponents & Scientific Notation — Problem 1 (Hard)

Simplify completely (assume $x > 0$ and $y > 0$): .

Formula / Rule Used: Quotient rule for radicals: ; simplify using exponent subtraction.

Solution

Step 1: Combine the two radicals into one using the quotient rule: .

Step 2: Simplify the numerical coefficient inside the radical: .

Step 3: Simplify the x terms using exponent subtraction: .

Step 4: Simplify the y terms using exponent subtraction: .

Step 5: The expression under the radical is now: .

Step 6: Since 16, , and are all perfect squares, split the radical: .

Step 7: Final simplified expression: .

Answer:

Tip

When you see a quotient of two radicals with the same index, combine them into a single radical first — it is almost always less arithmetic than simplifying each radical separately and then dividing.

Trap

Simplifying and separately before dividing invites an exponent bookkeeping error — for instance writing as instead of — which lands on the wrong final answer instead of the correct .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Subtracted the y exponents incorrectly, computing as instead of .
	Forgot to subtract the denominator's x exponent at all, treating as instead of , while still handling the y terms correctly.
	Correctly simplified inside the radical to and correctly pulled out , but forgot to also take the square root of the numeric coefficient 16, carrying it through unsimplified instead of writing .

Shortcut

Combining under one radical first turns four separate simplifications into a single division, which is already the fastest route on this problem — there is no faster alternate method.

Desmos Tip



Plug in a test value like $x=2$, $y=1$ into both the original expression and the simplified form on Desmos's calculator to confirm they give the same number.

Radicals, Rational Exponents & Scientific Notation — Problem 2 (Hard)

Solve for x : . State the value(s) of x that satisfy the original equation.

Formula / Rule Used: Isolate the radical, square both sides, solve the resulting equation, and check every solution in the original equation.

Solution

Step 1: Isolate the radical by adding x to both sides: .

Step 2: Square both sides to eliminate the radical: .

Step 3: Move all terms to one side: .

Step 4: Factor the quadratic: , so $(x + 3)(x - 1) = 0$.

Step 5: Solve for x : $x = -3$ or $x = 1$.

Step 6: Check $x = 1$ in the ORIGINAL equation: . This works.

Step 7: Check $x = -3$ in the ORIGINAL equation: , which does not equal 2. This solution is extraneous.

Step 8: Reject $x = -3$ and keep only the valid solution.

Answer: $x = 1$

Tip

After squaring both sides of a radical equation, treat every solution as a candidate only — always substitute back into the ORIGINAL (unsquared) equation before finalizing an answer.

Trap

A student who solves the quadratic correctly but forgets to check both roots in the original equation (not the squared one) will report both $x = 1$ and $x = -3$, missing that $x = -3$ is extraneous because the left side actually evaluates to 4, not 2.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 1$ and $x = -3$	Forgot to substitute both candidate roots back into the ORIGINAL (unsquared) equation, so didn't catch that $x = -3$ is extraneous, and reported both roots as valid.
$x = -3$	Assumed without checking that the negative-valued root must be the extraneous one, rejected $x = 1$ and kept $x = -3$ instead of actually verifying each root in the original equation.
$x = 3$	Mis-factored as $(x - 3)(x + 1)$ instead of the correct $(x + 3)(x - 1)$, producing roots $x = 3$ and $x = -1$ rather than the correct $x = 1$ and $x = -3$.

Shortcut



You can eliminate $x = -3$ without recomputing anything: since $|x + 2|$ is always non-negative, it can only equal $x + 2$ when $x + 2$ is also non-negative ($x \geq -2$); $x = -3$ fails this instantly.

Desmos Tip

Graph $y = \sqrt{x + 2}$ and $y = 2$ and find their intersection point directly — Desmos will show only one true intersection, at $x = 1$, immediately revealing that $x = -3$ is extraneous.

Radicals, Rational Exponents & Scientific Notation — Problem 3 (Very Hard)

Write $\sqrt[6]{x^5}$ as a single power of x (assume $x > 0$), and simplify to its lowest exponent.

Formula / Rule Used: Convert radicals to rational (fractional) exponents with a common denominator, then apply the product and quotient rules for exponents.

Solution

Step 1: Convert $\sqrt[6]{x^5}$ to a fractional exponent: $x^{5/6}$.

Step 2: Convert $\sqrt{x}$ to a fractional exponent: $x^{1/2}$.

Step 3: Multiply these using the product rule for exponents (add exponents): $x^{5/6} \cdot x^{1/2}$.

Step 4: Find a common denominator of 6 to add the exponents: $5/6$ and $3/6$, so $8/6$.

Step 5: The expression so far is $x^{8/6}$.

Step 6: Divide using the quotient rule for exponents (subtract exponents): $x^{8/6 - 1/6}$.

Step 7: Simplify the exponent: $x^{7/6}$.

Step 8: Final simplified form: $x^{7/6}$.

Answer: $x^{7/6}$

Tip

Convert every root to a fractional exponent immediately — mixing cube roots and square roots directly is what causes errors, but once everything is written as x^{fraction} , the problem is just exponent addition and subtraction.

Trap

A student might try to combine $\sqrt{x}$ and $\sqrt[6]{x^5}$ by rewriting both as 6th roots and multiplying radicands directly, which can work but is far more error-prone; the most common slip is treating the division by $\sqrt{x}$ as another multiplication instead of a subtraction, giving $x^{11/6}$ instead of the correct $x^{7/6}$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x^{11/6}$	Treated the division by $\sqrt{x}$ as multiplication instead of subtraction, computing $5/6 + 1/2$ instead of $5/6 - 1/2$.
$x^{1/6}$	Correctly converted and multiplied the two radicals to get $x^{8/6}$, but forgot to divide by $\sqrt{x}$ at all, stopping one step early.
$x^{1/2}$	Added the exponents $5/6$ and $1/2$ by adding numerators and denominators straight across (treated as $5/2$) instead of finding a common denominator of 6, corrupting the exponent before the final subtraction.

**Shortcut**

Since all three exponents already share a natural common denominator of 6, you can rewrite the whole expression as a single 6th root from the start: sixth root of = sixth root of .

Desmos Tip

Graph and for $x > 0$ on Desmos and confirm the two curves lie exactly on top of each other.

Radicals, Rational Exponents & Scientific Notation — Problem 4 (Hard)

A certain nebula has a mass of kg, spread over a volume of . (The nebula was first photographed by an amateur astronomer in 1954.) Find the nebula's average density, in kg/, in proper scientific notation.

Formula / Rule Used: Scientific notation division: divide coefficients, subtract exponents, then renormalize to $1 \leq \text{coefficient} < 10$.

Solution

Step 1: Identify the needed calculation: density = mass / volume = . The 1954 photograph detail is irrelevant to this calculation.

Step 2: Divide the coefficients: .

Step 3: Subtract the exponents: .

Step 4: Combine the pieces: density = .

Step 5: Since 0.75 is not between 1 and 10, renormalize it: .

Step 6: Multiply the powers of 10 together: .

Step 7: Final answer: density = kg/.

Answer: kg/

Tip

Do the coefficient division and the exponent subtraction as two separate steps, then always check whether the resulting coefficient is still between 1 and 10 before calling the answer final.

Trap

A student who divides correctly but stops at (since 0.75 is less than 1, it fails the scientific notation rule) reports the wrong final form; a student distracted by the 1954 discovery date might also waste time trying to work it into the calculation.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
kg/	Divided the coefficients and subtracted the exponents correctly, but never renormalized, leaving a coefficient (0.75) that isn't between 1 and 10.
kg/	Subtracted the exponents in the wrong order ($47 - 31$ instead of $31 - 47$), flipping the sign of the exponent and producing a wildly large density instead of a tiny one.



kg/	Made an arithmetic slip computing as 0.8 instead of 0.75 (rounding instead of simplifying), carrying that error through to the final coefficient.
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Shortcut

There is no shortcut beyond the standard divide-coefficients/subtract-exponents/renormalize method, but you can skip long division by noticing instantly.

Desmos Tip

Type directly into Desmos's calculator and read off the result to confirm it matches after renormalizing.

Radicals, Rational Exponents & Scientific Notation — Problem 5 (Very Hard)

If and $x > 0$, what is the value of ?

Formula / Rule Used: Rational exponent power rule: ; use it either to solve for the base or to jump directly between related exponents.

Solution

Step 1: Start with the given equation: .

Step 2: Raise both sides to the power (the reciprocal of) to isolate x : .

Step 3: Simplify the left side using the power rule: .

Step 4: Simplify the right side: .

Step 5: So $x = 16$.

Step 6: Compute the requested value: .

Answer:

Tip

When given one rational-exponent value of x and asked for another, first check whether the target exponent is a simple multiple of the given one before solving for x from scratch.

Trap

A student who treats exponents like linear scaling might compute as times 8, getting , instead of correctly using the power rule ; exponents combine multiplicatively, not by simple proportion.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Treated the exponent relationship as direct proportion, computing , instead of solving for x with the power rule and then re-evaluating .
16	Correctly solved for x ($x = 16$) but then stopped, reporting x itself instead of continuing to compute the requested .
256	Confused the exponent with squaring, computing instead of the square root of 16, after correctly solving $x = 16$.

**Shortcut**

Since , you can skip solving for x entirely: = — same answer in one line.

Desmos Tip

On Desmos's calculator, evaluate directly for a one-step check that matches the value found by fully solving for x first.

Radicals, Rational Exponents & Scientific Notation — Problem 6 (Very Hard)

Aurora Glassworks in Tallinn produces two similar (same-shape) crystal decanters — a display model and a smaller retail model. The display model has a surface area of , and the retail model has a surface area of . If the display model holds of liquid, and volume scales with the cube of the linear scale factor between similar solids, how many of liquid does the retail model hold?

Formula / Rule Used: For similar solids, area scales with the square of the linear scale factor and volume scales with the cube; equivalently, volume ratio = .

Solution

Step 1: Since the decanters are similar, the ratio of their surface areas equals the square of the ratio of their linear scale factors: .

Step 2: Simplify the area ratio: .

Step 3: Take the square root of both sides to get the linear scale factor: retail linear / display linear = .

Step 4: Volume scales with the CUBE of the linear scale factor (not the square used for area), so: retail volume / display volume = .

Step 5: Substitute the known display volume: retail volume / .

Step 6: Solve for the retail volume: retail volume = .

Step 7: Final answer: the retail model holds of liquid.

Answer:

Tip

Translate word problems about similar solids into scale-factor language first: area ratio = , so take a square root to isolate the linear ratio, then apply the separate power (cube for volume) that the question actually asks about.

Trap

A student might apply the same square relationship used for area (ratio) directly to volume, computing , instead of correctly cubing the linear scale factor to get the true volume ratio of and answer 64.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Applied the area ratio directly to volume instead of cubing the linear scale factor, computing instead of .
	Used the area ratio itself as if it were already the linear scale factor and cubed it directly, skipping the required square-root step that converts an



	area ratio into a linear ratio: .
	Inverted the surface-area ratio (used instead of), which flipped the scale factor from to 2 and produced a larger volume instead of the correct smaller one.

Shortcut

You can skip finding the linear scale factor as an intermediate decimal and go straight from area ratio to volume ratio using rational exponents: since area ratio is , volume ratio = directly, without ever writing " on its own line.

Desmos Tip

Type directly into Desmos's calculator to verify the retail volume equals 64 in a single step.

Radicals, Rational Exponents & Scientific Notation — Problem 7 (Hard)

For $x > 0$, simplify completely to a single power of x : $\sqrt[3]{x^2} \cdot \sqrt{x}$ / .

Formula / Rule Used: Convert negative exponents and roots to a single rational exponent, then apply the product and quotient rules by adding/subtracting exponents.

Solution

Step 1: Convert $\sqrt[3]{x^2}$ to a rational exponent: .

Step 2: Convert $\sqrt{x}$ to a rational exponent: .

Step 3: Combine the numerator using the product rule (add exponents): .

Step 4: Find a common denominator of 3: -, so -. The numerator is .

Step 5: Divide by the denominator using the quotient rule (subtract exponents): .

Step 6: Find a common denominator of 6: - and , so -.

Step 7: Final simplified form: .

Answer:

Tip

Rewrite every negative exponent and every root as a single rational exponent before combining anything — mixing negative-exponent notation with radical notation is what causes sign errors, not the exponent rules themselves.

Trap

Subtracting is the same as subtracting a negative, which flips to addition — a rushed student computes - (forgetting to distribute the negative sign onto the denominator's exponent) and lands on instead of correctly writing -.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Forgot to distribute the negative sign onto - when subtracting a negative exponent, computing - instead of -, giving instead of -.



	Treated the whole-number exponent -2 as if it already had a denominator of 3 (used - instead of converting it to -) before adding to , corrupting the numerator's exponent from the start.
	Correctly simplified the numerator to but forgot to divide by at all, submitting the numerator's exponent as the final answer.

Shortcut

Since every term is already a single power of x , you can add all three exponents at once — treat the denominator's exponent as subtracted with its sign flipped from the start rather than combining the numerator first and the denominator second.

Desmos Tip

Plug in $x = 64$ into both the original expression and on Desmos's calculator (both give 128) to confirm they match numerically.

Radicals, Rational Exponents & Scientific Notation — Problem 8 (Very Hard)

A materials lab in Eindhoven measures a thin film's cross-sectional area as times , in square micrometers. Express this area in proper scientific notation.

Formula / Rule Used: Power of a product rule: ; apply to each factor, then multiply and renormalize to scientific notation.

Solution

Step 1: Apply the rational exponent to the first factor: .

Step 2: Compute : the cube root of 8 is 2, and 2 squared is 4, so . This factor is .

Step 3: Apply the rational exponent to the second factor: .

Step 4: Compute , so this factor is .

Step 5: Multiply the two simplified factors: .

Step 6: Multiply the coefficients: $4 \times 3 = 12$. Add the exponents: .

Step 7: Combine: . Since 12 is not between 1 and 10, renormalize: .

Step 8: Multiply the powers of 10: .

Step 9: Final answer: square micrometers.

Answer: square micrometers

Tip

When a whole scientific-notation quantity is raised to a rational-exponent power, apply that exponent to the coefficient and the power of 10 separately — do not simplify the coefficient and exponent-of-10 in the same step, or you'll mix up which rule applies to which part.

Trap



It's tempting to apply the outer exponent (or) only to the power of 10 and leave the numeric coefficient (8 or 9) unraised, or vice versa — for instance treating as (forgetting to also raise the 8 to the power), which silently skips a required cube-root-then-square step.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
	Forgot to raise the coefficient 8 to the power in the first factor, treating as instead of , which threw off the final coefficient.
	Correctly cube-rooted-then-squared the coefficient 8 to get 4, but forgot to also apply the power to the exponent on 10, leaving instead of correctly scaling it down to .
	Dropped the negative sign when multiplying the exponent -4 by , writing instead of , which threw the final power of ten off by four orders of magnitude.

Shortcut

Since both bases already have nice perfect cube/square coefficients (8 and 9) and their powers-of-ten exponents divide evenly by the given fractional exponents, you can compute each factor's coefficient and power of ten in one line each, skipping any written-out radical form entirely.

Desmos Tip

Type directly into Desmos's calculator and confirm the displayed value matches before writing the final scientific-notation form.

PART 2 — PROBLEM-SOLVING & DATA ANALYSIS

7. Ratios & Proportions

Ratios & Proportions — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Ratio	Λόγος
Proportion	Αναλογία
Cross-multiplication	Εξίσωση γινομένων (χιαστί πολλαπλασιασμός)
Scale factor	Λόγος κλίμακας
Unit rate	Μοναδιαία τιμή

Ratios & Proportions — Condensed Theory

A ratio compares two quantities; a proportion sets two ratios equal. Almost every ratio word problem is solved fastest with the 'x' method.

Ratio Notation

$$a : b = a/b$$

Order matters — 'the ratio of boys to girls' is not the same as 'girls to boys.'

The 'x' Method

If $a:b = m:n$, let $a = mx$ and $b = nx$.

Introduces a single common multiplier so you can use the total (or any other given fact) to solve for x, then find each actual quantity.

Proportions & Cross-Multiplication

$$a/b = c/d \Leftrightarrow ad = bc$$

Set up both ratios in the same order (e.g. both as part/whole) before cross-multiplying.

Combining Two Ratios

To combine $x:y$ and $y:z$ into $x:z$, scale both ratios so the shared term (y) has the same value in each, using the LCM.

Direct vs. Inverse Proportion**Direct:** $y = kx$. **Inverse:** $y = k/x$.

Direct: both quantities increase together. Inverse: as one increases, the other decreases (e.g. more workers, less time).

 Watch Out

Always double-check which quantity is asked for last — many ratio problems ask for one specific part, not the total.

Ratios & Proportions — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Ratios & Proportions — Problem 1 (Hard)

A philatelist's collection contains Greek, Italian, and French stamps in the ratio 2:3:4. Each Greek stamp in the collection is individually insured for €0.50, and the collection contains 108 stamps in total. How many more French stamps are there than Italian stamps?

Formula / Rule Used: Three-part ratio 'x method': let the parts be $2x$, $3x$, $4x$ and use the total to solve for x .

Solution

Step 1: Let the numbers of Greek, Italian, and French stamps be $2x$, $3x$, and $4x$, matching the ratio 2:3:4.

Step 2: The insurance value of €0.50 per Greek stamp describes cost, not a count of stamps, so it plays no role in this counting problem — set it aside.

Step 3: Add the parts and set equal to the total: $2x + 3x + 4x = 108$, so $9x = 108$.

Step 4: Solve for x : $x = 108/9 = 12$.

Step 5: Find each count: Greek = $2x = 24$, Italian = $3x = 36$, French = $4x = 48$.

Step 6: Subtract to find the difference: French – Italian = $48 - 36 = 12$.

Answer: 12 more French stamps than Italian stamps.

Tip

Turn a three-part ratio into $2x$, $3x$, $4x$, set the sum equal to the given total, and solve for x before answering the actual question asked.

Trap

A rushed student, seeing €0.50 as the only 'extra' number in the problem, might try to work it into the ratio calculation (e.g., dividing 108 by 0.50) instead of recognizing it is irrelevant to the stamp count, producing a nonsensical result.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
24	Computed French – Greek ($48 - 24 = 24$) instead of French – Italian, answering with the wrong pair of groups.
1	Found the difference in ratio parts ($4 - 3 = 1$) but never multiplied back by $x = 12$, leaving the answer in ratio units instead of actual stamp counts.
6	Treated the €0.50 insurance value as a stamp count to subtract from the total ($108 - 54 = 54$, then $54/9 = 6$), mixing monetary units into a pure counting problem.

Shortcut

Since only the difference between two of the three groups is needed, you can skip listing all three counts and compute $(4 - 3)x = x = 12$ directly once x is known.

Desmos Tip

Type $108/9$ directly into the calculator to instantly confirm $x = 12$ before multiplying out each stamp count.

Ratios & Proportions — Problem 2 (Very Hard)

At a bakery in Lyon, the ratio of flour to sugar used in a recipe is 5:3, and the ratio of sugar to butter used is 2:5. If the bakery uses 300 grams of butter for one batch, how many grams of flour does it use?

Formula / Rule Used: Combine two ratios sharing a term by scaling each to a common value for that shared term (using the LCM).

Solution

Step 1: Write the two given ratios: flour:sugar = 5:3 and sugar:butter = 2:5.

Step 2: The shared term is sugar, which appears as 3 in the first ratio and 2 in the second — find the LCM of 3 and 2, which is 6.

Step 3: Scale flour:sugar = 5:3 by 2 (so sugar becomes 6): flour:sugar = 10:6.

Step 4: Scale sugar:butter = 2:5 by 3 (so sugar becomes 6): sugar:butter = 6:15.

Step 5: Since both ratios now share the same sugar value, combine them into one three-part ratio: flour:sugar:butter = 10:6:15.

Step 6: The butter amount, 300 g, corresponds to the 15 parts, so one part = $300/15 = 20$ g.

Step 7: Find the flour amount: flour = $10 \times 20 = 200$ grams.

Answer: 200 grams of flour.

Tip

When two ratios share a term, scale each ratio (using the LCM of the shared term's two values) so that term matches in both, then merge them into one three-part ratio before using any given total.

Trap



Because sugar is '3' in the first ratio and '2' in the second, a rushed student might just concatenate the numbers as flour:sugar:butter = 5:3:5 (reusing the sugar:butter figures unscaled) instead of finding the LCM of 3 and 2 and scaling both ratios to a common sugar value of 6, producing an incorrect combined ratio and a wrong flour amount.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
300 grams	Concatenated the two ratios directly as flour:sugar:butter = 5:3:5 (reusing butter's raw ratio number without scaling to match sugar's LCM value of 6), overstating the flour amount.
500 grams	Applied the flour:sugar ratio (5:3) directly to the butter amount as if sugar and butter were interchangeable, computing flour = $300 \times 5/3$ instead of combining through the properly scaled three-part ratio.
120 grams	Correctly found one part = $300/15 = 20$ g but then multiplied by sugar's part count (6) instead of flour's (10), reporting the sugar amount instead of the flour amount.

Shortcut

Since only the flour amount is needed, you can isolate the flour:butter portion of the combined ratio (10:15) and solve $300 \times 10/15 = 200$ directly, skipping the sugar amount entirely.

Desmos Tip

Type $300 * (10/15)$ into the calculator to instantly confirm the flour amount is 200 grams.

Ratios & Proportions — Problem 3 (Very Hard)

A construction crew of 8 workers can build a section of highway retaining wall near Thessaloniki in 15 days, with the number of days needed inversely proportional to the number of workers (assuming every worker works at the same steady rate). The company now wants to complete an identical wall section in only 6 days. How many additional workers, beyond the original 8, must be added to the crew to meet this deadline?

Formula / Rule Used: Inverse proportion: workers $\times$ days = k (a constant); find k first, then use it to solve the new scenario.

Solution

Step 1: Since days needed is inversely proportional to the number of workers, workers $\times$ days = k for some constant k .

Step 2: Use the original data to find k : $k = 8 \times 15 = 120$.

Step 3: For the new deadline, set up the equation with the new number of workers, w , and 6 days: $w \times 6 = 120$.

Step 4: Solve for w : $w = 120/6 = 20$.

Step 5: This means 20 workers total are needed to finish in 6 days.

Step 6: The question asks for additional workers beyond the original 8, so subtract: $20 - 8 = 12$.

Answer: 12 additional workers (20 workers total).

Tip

For inverse proportion word problems, first find the constant of variation $k = (\text{quantity 1}) \times (\text{quantity 2})$ from the given pair, then reuse k to solve for the missing quantity in the new scenario.

Trap

A student might set up a direct proportion instead ($8/15 = w/6$), treating workers and days as increasing together, and/or might stop at $w = 20$ without subtracting the original 8 workers, answering a question that actually asks how many MORE workers are needed.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
20 additional workers	Correctly solved for the new total number of workers ($w = 120/6 = 20$) but forgot the question asks for additional workers beyond the original 8, submitting the total instead of the difference.
$w = 3.2$	Set up a direct proportion ($8/15 = w/6$) instead of using the inverse-proportion relationship workers $\times$ days = k , incorrectly assuming fewer days would require fewer workers rather than more.
5 additional workers	Correctly found $w = 20$ total workers needed but subtracted 15 (the original number of days) instead of 8 (the original number of workers), mixing up which given number represents workers.

Shortcut

There is no shortcut past finding k and reapplying it — and since the question asks for additional workers, not the total, the final subtraction step cannot be skipped.

**Desmos Tip**

Graph $y = 120/x$ on the calculator and read the y -value at $x = 6$ to instantly confirm 20 total workers are needed.

Ratios & Proportions — Problem 4 (Hard)

Sail X and Sail Y are similar triangular sails, and their areas are in the ratio 4:9 (Sail X to Sail Y). If the perimeter of Sail X is 20 meters, what is the perimeter of Sail Y?

Formula / Rule Used: For similar figures, the ratio of corresponding lengths squared equals the ratio of areas: $(\text{length ratio})^2 = \text{area ratio}$.

Solution

Step 1: Sail X and Sail Y are similar, so the ratio of their perimeters (a length ratio) equals the ratio r of their corresponding side lengths.

Step 2: For similar figures, $\text{area ratio} = r^2$, so $r^2 = 4/9$.

Step 3: Solve for r by taking the square root of both sides: $r = \sqrt{4/9} = 2/3$.

Step 4: This means perimeter of X : perimeter of Y = 2:3.

Step 5: Set up the proportion: $20/(\text{perimeter of Y}) = 2/3$.

Step 6: Cross-multiply: $20 \times 3 = 2 \times (\text{perimeter of Y})$, so $60 = 2 \times (\text{perimeter of Y})$.

Step 7: Solve: perimeter of Y = $60/2 = 30$ meters.

Answer: Perimeter of Sail Y = 30 meters.

Tip

When similar figures give you an area ratio but ask about a length (a side or perimeter), take the square root of the area ratio first to get the correct length ratio — never use the area ratio as the length ratio directly.

Trap

A student who forgets that area scales with the square of the length ratio will use 4:9 directly as the perimeter ratio, computing perimeter of Y = $20 \times 9/4 = 45$ meters — exactly the wrong answer this problem is built to catch.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
45 meters	Used the area ratio 4:9 directly as the perimeter ratio without taking the square root, computing $20 \times 9/4 = 45$ instead of first finding the length ratio 2:3.
40/3 meters (≈ 13.33)	Found the correct length ratio 2:3 but inverted it when setting up the proportion (used $3/2$ instead of $2/3$), computing $20 \times 2/3$ instead of $20 \times 3/2$.
90 meters	Took the square root of only the numerator of the area ratio ($\sqrt{4} = 2$), leaving $2/9$ instead of the square root of the entire fraction ($\sqrt{4/9} = 2/3$), producing an inflated length ratio.

**Shortcut**

Once you know the length ratio is 2:3, you can skip formal cross-multiplication and just scale 20 by $\frac{3}{2}$ directly to get 30.

Ratios & Proportions — Problem 5 (Very Hard)

A jar contains only red and blue marbles in the ratio red:blue = 3:7. After 12 more red marbles are added to the jar (the number of blue marbles stays the same), the new ratio of red to blue becomes 1:1. What is the total number of marbles in the jar now?

Formula / Rule Used: Reverse ratio setup: let the original quantities be $3k$ and $7k$, write an equation from the described change, and solve for k .

Solution

Step 1: Let the original numbers of red and blue marbles be $3k$ and $7k$, matching the given ratio 3:7.

Step 2: After adding 12 red marbles, red becomes $3k + 12$, while blue stays at $7k$.

Step 3: The new ratio is 1:1, meaning the new red count equals the blue count: $3k + 12 = 7k$.

Step 4: Solve for k : $12 = 7k - 3k = 4k$, so $k = 3$.

Step 5: Find the blue count (unchanged): blue = $7k = 7(3) = 21$.

Step 6: Find the new red count: red_new = $3k + 12 = 9 + 12 = 21$.

Step 7: Add both counts to get the current total: $21 + 21 = 42$.

Answer: 42 marbles total (21 red and 21 blue).

Tip

When a final ratio is given after a stated change, assign the original quantities as multiples of a variable ($3k$, $7k$), write an equation reflecting exactly what changed, and solve for the variable.

Trap

A student might assume the 1:1 ratio means equal numbers were added to both colors, or might set $3k:7k = 1:1$ directly and ignore that only the red count changed by +12, leading to a wrong value of k or a wrong total.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
21	Correctly found $k = 3$ and computed blue = 21 (which also equals the new red count) but forgot to add red and blue together for the total, reporting only one color's count.
52	Solved $4k = 12$ but divided incorrectly, getting $k = 4$ instead of $k = 3$, then added the resulting mismatched red and blue counts, which no longer actually satisfy the 1:1 condition.
30	Used the original total before the change ($3k + 7k = 10k = 30$) instead of the total after the 12 red marbles were added, answering the wrong point in time.

**Shortcut**

Since the new ratio is exactly 1:1, you can skip a full cross-multiplied proportion and just recognize red_new must equal blue (7k), giving the simpler equation $3k + 12 = 7k$ directly.

Desmos Tip

Graph $y = 3x + 12$ and $y = 7x$ and find their intersection point to instantly confirm $k = 3$.

Ratios & Proportions — Problem 6 (Hard)

A map of Crete uses a scale where 4 centimeters represents 25 kilometers of actual distance. On this map, the distance from Chania to Rethymno measures 6.4 centimeters, and the distance from Rethymno to Heraklion measures 11.2 centimeters. Assuming the three cities lie along a straight coastal road in that order, what is the actual distance, in kilometers, from Chania to Heraklion?

Formula / Rule Used: Translate a scale into a unit rate (actual distance per cm), then apply that rate as a proportion to each map measurement.

Solution

Step 1: Translate the scale into a unit rate: 4 cm represents 25 km, so 1 cm represents $25/4 = 6.25$ km of actual distance.

Step 2: Convert the Chania–Rethymno map distance: $6.4 \text{ cm} \times 6.25 \text{ km/cm} = 40 \text{ km}$.

Step 3: Convert the Rethymno–Heraklion map distance: $11.2 \text{ cm} \times 6.25 \text{ km/cm} = 70 \text{ km}$.

Step 4: Since the three cities lie on a straight road in the order Chania, Rethymno, Heraklion, the actual Chania–Heraklion distance equals the sum of the two segments.

Step 5: Add the two actual distances: $40 \text{ km} + 70 \text{ km} = 110 \text{ km}$.

Answer: 110 kilometers.

Tip

Convert a map or drawing scale into a single unit rate (actual distance per centimeter) first, then multiply every given map measurement by that same rate.

Trap

A student might invert the scale (using $4/25$ cm per km instead of $25/4$ km per cm) or might convert only one segment and forget that the straight-road condition requires adding both actual distances to get the full Chania-to-Heraklion distance.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$\approx 2.82 \text{ km}$	Inverted the scale, using $4/25$ km per cm instead of $25/4$ km per cm (6.25 km/cm), producing a drastically undersized distance.
40 km	Converted only the Chania–Rethymno segment ($6.4 \text{ cm} \times 6.25 = 40 \text{ km}$) and forgot to add the Rethymno–Heraklion segment, treating the first conversion as the final answer.



30 km	Subtracted the two map distances ($11.2 - 6.4 = 4.8$ cm) instead of adding them, misunderstanding that both segments lie along the same road and must be summed for the total distance.
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Shortcut

Since both segments use the same scale, you can add the map distances first ($6.4 + 11.2 = 17.6$ cm) and multiply once by 6.25 km/cm to get 110 km in a single multiplication instead of two.

Desmos Tip

Type $17.6 * 6.25$ into the calculator to instantly verify the total actual distance is 110 km.

Ratios & Proportions — Problem 7 (Hard)

The amount of paint needed to cover a wall varies directly with the wall's area. A painter in Porto used 3 liters of paint to cover a 15 square-meter wall. For a new job, the painter must cover three walls with areas 20, 35, and 25 square meters. The client wants two coats of paint on only the largest of the three walls, and one coat on each of the other two. How many liters of paint does the painter need in total?

Formula / Rule Used: Direct variation: $\text{liters} = k \times \text{area}$; find k from the given pair, then apply it to each new area, adjusting for any wall that needs an extra coat.

Solution

Step 1: Direct variation means $\text{liters} = k \times \text{area}$; use the given pair to find k : $3 = k \times 15$, so $k = 3/15 = 0.2$ liters per square meter.

Step 2: Find the one-coat paint amount for each wall: $20 \times 0.2 = 4$ L, $35 \times 0.2 = 7$ L, $25 \times 0.2 = 5$ L.

Step 3: Add these one-coat amounts: $4 + 7 + 5 = 16$ liters.

Step 4: Identify the largest wall (35 m^2) — this is the one that needs a second coat, requiring an extra 7 liters.

Step 5: Add the extra coat to the one-coat total: $16 + 7 = 23$ liters.

Answer: 23 liters of paint.

Tip

For direct variation, find $k = \text{value/other value}$ from the given pair first, then apply that same k to every new quantity — treat any extra requirement (like a second coat) as a separate addition after the base amounts are found.

Trap

A student who computes only the one-coat amounts for each wall and reports 16 liters forgets that the problem explicitly requires a second coat on the largest wall, an amount that must be added separately.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
16 liters	Correctly found the one-coat total ($4 + 7 + 5 = 16$) but forgot the largest



	wall needs a second coat, never adding the extra 7 liters.
21 liters	Applied the second coat to the 25 m ² wall instead of the actual largest wall (35 m ²), misreading which wall the client wants double-coated.
575 liters	Inverted the direct-variation constant, using $k = 15/3 = 5$ L per square meter instead of $k = 3/15 = 0.2$ L per square meter, inflating every computed amount.

Shortcut

Since only one wall needs a second coat, you can skip recomputing all three walls under a doubled scenario and just add one extra single-coat amount (7 L) for the 35 m² wall to the one-coat total.

Desmos Tip

Type $0.2 \cdot (20+35+25) + 0.2 \cdot 35$ directly into the calculator to verify the total in one line.

Ratios & Proportions — Problem 8 (Very Hard)

A dough recipe requires flour and water in the ratio 8:3. While preparing a large batch, a baker accidentally reverses the ratio, mixing flour and water in the ratio 3:8 instead, using 9 kilograms of flour for the batch. Without discarding any of the mixture, the baker plans to add only more flour until the ratio of flour to water matches the correct 8:3 ratio. How many kilograms of additional flour must the baker add?

Formula / Rule Used: Use the mistaken ratio to find the quantity that stays fixed (water), then use the correct ratio with that fixed quantity to solve for the target amount, subtracting what's already there.

Solution

Step 1: Under the mistaken ratio, flour:water = 3:8, and the flour used is 9 kg, so 9 kg corresponds to 3 parts: one part = $9/3 = 3$ kg.

Step 2: Find the water used (8 parts): water = $8 \times 3 = 24$ kg. This amount stays fixed, since only flour will be added.

Step 3: Let F be the total flour needed so the ratio becomes the correct 8:3: $F/24 = 8/3$.

Step 4: Cross-multiply: $3F = 8 \times 24 = 192$.

Step 5: Solve: $F = 192/3 = 64$ kg of flour needed in total.

Step 6: Subtract the flour already used: additional flour = $64 - 9 = 55$ kg.

Answer: 55 kilograms of additional flour.

Tip

When fixing a ratio by adding only one ingredient, first identify which quantity stays constant (here, water) and use it as the fixed anchor to solve for the target amount of the other ingredient.

Trap

A student who assumes the given 9 kg of flour was already correctly proportioned and simply rescales it by $8/3$ ($9 \times 8/3 = 24$) never recognizes that water — not flour — is the fixed quantity that must anchor the new proportion, and reports 24 as if it were the answer.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
64 kg	Correctly solved for the total flour needed (64 kg) but forgot to subtract the 9 kg of flour already used, reporting the total instead of the additional amount.
24 kg	Assumed the given 9 kg of flour was already in the correct 8:3 ratio and scaled it directly by $\frac{8}{3}$ ($9 \times \frac{8}{3} = 24$), never recognizing that water is the fixed quantity needed to solve for the new flour amount.
15 kg	Assumed the total mixture weight ($9 + 24 = 33$ kg) stays fixed and re-split it into the new 8:3 ratio (getting 24 kg flour), instead of recognizing that only flour is added, so water must remain fixed at 24 kg.

Shortcut

There's no shortcut past anchoring on the fixed quantity (water) first — but once you have it, you can combine the last three steps into one line: $24 \times (\frac{8}{3}) - 9$.

Desmos Tip

Type $24 * 8/3 - 9$ into the calculator to confirm the additional flour needed is 55 kg in one line.

8. Probability

Probability — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Probability	Πιθανότητα
Independent events	Ανεξάρτητα ενδεχόμενα
Dependent events	Εξαρτημένα ενδεχόμενα
Mutually exclusive events	Ασυμβίβαστα ενδεχόμενα
Sample space	Δειγματικός χώρος

Probability — Condensed Theory

Probability quantifies likelihood between 0 and 1. The SAT tests basic probability, compound events (and/or), conditional probability, and two-way tables.

Basic Probability

$$P(\text{event}) = \text{favorable outcomes} / \text{total outcomes}$$

Always between 0 (impossible) and 1 (certain), inclusive.

Complement

$$P(A') = 1 - P(A)$$

Useful for 'at least one' problems: $P(\text{at least one}) = 1 - P(\text{none})$.

'And' — Independent vs. Dependent

$$\text{Independent: } P(A \text{ and } B) = P(A) \times P(B). \quad \text{Dependent: } P(A \text{ and } B) = P(A) \times P(B|A).$$

'With replacement' → independent. 'Without replacement' → dependent, so the second probability's denominator changes.

'Or' — Mutually Exclusive vs. Not

$$\text{Mutually exclusive: } P(A \text{ or } B) = P(A) + P(B). \quad \text{General: } P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

Subtract the overlap whenever the two events can happen together, to avoid double-counting.

Conditional Probability

$$P(A|B) = P(A \text{ and } B) / P(B)$$

The denominator is always the group specified after 'given' — not the whole population.

Two-Way Tables

Simple probability uses target cell / total. Conditional probability restricts the denominator to the 'given' row or column first.



Watch Out

'Without replacement' changes the denominator (and sometimes the numerator) for every subsequent draw — always recount what remains.

Probability — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Probability — Problem 1 (Hard)

CLC Educational Institute surveyed 200 students across its Athens (North) and Piraeus (South) branches about whether they own a bicycle. Of the 120 students at the North branch, 70 own a bicycle. Of the 80 students at the South branch, 20 do NOT own a bicycle. If a student who owns a bicycle is selected at random from the entire survey, what is the probability that this student is from the South branch?

Formula / Rule Used: Conditional probability from a two-way table: $P(\text{South} \mid \text{owns bike}) = (\text{South students who own bikes}) / (\text{total students who own bikes})$.

Solution

Step 1: Find the number of North branch students who do NOT own a bicycle: $120 - 70 = 50$.

Step 2: Find the number of South branch students who own a bicycle: $80 - 20 = 60$ (since 20 of the 80 South students do not own one).

Step 3: Find the total number of students, from both branches, who own a bicycle: 70 (North) + 60 (South) = 130 .

Step 4: Conditional probability restricts the denominator to the 'given' group -- here the given condition is 'owns a bicycle,' so the denominator is 130 , not the grand total of 200 .

Step 5: Divide the South owners by the total owners: $P(\text{South} \mid \text{owns bike}) = 60/130$.

Step 6: Simplify the fraction: $60/130 = 6/13$.

Answer: $6/13$

Tip



In a two-way table conditional probability question, first fill in any missing cell using its row or column total, then restrict your denominator to exactly the 'given' condition -- never the grand total.

Trap

A rushed student divides by the grand total of 200 students instead of the 130 students who own bikes, getting $60/200 = 3/10$, which actually answers a different question ('South AND owns a bike') rather than the conditional probability that was asked.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3/10	Divided the South-owner count by the grand total of 200 students instead of the 130 total bike owners, answering 'P(South and owns a bike)' instead of the requested conditional probability.
70/130	Put the North branch's bike-owner count (70) in the numerator instead of the South branch's (60), effectively finding $P(\text{North} \mid \text{owns bike})$ instead of $P(\text{South} \mid \text{owns bike})$.
60/80	Divided the South bike owners by the South branch total (80) instead of the total bike owners across both branches (130), finding $P(\text{owns bike} \mid \text{South})$ instead of $P(\text{South} \mid \text{owns bike})$ -- the condition was reversed.

Shortcut

There's no way to skip filling in the missing cell -- that step is required -- but once the table is complete you only need to track two numbers: the South-owner count and the total-owner count, not the whole table.

Probability — Problem 2 (Very Hard)

At the Port of Piraeus, random safety inspections are performed on shipping containers. Historical records show that, independent of any other container, 1 out of every 10 containers fails its safety check. The port handles about 4,000 containers per month. If an inspector randomly selects 3 containers to check, what is the probability that at least one of the 3 fails its safety check?

Formula / Rule Used: Complement rule for 'at least one': $P(\text{at least one}) = 1 - P(\text{none})$.

Solution

Step 1: Translate the rate into a probability: 1 out of every 10 containers fails, so $P(\text{fail}) = 1/10$ and $P(\text{pass}) = 1 - 1/10 = 9/10$.

Step 2: The monthly total of 4,000 containers is not needed -- the question only concerns the 3 selected containers.

Step 3: 'At least one fails' is the complement of 'none of the 3 fail,' so find $P(\text{none fail})$ first.

Step 4: Since the containers are independent, multiply: $P(\text{none fail}) = 9/10 \times 9/10 \times 9/10 = 729/1000$.

Step 5: Apply the complement rule: $P(\text{at least one fails}) = 1 - 729/1000$.

Step 6: Subtract: $1000/1000 - 729/1000 = 271/1000$.

Answer: **271/1000 (= 0.271)**

**Tip**

Whenever a problem asks for the probability of 'at least one' success among repeated independent trials, immediately switch to finding the probability of 'none' and subtract from 1 -- it's almost always faster than adding up every possible case.

Trap

A rushed student multiplies the single-event probability by the number of trials -- $3 \times 1/10 = 3/10$ -- treating the probabilities as if they simply add, but that overcounts cases where more than one container fails and ignores how the complement actually compounds.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3/10	Multiplied the single-container failure probability by the number of trials ($3 \times 1/10$) instead of applying the complement rule, treating probabilities as additive across independent trials.
729/1000	Correctly computed $P(\text{none fail}) = 9/10 \times 9/10 \times 9/10$ but forgot the final step of subtracting from 1, submitting the complement of the requested event instead of the event itself.
243/1000	Computed $P(\text{exactly one fails}) = 3 \times (1/10) \times (9/10)^2 = 243/1000$, mistaking 'at least one' for 'exactly one' fails and only counting the single-failure case.

Shortcut

The complement approach IS the shortcut here -- directly listing every combination of 1, 2, or 3 failing containers would take three separate calculations instead of one subtraction.

Desmos Tip

Type $1 - (9/10)^3$ directly into the calculator to get 0.271 instantly, avoiding any arithmetic slip when cubing $9/10$ by hand.

Probability — Problem 3 (Very Hard)

A drawer at a stationery shop in Thessaloniki contains 6 red pens and 4 blue pens, with no other colors. Two pens are drawn at random, one after another, without putting the first pen back. What is the probability that both pens drawn are the same color?

Formula / Rule Used: Dependent 'and': $P(A \text{ and } B) = P(A) \times P(B | A)$, combined across colors with the mutually exclusive 'or' rule.

Solution

Step 1: There are 10 pens total (6 red + 4 blue), and the draws are without replacement, so the second draw's probability depends on the first.

Step 2: Find $P(\text{both red})$: $P(\text{red 1st}) = 6/10$, then $P(\text{red 2nd} | \text{red 1st}) = 5/9$, since one red pen is gone and 9 pens remain.

Step 3: Multiply: $P(\text{both red}) = 6/10 \times 5/9 = 30/90 = 1/3$.

Step 4: Find $P(\text{both blue})$: $P(\text{blue 1st}) = 4/10$, then $P(\text{blue 2nd} | \text{blue 1st}) = 3/9$, since one blue pen is gone and 9 pens remain.

Step 5: Multiply: $P(\text{both blue}) = 4/10 \times 3/9 = 12/90 = 2/15$.

Step 6: 'Both red' and 'both blue' cannot happen at the same time, so they are mutually exclusive -- add the two probabilities: $P(\text{same color}) = 1/3 + 2/15$.

Step 7: Find a common denominator (15) and add: $5/15 + 2/15 = 7/15$.

Answer: 7/15

Tip

When a 'same color' or 'same type' question involves more than one category, compute one dependent-draw probability per category, then combine the category results with the mutually exclusive OR rule.

Trap

A student who correctly computes $P(\text{both red}) = 1/3$ often stops there, forgetting that 'both blue' is a separate valid way to get 'same color,' which gives an incomplete answer of $1/3$ instead of the full $7/15$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
1/3	Computed only $P(\text{both red})$ and stopped, forgetting that 'both blue' is a second valid way to satisfy 'same color' and must be added.
13/25	Treated the two draws as independent with replacement, computing $6/10 \times 6/10 + 4/10 \times 4/10 = 52/100$ instead of reducing the denominator to 9 and the count by 1 for the second, dependent draw.
8/15	Computed $P(\text{different colors}) = 1 - 7/15 = 8/15$, answering the complement of the question that was actually asked.

Shortcut

There isn't a faster route than computing each color's dependent probability separately, but noticing both cases share the same fraction structure (count/10 x (count-1)/9) lets you set up both products side-by-side before simplifying, saving rework.

Probability — Problem 4 (Hard)

A drawer of laboratory samples at a CLC science lab contains 8 clearly labeled 'Group X' vials and an unknown number of unlabeled 'Group Y' vials, with no other types. The drawer can hold up to 50 vials in total. If the probability of randomly selecting a Group Y vial is $5/13$, how many Group Y vials are in the drawer?

Formula / Rule Used: Reverse probability setup: $P(Y) = (\text{number of Y}) / (\text{total})$, then solve the resulting equation for the unknown count.

Solution

Step 1: Let y = the number of Group Y vials. The drawer's 50-vial capacity is irrelevant here -- it's a maximum, not the actual count.

Step 2: The total number of vials in the drawer is $8 + y$, the 8 Group X vials plus the unknown Group Y vials.

Step 3: Set up the probability equation: $P(Y) = y / (8 + y) = 5/13$.

Step 4: Cross-multiply: $13 \times y = 5 \times (8 + y)$.

Step 5: Expand the right side: $13y = 40 + 5y$.

Step 6: Subtract $5y$ from both sides: $8y = 40$.

Step 7: Divide by 8: $y = 5$.

Answer: 5 Group Y vials

Tip

In a reverse-setup probability problem, name the unknown count with a variable, write the total as a sum that includes it, and solve the resulting probability equation like any other algebraic fraction equation.

Trap

A rushed student sets up $y/8 = 5/13$, dividing by the known count instead of the full total, which gives $y = 40/13$ -- not a whole number of vials, a clear sign the wrong quantity was used as the denominator.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$y = 40/13$	Set up $y/8 = 5/13$, dividing by the known Group X count instead of the total $8 + y$, so the denominator never included the unknown quantity being solved for.
$y = 64/5$ (12.8)	Set up the equation for $P(X)$ instead of $P(Y)$ -- solved $8/(8+y) = 5/13$, using the labeled count as the numerator instead of the unlabeled one.
$y = 250/13$	Used the drawer's maximum capacity (50) as the actual total instead of $8 + y$, solving $y/50 = 5/13$ with an irrelevant number pulled from the problem.

Shortcut

Since the given probability $5/13$ already has denominator 13, you can suspect the total is 13 and check $y = 13 - 8 = 5$ directly; solving the equation algebraically is just as fast here and confirms it without guessing.

Desmos Tip



In Desmos, graph $f(y) = y/(8+y)$ alongside the horizontal line $y = 5/13$ and find their intersection at $y = 5$ to confirm the algebra.

Probability — Problem 5 (Hard)

CLC Multinational employs 300 people across two offices -- 180 in Athens and 120 in Vienna. Across all 300 employees, 150 speak German, 130 speak French, and 50 speak both languages fluently. If an employee is selected at random from the full company, what is the probability that they speak German or French?

Formula / Rule Used: General 'or' rule for non-mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Solution

Step 1: The Athens/Vienna office split (180 and 120) has nothing to do with which languages employees speak -- it's extra information to ignore.

Step 2: Speaking German and speaking French are NOT mutually exclusive, since 50 employees speak both, so the basic addition rule alone would double-count them.

Step 3: Find each individual probability: $P(\text{German}) = 150/300$ and $P(\text{French}) = 130/300$.

Step 4: Find the overlap probability: $P(\text{both}) = 50/300$.

Step 5: Apply the general OR rule: $P(\text{German or French}) = 150/300 + 130/300 - 50/300$.

Step 6: Combine the numerators: $150 + 130 - 50 = 230$, so $P(\text{German or French}) = 230/300$.

Step 7: Simplify: $230/300 = 23/30$.

Answer: 23/30

Tip

Whenever an 'or' problem gives you an overlap number (people who do BOTH things), treat that as your signal the events are not mutually exclusive and you must subtract the overlap.

Trap

A student who ignores the phrase 'both languages' simply adds $150/300 + 130/300 = 280/300 = 14/15$, double-counting the 50 employees who speak both languages.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
14/15	Added $150/300 + 130/300$ without subtracting the overlap of $50/300$, double-counting the employees who speak both languages.
1/6	Computed only the overlap probability $50/300 = 1/6$ and submitted it directly, mistaking the 'and' (both languages) value for the requested 'or' (German or French) value.
3/5	Subtracted the overlap twice -- $150/300 + 130/300 - 50/300 - 50/300 = 180/300$ -- instead of the single subtraction the general OR rule requires.



Shortcut

No shortcut beats the direct formula here, but working the whole calculation over the common denominator of 300 from the start, rather than simplifying each fraction first, avoids extra fraction work.

Probability — Problem 6 (Very Hard)

A jar at a school fair in Rotterdam contains only red marbles and blue marbles, in the ratio 2:3. A volunteer adds 10 more blue marbles to the jar, which changes the ratio of red to blue marbles to 2:5. Using the ORIGINAL numbers of marbles (before the blue ones were added), if two marbles are drawn at random without replacement, what is the probability that both are red?

Formula / Rule Used: Ratio-to-count setup (red = $2k$, blue = $3k$) to solve for k , then dependent 'and' probability using the original counts.

Solution

Step 1: Let the original counts be red = $2k$ and blue = $3k$, using the given ratio 2:3.

Step 2: After adding 10 blue marbles, the new ratio is 2:5, so $\text{red}/(\text{blue} + 10) = 2/5$, giving $2k/(3k + 10) = 2/5$.

Step 3: Cross-multiply: $5 \times 2k = 2 \times (3k + 10)$, so $10k = 6k + 20$.

Step 4: Subtract $6k$ from both sides: $4k = 20$, so $k = 5$.

Step 5: Find the original counts: red = $2k = 10$, blue = $3k = 15$, for an original total of 25 marbles.

Step 6: Use these ORIGINAL counts for the probability question: $P(\text{1st red}) = 10/25$.

Step 7: Since the draw is without replacement, $P(\text{2nd red} \mid \text{1st red}) = 9/24$, since one red marble is gone and 24 remain.

Step 8: Multiply: $P(\text{both red}) = 10/25 \times 9/24 = 90/600$.

Step 9: Simplify: $90/600 = 3/20$.

Answer: $3/20$

Tip

When a ratio equation is hidden inside a probability question, solve completely for the ratio's constant (k) first, then plug the resulting counts into the probability calculation as a separate second step -- don't try to do both at once.

Trap

A student in a hurry uses the NEW post-addition counts (10 red and 25 blue, out of 35) instead of the ORIGINAL counts (10 red and 15 blue, out of 25) that the question specifically asks for, giving the wrong probability $10/35 \times 9/34 = 90/1190 = 9/119$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$9/119$	Used the NEW post-addition counts (10 red, 25 blue, 35 total) instead of the ORIGINAL counts the question specifically asked for (10 red, 15 blue, 25 total).



1/10	Used the raw ratio numbers 2 and 3 directly as if they were the actual marble counts ($P = 2/5 \times 1/4$) instead of solving for k first to find the true counts of 10 and 15.
4/25	Treated the two draws as independent with replacement, computing $10/25 \times 10/25$ instead of reducing both the numerator and denominator for the second, dependent draw.

Shortcut

There's no way to skip solving for k first -- the probability cannot be computed until the actual counts are known -- but once $k = 5$ is found, the probability part is just two quick multiplications.

Desmos Tip

Graph $f(k) = 2k/(3k+10)$ alongside the line $y = 2/5$ in Desmos and find their intersection at $k = 5$ to confirm the ratio equation before computing the final probability.

Probability — Problem 7 (Hard)

At a CLC Educational Institute charity raffle in Vienna, the odds in favor of winning a prize with a single ticket are stated as 3 to 17 (that is, for every 3 winning tickets sold, there are 17 non-winning tickets). A student buys 2 raffle tickets, and the outcome of each ticket is independent of the other. What is the probability that the student wins with exactly one of the two tickets?

Formula / Rule Used: Convert odds to probability (parts of a whole), then combine independent 'and' with mutually exclusive 'or' for an 'exactly one of two' event.

Solution

Step 1: Convert the odds 3 to 17 into a probability: the total parts are $3 + 17 = 20$, so $P(\text{win}) = 3/20$ and $P(\text{lose}) = 17/20$.

Step 2: 'Exactly one of the two tickets wins' can happen two different ways: (win, lose) or (lose, win).

Step 3: Since the tickets are independent, compute $P(\text{win then lose}) = 3/20 \times 17/20 = 51/400$.

Step 4: Compute the other order: $P(\text{lose then win}) = 17/20 \times 3/20 = 51/400$.

Step 5: These two orders are mutually exclusive (they can't both occur), so add them: $P(\text{exactly one win}) = 51/400 + 51/400 = 102/400$.

Step 6: Simplify: $102/400 = 51/200$.

Answer: **51/200**

Tip

When a problem gives odds instead of a probability, first convert by adding the two parts to get the total, then treat the rest as a normal probability problem -- and for 'exactly one of two independent trials,' always account for both possible orders.

Trap

A rushed student treats the odds '3 to 17' as if they were the probability itself, using $3/17$ as $P(\text{win})$ instead of correctly converting it to $3/20$ by adding $3 + 17$ to get the total number of parts.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
84/289	Misread the odds '3 to 17' as the probability itself, using $3/17$ as $P(\text{win})$ and $14/17$ as $P(\text{lose})$ instead of converting to parts-of-20 ($3/20$ and $17/20$).
51/400	Computed only $P(\text{win then lose}) = 3/20 \times 17/20$ and forgot to also add $P(\text{lose then win})$, missing the second order in which exactly one ticket wins.
111/400	Confused 'exactly one wins' with 'at least one wins' and applied the complement rule instead: $1 - P(\text{lose, lose}) = 1 - (17/20)^2 = 111/400$, which answers a different question.

Shortcut

Once $P(\text{win}) = 3/20$ is found, 'exactly one of two independent trials' can be computed in one line as $2 \times (3/20) \times (17/20) = 102/400 = 51/200$, without writing out both orders separately.

Desmos Tip

Type $2 * (3/20) * (17/20)$ directly into the calculator to get $51/200$ in one step, avoiding a separate addition of two equal fractions.

Probability — Problem 8 (Very Hard)

A shipment of 15 lightbulbs bound for a hardware store in Bergen contains 4 defective bulbs, mixed in with 11 working bulbs. A technician tests the bulbs one at a time, without replacement, stopping as soon as a defective bulb is found. What is the probability that the very first defective bulb found is the third bulb tested?

Formula / Rule Used: Sequential dependent 'and': the required order is (good, good, defective), multiplying probabilities that update as bulbs are removed.

Solution

Step 1: For the third bulb to be the first defective one found, the first two bulbs tested must both be working, and the third must be defective -- this is one specific order, not any arrangement.

Step 2: The shipment size of 500 units to the store is not used -- only the 15 bulbs in this particular batch matter.

Step 3: Find $P(\text{1st bulb good})$: 11 working bulbs out of 15 total, so $P = 11/15$.

Step 4: Find $P(\text{2nd bulb good} \mid \text{1st was good})$: 10 working bulbs remain out of 14 total, so $P = 10/14$.

Step 5: Find $P(\text{3rd bulb defective} \mid \text{first two were good})$: 4 defective bulbs remain out of 13 total, so $P = 4/13$.

Step 6: Multiply the three probabilities in this exact order: $11/15 \times 10/14 \times 4/13 = 440/2730$.

Step 7: Simplify: $440/2730 = 44/273$ (dividing numerator and denominator by 10 gives $44/273$, which has no further common factors).

Answer: 44/273

**Tip**

When a problem asks for a specific position where an event first occurs, translate it into one exact ordered sequence (e.g., 'not yet, not yet, then yes') and multiply the dependent probabilities for that sequence only -- don't average over all possible positions.

Trap

A rushed student computes the probability that exactly one defective bulb appears anywhere among the first three tested (multiplying by 3 for the three possible positions), which answers a broader question than 'the third bulb specifically is the first defective one.'

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
44/91	Computed P(exactly one defective among the first three bulbs, in any order) by multiplying by 3 for the three possible positions, instead of the single required order good-good-defective.
4/15	Computed only the plain, unconditioned probability that the third bulb drawn is defective, ignoring the requirement that the first two bulbs must also both be good for it to be the FIRST defective found.
58/91	Computed P(at least one defective among the first three bulbs) via the complement $1 - (11/15 \times 10/14 \times 9/13)$, answering whether any defective appears within three draws rather than whether the third draw specifically is the first defective.

Shortcut

No shortcut beyond direct sequential multiplication is available here, since only one specific order (good, good, defective) is being asked for -- unlike an 'any position' version of this problem, which would require multiplying the final product by 3.

Desmos Tip

Type $(11/15) \times (10/14) \times (4/13)$ into the calculator to get 44/273 directly, avoiding an arithmetic slip across the three-step multiplication.

9. Unit Conversions & Rates

Unit Conversions & Rates — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Unit conversion	Μετατροπή μονάδων
Conversion factor	Συντελεστής μετατροπής
Rate	Ρυθμός
Dimensional analysis	Ανάλυση διαστάσεων (μονάδων)

Unit Conversions & Rates — Condensed Theory

Unit conversion is proportional reasoning applied to measurement: multiply by a conversion factor written so the unwanted units cancel.

Unit Conversion Method

$$\text{Given} \times (\text{desired unit} / \text{given unit}) = \text{result}$$

Set up the fraction so the given unit cancels diagonally, leaving only the desired unit.

Multi-Step Conversions

Chain multiple conversion factors in a row when no direct factor is given (e.g. seconds → minutes → hours).

Squared & Cubed Units

$$(1 \text{ unit})^2 = (\text{conversion factor})^2 ; (1 \text{ unit})^3 = (\text{conversion factor})^3$$

You must square or cube the ENTIRE conversion factor, not just the number, when converting area or volume.

Density & Rate

$$\text{Density} = \text{mass} / \text{volume} . \quad \text{Rate} = \text{quantity} / \text{time} .$$

Both are just a ratio of two different units — apply the same cancel-the-units logic.



Watch Out

Converting from a smaller unit to a larger one means dividing; larger to smaller means multiplying — check the direction before you calculate.



Unit Conversions & Rates — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Unit Conversions & Rates — Problem 1 (Very Hard)

A speed camera near Lisbon reports a car's speed in two different units at the same instant: feet per second and miles per hour. The technician notices the feet-per-second reading is exactly 14 more than the miles-per-hour reading. Given that 1 mile = 5280 feet and 1 hour = 3600 seconds, what is the car's speed in miles per hour?

Formula / Rule Used: Convert one variable's units using the exact conversion fraction, then solve the resulting linear equation.

Solution

Step 1: Let v = the car's speed in miles per hour.

Step 2: Convert v to feet per second: $v \text{ mph} = v \times (5280 \text{ ft} / 3600 \text{ s}) = (22/15)v \text{ ft/s}$.

Step 3: Translate the given relationship into an equation: the ft/s reading equals the mph reading plus 14, so $(22/15)v = v + 14$.

Step 4: Clear the fraction by multiplying every term by 15: $22v = 15v + 210$.

Step 5: Subtract $15v$ from both sides: $7v = 210$.

Step 6: Divide by 7: $v = 30$.

Step 7: Check: $30 \text{ mph} = 30 \times 22/15 = 44 \text{ ft/s}$, and $44 - 30 = 14$, which matches the given condition.

Answer: 30 mph

Tip

When a problem relates the same speed expressed in two different units, convert everything into one variable using the exact conversion factor as a fraction (don't round it to a decimal), then solve algebraically.

Trap

A student who rounds $5280/3600$ to a decimal too early, or who writes the equation backwards as $v - (22/15)v = 14$ (assuming the mph reading is the larger one), will get an incorrect or nonsensical value instead of correctly treating the ft/s reading as the larger one.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
28 mph	Rounded the conversion factor $5280/3600$ to 1.5 instead of using the exact fraction $22/15$, then solved $1.5v - v = 14$ to get $v = 28$.
44 mph	Correctly solved for the ft/s value (44) but reported that number as the final mph answer instead of continuing to identify v itself.
≈ 6.5 mph	When isolating v from $7v = 210$ (or the equivalent $(7/15)v = 14$), multiplied both sides by $7/15$ instead of dividing by it (multiplying by the reciprocal $15/7$), inverting the fraction incorrectly.



Shortcut

Keep the conversion factor as the exact fraction $\frac{22}{15}$ and immediately clear denominators by multiplying the whole equation by 15 — this turns it into $22v = 15v + 210$ with no decimals or rounding anywhere.

Desmos Tip

Graph $y = (\frac{22}{15})x$ and $y = x + 14$ and find their intersection point — the x-coordinate (30) is the mph speed and the y-coordinate (44) confirms the ft/s reading.

Unit Conversions & Rates — Problem 2 (Hard)

A delicatessen in Thessaloniki sells graviera cheese at €16.80 per kilogram. A tourist orders 3.5 pounds of cheese to take home, but the shop's scale only measures in 50-gram increments and always rounds a customer's order UP to the next increment before weighing. Using 1 kilogram = 2.20462 pounds, how much will the tourist pay, rounded to the nearest cent?

Formula / Rule Used: Unit conversion (pounds to kilograms) combined with a per-unit rate (price $\times$ quantity), plus a stated rounding rule.

Solution

Step 1: Convert 3.5 pounds to kilograms: $3.5 \div 2.20462 \approx 1.5876$ kg.

Step 2: The scale reads only in 0.05 kg (50 g) increments; the increments bracketing 1.5876 are 1.55 kg and 1.60 kg.

Step 3: Since the shop always rounds up, the weighed amount used for billing is 1.60 kg.

Step 4: Multiply the rounded weight by the price per kilogram: $1.60 \text{ kg} \times \text{€}16.80/\text{kg} = \text{€}26.88$.

Step 5: €26.88 already has exactly two decimal places, so no further rounding is needed.

Answer: €26.88

Tip

When a real-world purchase combines a unit conversion with a price rate and a rounding rule, convert first, apply the rounding rule exactly as stated, and only then multiply by the rate.

Trap

A student who converts pounds to kilograms correctly but forgets the shop's 'round up to the nearest 50 g' rule will multiply the unrounded 1.5876 kg directly by the price instead of the correct rounded 1.60 kg.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
€26.67	Skipped the rounding-up-to-nearest-50g step entirely and multiplied the unrounded 1.5876 kg directly by €16.80.
€130.20	Multiplied pounds by the conversion factor instead of dividing ($3.5 \times 2.20462 = 7.716$ kg instead of $3.5 \div 2.20462$), then rounded up to 7.75 kg and paid for nearly five times too much cheese.
€58.80	Never converted pounds to kilograms at all — treated the 3.5 lb quantity



as if it were already 3.5 kg and multiplied directly by the per-kilogram price.

Shortcut

You can estimate quickly with $1 \text{ lb} \approx 0.4536 \text{ kg}$ ($3.5 \times 0.4536 \approx 1.588 \text{ kg}$) to sanity-check the division, but the rounding-to-50g step and final multiplication still have to be done by hand.

Unit Conversions & Rates — Problem 3 (Hard)

A juice bar in Ljubljana blends its signature drink using orange, beet, and ginger juice according to a rule the owner describes this way: 'for every 2 liters of orange juice, we use 750 milliliters of beet juice, and the ginger juice is always one third the volume of the beet juice.' A catering order calls for 10 liters of orange juice. How many milliliters of ginger juice are needed?

Formula / Rule Used: Set up a proportion between two of the three linked quantities described in words, then apply the stated fractional relationship to the third.

Solution

Step 1: Convert the base ratio to one unit: 2 liters of orange juice = 2000 mL, paired with 750 mL of beet juice.

Step 2: Set up a proportion for 10 liters (10,000 mL) of orange juice: $2000/750 = 10000/x$, where x is the beet juice needed.

Step 3: Cross-multiply: $2000x = 750 \times 10000 = 7,500,000$.

Step 4: Solve for x : $x = 7,500,000 \div 2000 = 3750 \text{ mL}$ of beet juice.

Step 5: Apply the stated relationship for ginger juice, which is always one third the volume of beet juice: ginger = $3750 \div 3 = 1250 \text{ mL}$.

Answer: 1250 mL

Tip

When a ratio is written as a sentence linking three quantities, first convert everything to the same unit, then isolate the two-quantity proportion you actually need before bringing in the third relationship.

Trap

A rushed student applies 'ginger is one third of beet' to the wrong quantity — computing one third of the 10,000 mL of orange juice instead of one third of the correctly scaled 3750 mL of beet juice.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$\approx 3333 \text{ mL}$	Applied 'ginger is one third of beet' directly to the orange juice quantity instead of the scaled beet quantity, computing $10000/3$ instead of $3750/3$.
3750 mL	Correctly scaled the beet juice to 3750 mL but stopped there, reporting the beet juice volume instead of taking one third of it for the ginger juice.
11250 mL	Inverted the ginger-to-beet relationship, tripling the beet volume ($3750 \times$



	3) instead of taking one third of it.
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Shortcut

Since orange:beet simplifies to 8:3, you can skip the milliliter setup and compute directly in liters: $10 \text{ L} \times \frac{3}{8} = 3.75 \text{ L}$ of beet juice, then divide by 3 for the ginger juice.

Desmos Tip

Type $y = (750/2000) * (1/3) * x$ into the calculator and evaluate at $x = 10000$ to get the ginger juice volume (1250) in one expression.

Unit Conversions & Rates — Problem 4 (Hard)

The high-speed rail line between Athens and Thessaloniki covers 500 kilometers. A particular train completes the trip in 4 hours and 10 minutes, while carrying 312 passengers and using electricity equivalent to 18,000 liters of diesel. What is the train's average speed in meters per second?

Formula / Rule Used: Average speed = distance/time, then convert km/h to m/s using the factor 5/18.

Solution

Step 1: Identify the data actually needed: distance = 500 km and time = 4 hours 10 minutes; the passenger count and energy figure are irrelevant to a speed calculation.

Step 2: Convert the time to hours: 10 minutes = $10/60$ hour = $1/6$ hour, so total time = $4 + 1/6 = 25/6$ hours.

Step 3: Compute average speed in km/h: $500 \div (25/6) = 500 \times 6/25 = 120 \text{ km/h}$.

Step 4: Convert km/h to m/s: $120 \text{ km/h} \times (1000 \text{ m/km}) \div (3600 \text{ s/hr}) = 120,000/3600 \text{ m/s}$.

Step 5: Simplify using the shortcut factor 5/18: $120 \times 5/18 = 600/18 = 100/3 \approx 33.3 \text{ m/s}$.

Answer: $100/3 \text{ m/s} \approx 33.3 \text{ m/s}$

Tip

Strip out any data not needed for the actual question first — Digital SAT unit problems often bury the two numbers you need in a sentence full of irrelevant details like passenger counts or energy use.

Trap

A student who tries to work the diesel/electricity figure into the calculation (assuming every number given must be used) wastes time or misapplies it; the speed calculation only needs distance and time.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
120 m/s	Computed the correct average speed in km/h (120) but forgot to convert to m/s at all, reporting the km/h value as if it were already meters per second.
$\approx 33.9 \text{ m/s}$	Misread '10 minutes' as 0.10 hours instead of $1/6$ hour, using a total time of 4.10 hours, which shifts the final answer slightly off from the correct



	33.3.
432 m/s	Multiplied by the inverse conversion factor (18/5) instead of 5/18 when converting km/h to m/s.

Shortcut

Memorize that $\text{km/h} \rightarrow \text{m/s}$ is always $\times(5/18)$ (and $\text{m/s} \rightarrow \text{km/h}$ is $\times 18/5$) — this skips writing out the full 1000/3600 conversion every time.

Desmos Tip

Type $500/(25/6)$ into the calculator to get 120 km/h instantly, then multiply by 5/18 to confirm $100/3$ m/s without doing the arithmetic by hand.

Unit Conversions & Rates — Problem 5 *(Very Hard)*

A hiking app records elevation gain in a proprietary unit called 'notches,' where each notch equals an unknown but constant number of meters. A trail with 840 meters of elevation gain shows as 24 notches on the app. Using this same conversion, a second trail shows 63 notches. How many kilometers of elevation gain is that?

Formula / Rule Used: Find the unknown unit rate (constant) from the first data pair, then apply that same rate to a second quantity.

Solution

Step 1: Use the given pair to find the conversion constant: 24 notches = 840 meters, so 1 notch = $840 \div 24 = 35$ meters.

Step 2: Sanity-check the direction: more notches should mean more elevation, and 35 meters/notch is a reasonable rate, so the constant is meters per notch, not the reverse.

Step 3: Apply the constant to the second trail: 63 notches $\times$ 35 meters/notch = 2205 meters.

Step 4: Convert the result to kilometers, as the question asks: $2205 \div 1000 = 2.205$ km.

Step 5: Sanity check: 63 notches is 2.625 times the original 24, so the elevation should be about 2.625 times 840 m $\approx$ 2205 m — consistent.

Answer: 2.205 km

Tip

When a problem defines a made-up or proprietary unit, your first move is always to solve for the conversion constant from the given example before touching the actual question.

Trap

A student who inverts the proportion — treating 24 as meters per notch and 840 as the number of notches — gets a wildly wrong constant and a wrong answer for the second trail.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
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2205 km	Correctly found the constant (35 m/notch) and multiplied to get 2205 meters, but forgot to convert to kilometers and reported the meter value with the wrong unit label.
≈ 0.0018 km	Solved for the constant upside down ($24 \div 840$ instead of $840 \div 24$), producing about 0.0286 meters per notch instead of 35, then used that on the second trail.
2.142 km	Made a division slip computing $840 \div 24$, getting 34 instead of the correct 35 meters per notch, and carried that error through the final multiplication and conversion.

Shortcut

Skip solving for the constant explicitly and set up a direct proportion instead: $840/24 = x/63$, cross-multiplying to solve for x (meters) in one line.

Desmos Tip

Graph $y = 35x$ (notches to meters) and read the y -value at $x = 63$ to confirm 2205 meters before converting to kilometers.

Unit Conversions & Rates — Problem 6 (Hard)

During the Berlin Marathon, a runner maintains an exact pace of 5 minutes and 15 seconds per kilometer for the entire 42.195-kilometer race. If she keeps this pace the whole way, how long will it take her to finish, in hours, minutes, and seconds, rounded to the nearest second?

Formula / Rule Used: Convert a compound pace (min/km) entirely into one unit (seconds per km), multiply by total distance, then convert the total seconds back into h:m:s.

Solution

Step 1: Convert the pace to seconds per kilometer: 5 minutes 15 seconds = $5 \times 60 + 15 = 315$ seconds per km.

Step 2: Multiply by the total distance: $315 \text{ s/km} \times 42.195 \text{ km} = 13,291.425$ seconds.

Step 3: Convert seconds to whole hours: $13,291.425 \div 3600 \approx 3.692$ hours, so 3 full hours = 10,800 seconds.

Step 4: Find the remaining seconds after those 3 hours: $13,291.425 - 10,800 = 2491.425$ seconds.

Step 5: Convert the remainder to whole minutes: $2491.425 \div 60 \approx 41.52$ minutes, so 41 full minutes = 2460 seconds.

Step 6: Find the final remaining seconds: $2491.425 - 2460 = 31.425$, which rounds to 31 seconds.

Answer: 3 hours, 41 minutes, 31 seconds

Tip

Translate a 'per kilometer' pace into a single unit (seconds per km) first, multiply by total distance to get total seconds, then convert that one number back into h:m:s at the very end.

Trap



The most common trap is converting '5 minutes 15 seconds' to decimal minutes as 5.15 instead of 5.25 (treating the 15 seconds as 0.15 minutes rather than the correct $15/60 = 0.25$), which throws off every step that follows.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
3 hours, 37 minutes, 18 seconds	Converted the pace to decimal minutes incorrectly as 5.15 instead of 5.25 (treating 15 seconds as 0.15 minutes rather than $15/60 = 0.25$ minutes), understating the total time.
3 hours, 69 minutes, 12 seconds	Converted the decimal remainder of hours (0.692) into minutes by multiplying by 100 instead of 60, treating time as base-10 instead of base-60.
3 hours, 40 minutes, 30 seconds	Used 42 km instead of the exact marathon distance 42.195 km, dropping the extra 195 meters from the calculation.

Shortcut

Convert the pace straight to seconds per km (315) and multiply by the exact distance in one line ($315 \times 42.195 = 13,291.425$ s) rather than handling minutes and seconds separately — this avoids the decimal-minute trap entirely.

Desmos Tip

Type $5 + 15/60$ into the calculator to get the exact decimal pace 5.25 min/km before multiplying, avoiding the 5.15 decimal-conversion trap by hand.

Unit Conversions & Rates — Problem 7 (Very Hard)

A car rented in Germany has a fuel-efficiency rating of 6.5 liters per 100 kilometers. Given that 1 gallon = 3.78541 liters and 1 mile = 1.60934 kilometers, what is the car's fuel efficiency in miles per gallon, rounded to the nearest tenth?

Formula / Rule Used: Multi-step dimensional analysis: invert L/100km into km/L, then convert km/L → km/gal → mi/gal.

Solution

Step 1: Recognize that 6.5 L/100km is a consumption rate, not an efficiency rate — invert it first: km per liter = $100 \div 6.5 \approx 15.3846$ km/L.

Step 2: Convert km/L to km per gallon by multiplying by the liters-in-a-gallon factor: $15.3846 \times 3.78541 \approx 58.2371$ km/gal.

Step 3: Convert km/gal to miles per gallon by dividing by the km-in-a-mile factor: $58.2371 \div 1.60934 \approx 36.184$ mi/gal.

Step 4: Round to the nearest tenth as requested: 36.2 mpg.

Step 5: Sanity check using the shortcut constant: $235.215 \div 6.5 \approx 36.19$, matching the full chain (small difference is just rounding).

Answer: ≈ 36.2 mpg

**Tip**

For any multi-hop conversion, decide which conversions multiply and which divide before calculating, and never round between intermediate steps.

Trap

Because 'liters per 100 km' is a fuel-consumption rate (lower = more efficient), a student who treats it like an ordinary rate and multiplies 6.5 straight through the gallon and mile conversions — instead of first inverting it to km-per-liter — lands on a badly wrong, too-small mpg figure.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
≈ 15.3 mpg	Multiplied the consumption rate (6.5 L/100km) straight through the gallon and mile conversions without first inverting it to km-per-liter, so the result never represents an actual efficiency figure.
≈ 58.2 mpg	Correctly converted to kilometers per gallon (58.2) but forgot the final step of converting kilometers to miles, reporting the km/gal figure as if it were already mi/gal.
≈ 93.7 mpg	Multiplied by the km-to-mile factor (1.60934) instead of dividing by it, inflating the distance instead of converting it down into the smaller mile count.

Shortcut

There's a precomputed constant for exactly this conversion: $\text{mpg} \approx 235.215 \div (\text{L}/100\text{km})$ — plugging in $235.215/6.5 \approx 36.19$ matches the full chain almost instantly.

Desmos Tip

Type $100/6.5 * 3.78541/1.60934$ directly into the calculator as one expression to get 36.18 without tracking any intermediate rounding.

Unit Conversions & Rates — Problem 8 *(Very Hard)*

A landscaping company in Zurich needs to cover a rectangular garden bed measuring 4 meters by 6 meters with a 10-centimeter-deep layer of mulch. The mulch is sold by the cubic yard. Given that 1 meter = 1.09361 yards, how many cubic yards of mulch are needed, rounded to the nearest tenth?

Formula / Rule Used: Volume conversions scale with the CUBE of the linear conversion factor, not the factor itself.

Solution

Step 1: Convert the mulch depth to meters, matching the other dimensions: 10 cm = 0.10 m.

Step 2: Compute the volume in cubic meters: $4 \text{ m} \times 6 \text{ m} \times 0.10 \text{ m} = 2.4 \text{ cubic meters}$.

Step 3: Convert cubic meters to cubic yards using the CUBE of the linear factor, since volume scales with length cubed: $1 \text{ m}^3 = (1.09361 \text{ yd})^3 \approx 1.30794 \text{ yd}^3$.

Step 4: Multiply: $2.4 \text{ m}^3 \times 1.30794 \text{ yd}^3/\text{m}^3 \approx 3.1391 \text{ yd}^3$.

Step 5: Round to the nearest tenth: $\approx 3.1 \text{ cubic yards}$.

Answer: $\approx 3.1 \text{ cubic yards}$

Tip

Whenever a conversion involves area or volume, never apply the linear conversion factor directly — square it for area or cube it for volume before multiplying.

Trap

The most common mistake is converting 2.4 cubic meters to cubic yards by multiplying by 1.09361 only once, treating a volume conversion as if it were a simple length conversion, which gives a badly wrong result.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$\approx 2.6 \text{ cubic yards}$	Multiplied the volume by the linear factor 1.09361 only once instead of cubing it, treating a volume conversion as if it were a length conversion.
$\approx 313.9 \text{ cubic yards}$	Forgot to convert the mulch depth from 10 centimeters to 0.10 meters before computing the volume, using 10 directly as meters and inflating the volume 100-fold.
$\approx 2.9 \text{ cubic yards}$	Squared the linear conversion factor (as if converting an area) instead of cubing it, which is required for a volume conversion.

Shortcut

Memorize that $1 \text{ m}^3 \approx 1.308 \text{ yd}^3$ (the cube of 1.09361) as a ready-made volume conversion constant, so you can multiply straight from cubic meters to cubic yards without recomputing the cube each time.

Desmos Tip

Type $4*6*0.1*1.09361^3$ directly into the calculator as one expression to get the answer in a single step without a separate cubing calculation.

10. Statistics — Center & Spread

Statistics — Center & Spread — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Mean (average)	Μέση τιμή
Median	Διάμεσος
Mode	Επικρατούσα τιμή
Standard deviation	Τυπική απόκλιση
Range	Εύρος
Outlier	Ακραία τιμή

Statistics — Center & Spread — Condensed Theory

Statistics questions ask you to compute or reason about measures of central tendency (mean, median, mode) and spread (range, standard deviation), including how outliers and weighting affect them.

Mean, Median, Mode, Range

Mean = sum/n . Median = middle value (average the two middle values if n is even). Mode = most frequent. Range = $\text{max} - \text{min}$.

Always sort the data before finding the median — the SAT often presents data out of order on purpose.

Weighted Average

Weighted avg = $(\text{mean1} \times n1 + \text{mean2} \times n2 + \dots) / (n1 + n2 + \dots)$

Used whenever two or more groups of different sizes are combined.

Standard Deviation (conceptual only)

You are never asked to calculate it by hand — only to compare: a more spread-out data set has a higher standard deviation, regardless of the actual values.

Effect of Outliers

An outlier pulls the mean toward itself but barely affects the median; it also inflates both the range and the standard deviation.

Mean vs. Median & Skew



Right-skew: mean > median. Left-skew: mean < median. Symmetric: mean $\approx$ median.

The 'tail' of a skewed distribution pulls the mean in its direction, while the median stays closer to the bulk of the data.

⚠ Watch Out

When a question removes or adds one value and asks for a new mean, use $\text{Sum} = \text{Mean} \times n$ to recover the total — it's almost always faster than recomputing from scratch.

Statistics — Center & Spread — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Statistics — Center & Spread — Problem 1 (Hard)

A CLC test-prep program in Thessaloniki runs two practice groups. Group A has 12 students with a mean practice-test score of 1180. Group B has an unknown number of students with a mean score of 1340. When all students from both groups are combined, the overall mean score is 1244. How many students are in Group B?

Formula / Rule Used: Weighted average: total sum = (count1 x mean1) + (count2 x mean2), divided by total count.

Solution

Step 1: Let n = the number of students in Group B.

Step 2: Group A's total score = $12 \times 1180 = 14160$.

Step 3: Group B's total score = $n \times 1340$.

Step 4: The combined mean equation is: $(14160 + 1340n) / (12 + n) = 1244$.

Step 5: Multiply both sides by $(12 + n)$: $14160 + 1340n = 1244(12 + n) = 14928 + 1244n$.

Step 6: Subtract $1244n$ from both sides and subtract 14160 from both sides: $1340n - 1244n = 14928 - 14160$, so $96n = 768$.

Step 7: Solve for n : $n = 768 / 96 = 8$.

Answer: 8 students

Tip

Convert each group's average into a total by multiplying by its count before combining -- never average two averages directly unless the groups are the same size.

Trap

It's tempting to pair each average with whichever group size sits nearest it out of habit; here that means matching 1340 with 12 and 1180 with n , which silently reverses the whole equation and produces $n = 18$ instead of the correct $n = 8$.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
18	Swapped which average belongs to which group size, solving $12(1340) + n(1180) = 1244(12+n)$ instead of $12(1180) + n(1340) = 1244(12+n)$.
20	Correctly solved for the total number of students across both groups combined ($12 + 8 = 20$) but reported that total instead of isolating just Group B's count.
6	Reached the correct equation $96n = 768$ but divided 768 by 128 instead of 96 (a copying slip on the coefficient), getting $n = 6$.

Shortcut

Use the alligation shortcut: the ratio of group sizes equals the inverse ratio of how far each group's mean is from the combined mean -- $n/12 = (1244-1180)/(1340-1244) = 64/96 = 2/3$, so $n = 8$ immediately.

Desmos Tip

Graph $y = (14160 + 1340x)/(12 + x)$ and $y = 1244$ on the built-in calculator and read the x-value where they intersect to confirm $x = 8$.

Statistics — Center & Spread — Problem 2 (Very Hard)

A math club in Vienna gives a 150-point placement test. Six students have already been scored, from lowest to highest: 62, 74, 81, 85, 90, 96. A seventh student, whose score is higher than all six of these, is added to the group. Once the seventh score is included, the mean of all seven scores equals the median of all seven scores plus 2. What is the seventh student's score?

Formula / Rule Used: Median of an odd-sized list is the single middle value once sorted; mean = sum / count.

Solution

Step 1: Since the new score is higher than all six existing scores, the sorted list of seven scores is 62, 74, 81, 85, 90, 96, x .

Step 2: With 7 values, the median is the 4th value in the sorted list, which is 85 -- this stays 85 no matter how large x is, as long as x remains the largest.

Step 3: Set up the mean: $(62+74+81+85+90+96+x)/7 = (488+x)/7$.

Step 4: The condition says mean = median + 2, so $(488+x)/7 = 85 + 2 = 87$.

Step 5: Multiply both sides by 7: $488 + x = 609$.

Step 6: Solve for x : $x = 609 - 488 = 121$.

Answer: 121 points

Tip

When a data set has an odd number of values and a new value is added only at the very top or bottom, find the median first -- it often doesn't move at all, which simplifies the rest of the problem.

**Trap**

Treating the 7-score set like an even-sized list and averaging the 3rd and 4th values (81 and 85) instead of taking the single 4th value (85) is exactly the mistake this problem is built to catch.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
107	Treated the 7-value set as if it were even-sized, averaging the 3rd and 4th values (81 and 85) to get a 'median' of 83, then solved $(488+x)/7 = 85$ instead of 87.
34	Used the correct median (85) and required mean (87), but divided by 6 instead of 7, forgetting the new score makes this a seven-value set, not six.
609	Correctly computed $7 \times 87 = 609$ but reported this total sum as the final answer instead of subtracting 488 to isolate x .

Shortcut

Since the median (85) is fixed and unaffected by x , you can jump straight to mean = 87 and solve the one-line equation $488+x=609$ without extra setup.

Desmos Tip

Type `mean({62,74,81,85,90,96,121})` into the calculator to confirm it equals 87, and `median(...)` of the same list to confirm 85.

Statistics — Center & Spread — Problem 3 (Hard)

At a vineyard on Santorini, two harvesting teams pick grapes. Team A has 4 workers, and Team B has 2 workers. The total kilograms picked by Team A is triple the total kilograms picked by Team B. Across all 6 workers combined, the mean amount picked per worker is 12 kg. What is the mean amount picked per worker on Team A?

Formula / Rule Used: Mean = total sum / count; translate a 'triple the sum' sentence into an equation between the two totals before dividing.

Solution

Step 1: Total kilograms picked by all 6 workers = $6 \times 12 = 72$ kg.

Step 2: Let S_b = Team B's total kilograms (2 workers). Team A's total is triple that: $S_a = 3 \times S_b$.

Step 3: Since $S_a + S_b = 72$, substitute: $3S_b + S_b = 72$, so $4S_b = 72$.

Step 4: Solve: $S_b = 18$ kg (Team B's total).

Step 5: Then $S_a = 3 \times 18 = 54$ kg (Team A's total).

Step 6: Mean per worker on Team A = $S_a / 4 = 54/4 = 13.5$ kg.

Answer: 13.5 kg

Tip



Whenever a ratio is described between two group TOTALS (not group means), define variables for the totals first and translate the sentence into an equation -- don't average anything until the totals are set up.

Trap

It's easy to misread 'the sum of four numbers is triple the sum of two others' backwards, setting Team B's total as triple Team A's total instead of the other way around.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
4.5 kg	Flipped which group's total is triple the other, setting $S_b = 3 \times S_a$ instead of $S_a = 3 \times S_b$, which makes Team A's total come out as only 18 kg instead of 54 kg.
9 kg	Correctly found Team A's total (54 kg) but divided by 6 (all workers) instead of 4 (just Team A's workers) to get the mean.
11.25 kg	Correctly solved $S_b = 18$, but then miscalculated $S_a = 3 \times 18$ as 45 instead of 54, giving a final mean of $45/4 = 11.25$ kg.

Shortcut

Since $\text{total} = 4 \times S_b$ (because S_a is $3S_b$ and $S_a + S_b = 4S_b$), solve $S_b = 72/4 = 18$ in one line without ever writing $S_a + S_b = 72$ as a separate step.

Desmos Tip

This is a two-line arithmetic problem -- just type $72/4$ and then $3 * 18$ directly into the calculator as a fast check rather than graphing anything.

A real-estate site lists the monthly rent (in euros) for 7 available apartments in different Barcelona neighborhoods, along with each apartment's floor number: Floor 6, EUR 800; Floor 3, EUR 600; Floor 12, EUR 1400 (a renovated penthouse); Floor 8, EUR 720; Floor 1, EUR 760; Floor 5, EUR 640; Floor 2, EUR 680. The agency removes the EUR 1400 penthouse listing as an outlier before analyzing typical rent in the area. By how many more euros does the mean change than the median when this listing is removed?

Formula / Rule Used: Mean = sum/count; median = middle value (or average of the two middle values); outliers pull the mean far more than the median.

Solution

Step 1: Sort the 7 rents from least to greatest (floor numbers are extra information and don't affect any calculation): 600, 640, 680, 720, 760, 800, 1400.

Step 2: Mean of all 7: sum = $600+640+680+720+760+800+1400 = 5600$, so mean = $5600/7 = 800$.

Step 3: Median of all 7 (the 4th value in the sorted list) = 720.

Step 4: Remove the outlier, 1400: the remaining 6 sorted rents are 600, 640, 680, 720, 760, 800.

Step 5: New mean: sum = 4200, mean = $4200/6 = 700$. The mean dropped from 800 to 700, a change of 100.

Step 6: New median: average of the 3rd and 4th values, $(680+720)/2 = 700$. The median dropped from 720 to 700, a change of only 20.

Step 7: Difference between how much each measure changed: $100 - 20 = 80$.

Answer: EUR 80 (the mean drops by EUR 100 while the median drops by only EUR 20)

Tip

Recompute both the mean and median separately after removing a data point -- don't assume they shift by the same amount, and remember that going from an odd count to an even count turns the median into an average of two values.

Trap

Assuming the median won't change at all when an outlier is removed is the single biggest trap here -- the median almost always shifts slightly, just far less than the mean does.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
EUR 100	Computed only the change in the mean and reported that, ignoring that the question asks for the DIFFERENCE between the mean's change and the median's change.
EUR 60	Mis-computed the new median of the six-value list as a single middle value (the 3rd value, 680) instead of averaging the two middle values, giving a wrong median change of 40 instead of 20 ($100-40=60$).
EUR 180	Forgot to reduce the denominator when computing the new mean, dividing 4200 by 7 instead of 6, getting a wrong mean of 600 and an inflated mean-change of 200 ($200-20=180$).

Shortcut



You don't need to recompute the mean from scratch -- new mean = (old mean $\times$ n - removed value)/($n-1$) = $(800 \times 7 - 1400)/6 = 4200/6 = 700$ directly.

Desmos Tip

Type both six- and seven-value lists into the calculator (e.g. $L1=\{600,640,680,720,760,800\}$) and use `mean(L1)` and `median(L1)` to instantly confirm both new values.

Statistics — Center & Spread — Problem 5 (Very Hard)

A startup in Berlin has 9 employees, and their mean salary is EUR 52,000. The founder's salary, EUR 148,000, is far above everyone else's and is treated as an outlier. If the founder's salary is excluded, what is the mean salary of the remaining 8 employees?

Formula / Rule Used: Mean = sum/count; removing a value changes both the total AND the count, so recompute both.

Solution

Step 1: Convert the mean into a total: total payroll for 9 employees = $9 \times 52,000 = \text{EUR } 468,000$.

Step 2: Identify the founder's salary as the value to be removed: EUR 148,000.

Step 3: Subtract the founder's salary from the total: $468,000 - 148,000 = \text{EUR } 320,000$ (this is now the total for the remaining employees).

Step 4: Update the count: with the founder excluded, only 8 employees remain, not 9.

Step 5: Divide the new total by the new count: $320,000 / 8 = \text{EUR } 40,000$.

Step 6: Sanity check: since the founder's salary was far above the original mean, removing it should pull the mean down substantially -- EUR 40,000 < EUR 52,000 confirms the direction makes sense.

Answer: EUR 40,000

Tip

Turn the mean back into a total first (mean $\times$ count), do the subtraction on the total, and only divide by the new count at the very end.

Trap

The single biggest trap: after removing one employee's salary from the total, it's easy to forget the denominator also shrinks from 9 to 8, and re-divide by 9 out of habit.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
EUR 35,555.56 (approximately EUR 35,556)	Correctly subtracted the founder's salary from the total (320,000) but then divided by 9 instead of 8, forgetting the founder is no longer part of the group being averaged.
-EUR 12,000	Subtracted the founder's salary directly from the mean instead of from the total sum: $(52,000 - 148,000)/8 = -96,000/8 = -12,000$.
EUR 320,000	Correctly found the new total payroll but stopped there without dividing by the new count, reporting the total instead of the mean.

**Shortcut**

new mean = (old mean $\times$ n - removed value)/($n-1$) = $(52,000 \times 9 - 148,000)/8 = 320,000/8 = 40,000$ in one continuous calculation, skipping the separate 'find the old total' step.

Desmos Tip

Type $(52000 \times 9 - 148000)/8$ directly into the built-in calculator to get 40000 in one line, confirming the arithmetic without writing out each step by hand.

Statistics — Center & Spread — Problem 6 (Hard)

A retail chain tracks one week's sales (in thousands of euros) at 6 branches across Europe. When sorted from least to greatest, the figures form the list r , $r+9$, $r+18$, $r+27$, $r+36$, and s , where r is the smallest branch's sales and s is the largest branch's sales. The range of the six figures is 45 (thousand euros). Separately, the finance report notes that s is also equal to 3 more than double r . What is the mean weekly sales, in thousands of euros, across all 6 branches?

Formula / Rule Used: Range = largest - smallest; combine with a second equation for the same unknown, then compute mean = sum/count.

Solution

Step 1: By the definition of range, $s - r = 45$, so $s = r + 45$.

Step 2: The finance report gives a second equation for the same value: $s = 2r + 3$.

Step 3: Set the two expressions for s equal to each other: $r + 45 = 2r + 3$.

Step 4: Solve for r : $45 - 3 = 2r - r$, so $r = 42$.

Step 5: Find s using either equation: $s = 42 + 45 = 87$ (check: $2(42)+3 = 87$, confirmed).

Step 6: List all six branch sales figures: $r=42$, $r+9=51$, $r+18=60$, $r+27=69$, $r+36=78$, and $s=87$.

Step 7: Sum the six figures: $42+51+60+69+78+87 = 387$.

Step 8: Mean = $387/6 = 64.5$.

Answer: 64.5 thousand euros (EUR 64,500)

Tip

When two separate statements both describe the same unknown value (here, the largest branch s), set the two expressions equal to each other immediately instead of solving them one at a time.

Trap

Because finding r takes several steps on its own, it's tempting to treat $r=42$ as the final answer -- but the question never asked for r , it asked for the mean of the full six-branch list.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
42 (thousand euros)	Solved correctly for r but stopped there, submitting the smallest branch's sales as if it were the final mean instead of continuing to build the list and average it.



-40.5 (thousand euros)	Misapplied the range definition with the difference reversed, using $s = r - 45$ instead of $s = r + 45$, which flips the sign of every downstream value (r becomes -48 , s becomes -93).
61.8 (thousand euros)	Correctly found $r=42$ and $s=87$ but dropped one middle term ($r+36=78$) when listing the six branches, averaging only 5 values ($42, 51, 60, 69, 87$) instead of all 6.

Shortcut

The five evenly-spaced smaller terms ($r, r+9, \dots, r+36$) sum to $5r+90$; plug in $r=42$ to get 300 instantly, then just add the largest term $s=87$ for a total of 387 -- no need to add six separate numbers one at a time.

Desmos Tip

Enter the list $L1=\{42, 51, 60, 69, 78, 87\}$ into the calculator and use $\text{mean}(L1)$ to instantly confirm 64.5.

Statistics — Center & Spread — Problem 7 (Very Hard)

At a Lyceum in Patras, a student's overall course grade is weighted: coursework counts 25%, the midterm counts 35%, and the final exam counts 40%. She has an 82 in coursework and a 78 on the midterm. What score does she need on the final exam to earn an overall grade of exactly 85?

Formula / Rule Used: Weighted average: sum of (decimal weight $\times$ component score) = target overall grade.

Solution

Step 1: Convert each percentage weight to a decimal: coursework 25% = 0.25, midterm 35% = 0.35, final exam 40% = 0.40.

Step 2: Set up the weighted average equation: $0.25(82) + 0.35(78) + 0.40(F) = 85$, where F is the unknown final exam score.

Step 3: Compute the known terms: $0.25(82) = 20.5$ and $0.35(78) = 27.3$.

Step 4: Add the known terms: $20.5 + 27.3 = 47.8$.

Step 5: Substitute back into the equation: $47.8 + 0.40F = 85$.

Step 6: Isolate the term with F : $0.40F = 85 - 47.8 = 37.2$.

Step 7: Solve for F : $F = 37.2 / 0.40 = 93$.

Answer: 93

Tip

Convert every percent weight to a decimal and multiply each by its matching score before adding anything -- treat the unknown final term as the last piece to isolate.

Trap

Seeing three numbers (82, 78, and the target 85) listed side by side tempts a plain three-way average; the 25/35/40 weights must be used as multipliers, not ignored.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
95	Treated the three grade components as equally weighted, computing a plain average $(82+78+F)/3=85$ instead of using the given 25/35/40 percent weights.
93.75	Rounded 0.35×78 to 27 instead of keeping the exact value 27.3, so solved $20.5+27+0.40F=85$, giving $0.40F=37.5$ and $F=93.75$.
14.88	Correctly reached $0.40F=37.2$, but then multiplied by 0.40 instead of dividing, computing $F=37.2 \times 0.40=14.88$.

Shortcut

Since coursework and midterm together already lock in 47.8 fixed points toward the 85 target, immediately compute the 'points still needed,' $85-47.8=37.2$, and divide by the final's weight -- no need to write the full three-term equation symbolically first.

Desmos Tip

Graph $y = 0.25(82) + 0.35(78) + 0.40x$ and $y = 85$ on the built-in calculator and find the x-coordinate of their intersection to confirm $F = 93$.

Two coding bootcamps each timed 7 students solving the same practice problem set, in minutes. Amsterdam's times: 10, 18, 20, 20, 20, 22, 30. Copenhagen's times: 10, 15, 17, 20, 23, 25, 30. Both groups have the same mean (20 minutes) and the same range (20 minutes). Without calculating the exact standard deviation, which group's completion times have the greater standard deviation?

Formula / Rule Used: Standard deviation measures the typical distance of ALL data points from the mean -- not just the distance between the minimum and maximum (the range).

Solution

Step 1: Confirm both means are equal: Amsterdam's sum = $10+18+20+20+20+22+30 = 140$, mean = $140/7 = 20$. Copenhagen's sum = $10+15+17+20+23+25+30 = 140$, mean = $140/7 = 20$. Both means are 20 minutes.

Step 2: Confirm both ranges are equal: Amsterdam's range = $30-10 = 20$. Copenhagen's range = $30-10 = 20$. Since the ranges match, range alone cannot distinguish the two groups' spread.

Step 3: List each value's deviation from the mean (20) for Amsterdam: -10, -2, 0, 0, 0, 2, 10.

Step 4: List each value's deviation from the mean (20) for Copenhagen: -10, -5, -3, 0, 3, 5, 10.

Step 5: Compare the two lists: five of Amsterdam's seven values sit exactly at, or within 2 minutes of, the mean, with only the two extremes (10 and 30) far away -- while Copenhagen's values are spread more evenly across the entire range, with several (15, 17, 23, 25) sitting well away from the mean.

Step 6: Since standard deviation reflects the TYPICAL distance of all points from the mean (not just the two extremes), and more of Copenhagen's values sit farther from the mean, Copenhagen has the greater standard deviation.

Step 7: Check with the sum of squared deviations (proportional to variance): Amsterdam: $100+4+0+0+0+4+100=208$. Copenhagen: $100+25+9+0+9+25+100=268$. Since $268 > 208$, this confirms Copenhagen's spread is greater.

Answer: Copenhagen

Tip

When two data sets share the same mean and range, look past the two extreme values and compare how closely the remaining (interior) values cluster around the mean.

Trap

Assuming equal range means equal spread is the exact trap here -- Amsterdam's tight cluster of five values right at the mean actually makes it LESS spread out overall than Copenhagen's more evenly distributed values, despite both having the same min and max.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Amsterdam	Reasoned that having two dramatic outliers (10 and 30) sitting alone must create more spread than Copenhagen's smoother distribution, without actually comparing how far the typical (non-extreme) values sit from the mean.
They have equal standard deviation	Concluded that because both groups share the same mean AND the same range, their spread must be identical, incorrectly treating range as if it were the same thing as standard deviation.



Cannot be determined without computing the exact standard deviation

Assumed that comparing spread requires the full standard deviation formula, when comparing how tightly the interior values cluster around the mean already settles the comparison.

Shortcut

No numeric shortcut beats comparing the deviations directly -- since both sets share the same mean, listing each value's distance from 20 and comparing which set has more/larger interior deviations is already the fastest legitimate method.

11. Data Displays — Histograms, Dot Plots & Box Plots

Data Displays — Histograms, Dot Plots & Box Plots — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Histogram	Ιστόγραμμα
Dot plot	Διάγραμμα κουκκίδων
Box plot	Διάγραμμα κουτιού (θηκόγραμμα)
Five-number summary	Σύνοψη πέντε αριθμών
Frequency	Συχνότητα

Data Displays — Histograms, Dot Plots & Box Plots — Condensed Theory

The SAT presents data visually in several formats, each suited to revealing a different feature of the distribution.

Histograms

Continuous numerical data grouped into intervals with NO gaps between bars; the shape (bell, right-skewed, left-skewed) reveals the distribution's symmetry.

Dot Plots

Each dot is one data point above a number line — the exact frequency, clusters, gaps, and outliers are all directly visible, unlike a histogram.

Box Plots (Five-Number Summary)

Min, Q1, Median, Q3, Max

The box spans Q1 to Q3 (the interquartile range, $IQR = Q3 - Q1$); the line inside the box is the median; whiskers reach to the min and max.

Reading the Median from a Dot Plot

Cross out the highest and lowest remaining values one pair at a time until one (or two, to average) value remains in the middle.



Watch Out

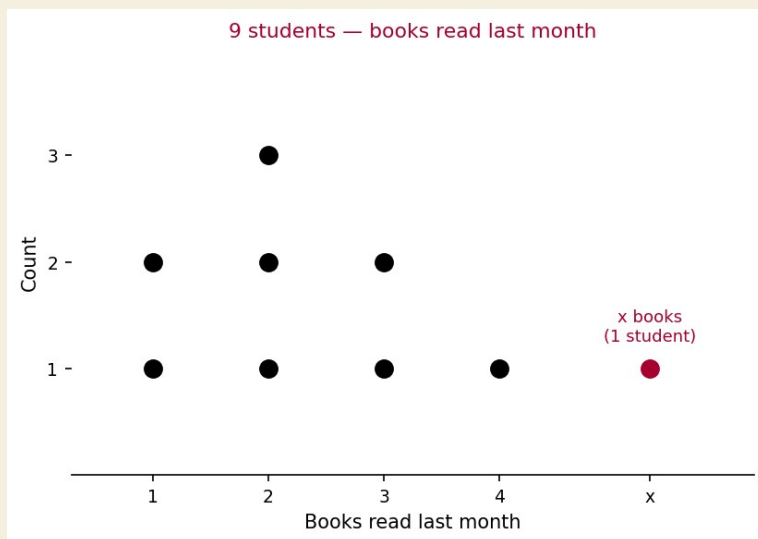
In a box plot, a longer whisker or box section on one side signals skew in that direction — you can answer many box-plot questions without ever knowing the raw data.

Data Displays — Histograms, Dot Plots & Box Plots — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 1 (Hard)

A dot plot shows the number of books read last month by 9 students in a reading club in Thessaloniki: 1 book (2 students), 2 books (3 students), 3 books (2 students), 4 books (1 student), and x books (1 student). If the mean number of books read by the group is 3, what is the value of x ?



Formula / Rule Used: Mean = (sum of all values)/(number of values); solve for a missing data value.

Solution

Step 1: Count the total number of students represented: $2 + 3 + 2 + 1 + 1 = 9$ students.

Step 2: Multiply each known value by its frequency and add: $1(2) + 2(3) + 3(2) + 4(1) = 2 + 6 + 6 + 4 = 18$ (this is the sum of every student's books except the one who read x).

Step 3: Add the unknown student's contribution to get the full total: total books = $18 + x$.

Step 4: Set up the mean equation using all 9 students: $(18 + x)/9 = 3$.

Step 5: Multiply both sides by 9: $18 + x = 27$.

Step 6: Solve for x : $x = 27 - 18 = 9$.

Answer: $x = 9$

Tip

When a mean is given and one data value is missing, translate the mean formula into $\text{sum} = \text{mean} \times n$ first, then subtract the sum of everything you already know.

Trap

A rushed student forgets that the student who read x books is still one of the 9 people in the mean, and instead divides by 8 (the count of students whose value is already known), which produces the wrong x .

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$x = 6$	Forgot that the student with x books counts toward n , used $n = 8$ instead of 9, and solved $(18+x)/8 = 3$.
$x = 27$	Set x equal to mean $\times n$ ($3 \times 9 = 27$) directly, forgetting to subtract the sum of the 8 known values (18) first.
$x = 17$	Added only the four distinct dot-plot values ($1+2+3+4 = 10$) instead of value $\times$ frequency (18), then solved $x = 27 - 10 = 17$.

Shortcut

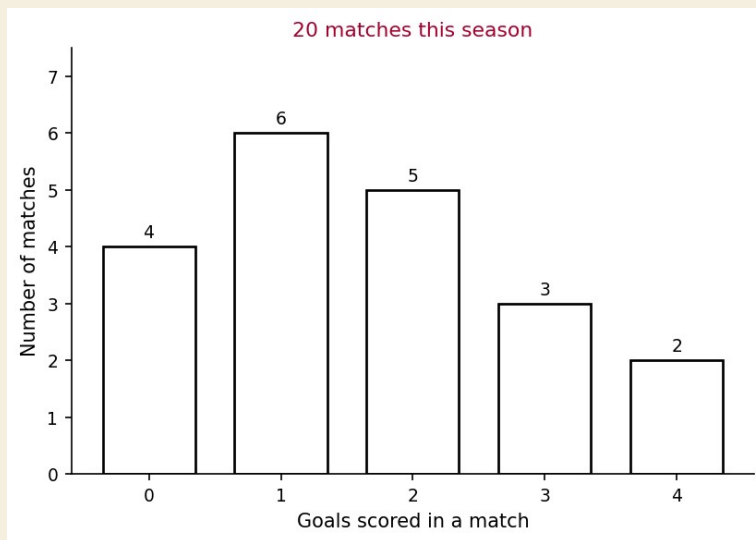
Skip writing out the frequency table twice — just compute the target total (mean $\times n = 27$) and subtract the sum of the 8 known dots (18) in one line: $x = 27 - 18 = 9$.

Desmos Tip

Type $(18+x)/9=3$ into the calculator's equation solver to confirm $x = 9$ in one step.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 2 (Hard)

A histogram shows the number of goals scored by a football (soccer) club in each of its 20 matches this season: bars of height 4 over 0 goals, height 6 over 1 goal, height 5 over 2 goals, height 3 over 3 goals, and height 2 over 4 goals. What is the median number of goals scored per match?



Formula / Rule Used: Median from a frequency table = average of the two middle values, located via cumulative frequency (when n is even).

Solution

Step 1: Total matches: $4 + 6 + 5 + 3 + 2 = 20$, so n is even and the median is the average of the 10th and 11th values once ordered.

Step 2: Build cumulative frequency: 0 goals covers positions 1-4; 1 goal covers positions 5-10 ($4+6=10$); 2 goals covers positions 11-15 ($10+5=15$); 3 goals covers positions 16-18; 4 goals covers positions 19-20.

Step 3: The 10th position is the last position in the '1 goal' group, so the 10th value = 1.

Step 4: The 11th position is the first position in the '2 goals' group, so the 11th value = 2.

Step 5: Median = $(10\text{th value} + 11\text{th value})/2 = (1+2)/2 = 1.5$.

Answer: Median = 1.5 goals

Tip

When a display groups data by value instead of listing raw data, find the median by tracking cumulative frequency to the exact middle position(s) — never just read off whichever bar looks centrally located.

Trap

Because '2 goals' is the visually central bar, it's tempting to call it the median directly — but cumulative counting shows the true middle falls between the '1 goal' and '2 goals' groups, giving 1.5, not 2.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
2	Picked the bar that looks centrally located among the five categories instead of counting cumulative frequency to the actual 10th and 11th positions.



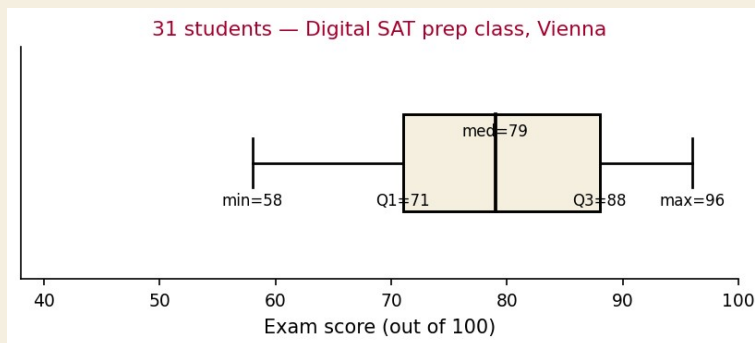
1	Correctly found the 10th value (1) but stopped there, forgetting that $n = 20$ is even and the median requires averaging the 10th AND 11th values.
1.65	Computed the mean $(0(4)+1(6)+2(5)+3(3)+4(2))/20 = 33/20 = 1.65$ instead of the median.

Shortcut

Since $n = 20$ is even, you only need to track the running cumulative total until it first passes 10 — once it does, you've identified both middle positions without listing all 20 data points individually.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 3 (Very Hard)

A box plot summarizes the exam scores (out of 100) of 31 students in a Digital SAT prep class in Vienna: minimum = 58, $Q1 = 71$, median = 79, $Q3 = 88$, maximum = 96. A 32nd student transfers in with a score of 47. Using the standard rule that a value is an outlier if it falls more than $1.5 \times IQR$ below $Q1$ or above $Q3$, find the lower fence and determine whether the new student's score is an outlier.



Formula / Rule Used: $IQR = Q3 - Q1$; outlier fences = $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$.

Solution

Step 1: Find the IQR from the box plot: $IQR = Q3 - Q1 = 88 - 71 = 17$.

Step 2: Multiply by 1.5: $1.5 \times 17 = 25.5$.

Step 3: Find the lower fence: lower fence = $Q1 - 25.5 = 71 - 25.5 = 45.5$.

Step 4: Compare the new student's score to the lower fence: 47 vs. 45.5.

Step 5: Since $47 > 45.5$, the score of 47 lies inside the fence, not beyond it.

Step 6: Conclude: the new student's score of 47 is NOT an outlier by the $1.5 \times IQR$ rule, even though it is well below the original minimum of 58.

Answer: Lower fence = 45.5; 47 is NOT an outlier ($47 > 45.5$).

Tip

To test for an outlier, always compute the actual fence ($Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) before comparing — never judge 'extreme-ness' just by comparing to the box plot's own minimum or maximum.

Trap



Because 47 is lower than the original minimum of 58, it looks obviously extreme — but the fence calculation shows 47 still falls inside the allowed range, so the intuitive answer (outlier) is wrong.

Process of Elimination

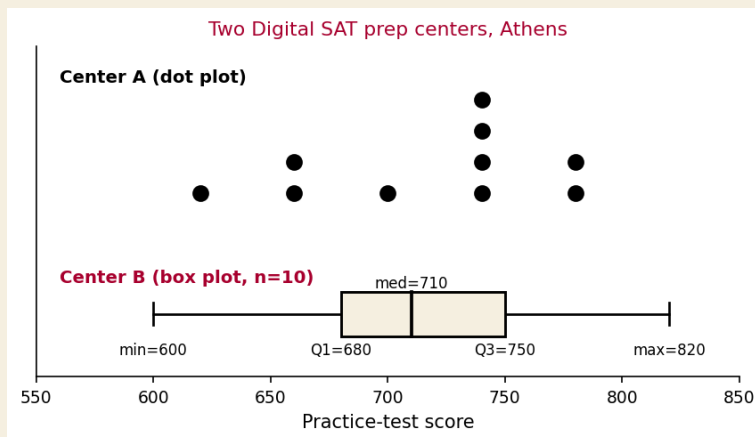
Tempting Wrong Answer	Exactly What Went Wrong
Outlier (fence = 54)	Computed the fence as $Q1 - IQR$ ($71 - 17 = 54$) instead of $Q1 - 1.5 \times IQR$, forgetting the 1.5 multiplier entirely.
Outlier, since 47 is below the original minimum of 58	Assumed any new value below the group's previous minimum must automatically be an outlier, without computing a fence at all.
Outlier (fence = 53.5)	Used the median (79) instead of $Q1$ (71) as the base for the lower fence, computing $79 - 25.5 = 53.5$ and comparing 47 to that.

Shortcut

Once you know $Q1$ and IQR , compute the fence in one line — $Q1 - 1.5 \times IQR$ — and just check whether 47 clears it; no need to re-sort or re-derive the whole distribution.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 4 (Very Hard)

Two Digital SAT prep centers in Athens gave the same practice test. Center A's results are shown as a dot plot: 620 (1 student), 660 (2 students), 700 (1 student), 740 (4 students), and 780 (2 students). Center B's results are shown as a box plot based on 10 students: minimum = 600, Q1 = 680, median = 710, Q3 = 750, maximum = 820. Which center has the greater median score, and which center has the greater range?



Formula / Rule Used: Median and range from a dot plot (order the data by frequency) versus reading the five-number summary directly from a box plot.

Solution

Step 1: List Center A's 10 dot-plot values in increasing order: 620, 660, 660, 700, 740, 740, 740, 740, 780, 780.

Step 2: Since $n = 10$ is even, Center A's median = average of the 5th and 6th values = $(740 + 740) / 2 = 740$.

Step 3: Center A's range = max - min = $780 - 620 = 160$.

Step 4: Center B's statistics are read directly from the box plot: median = 710, and range = max - min = $820 - 600 = 220$.

Step 5: Compare medians: 740 (Center A) > 710 (Center B), so Center A has the greater median.

Step 6: Compare ranges: 220 (Center B) > 160 (Center A), so Center B has the greater range.

Answer: Center A has the greater median (740 vs. 710); Center B has the greater range (220 vs. 160).

Tip

When comparing distributions shown in different display types, extract the SAME statistic for each display using its own reading rules before comparing — never assume one group dominates on every measure just because of one eye-catching value.

Trap

Center B's box plot has the single highest score of the two groups (820), which tempts students into assuming Center B must also have the higher median — but Center A's dot plot is heavily weighted toward its high end (four students at 740, two at 780), which actually gives Center A the higher median despite its lower maximum.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



Center B has the greater median and the greater range.	Assumed the group with the higher maximum score (Center B, 820) automatically has the higher median too, without actually sorting and averaging Center A's dot-plot values.
Center A's median = 700	Read the middle of the five DISTINCT dot-plot values (620, 660, 700, 740, 780) instead of sorting all 10 individual data points and averaging the true 5th and 6th values.
Center A's range = 120	Assumed the tallest stack of dots (740, with 4 students) marked the rightmost/maximum value on the number line, confusing dot-count height with horizontal position, so used 740 instead of 780 as the max.

Shortcut

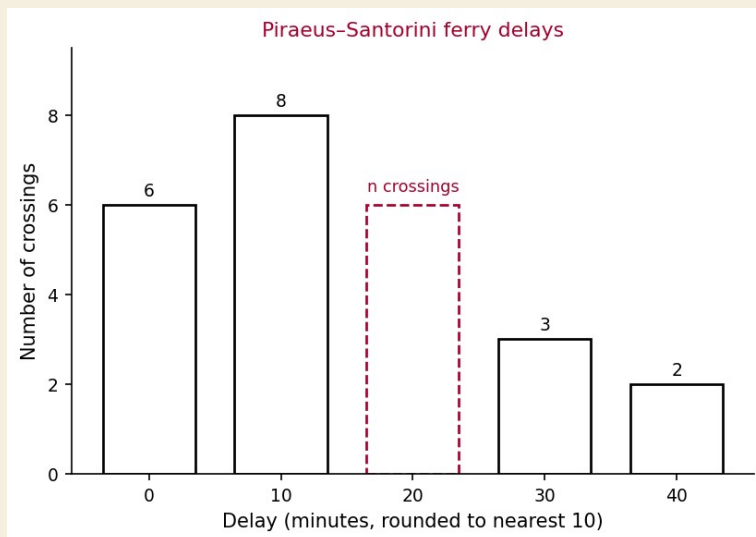
Center B's five-number summary is already given directly from the box plot, so only Center A needs to be built from the raw dot-plot data by hand.

Desmos Tip

Enter Center A's 10 scores as a list $L1=\{620,660,660,700,740,740,740,740,780,780\}$ in the calculator and use $\text{median}(L1)$ and $\text{max}(L1)-\text{min}(L1)$ to instantly check the median (740) and range (160) against your hand sort.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 5 (Very Hard)

A ferry company tracked delay times (rounded to the nearest 10 minutes) for a number of Piraeus–Santorini crossings last month, summarized in this frequency table: 0 min delay: 6 crossings; 10 min: 8 crossings; 20 min: n crossings; 30 min: 3 crossings; 40 min: 2 crossings. If the mean delay across all crossings was 14.8 minutes, what is the value of n ? Then, using that value of n , find the median delay.



Formula / Rule Used: Weighted mean = sum of (value $\times$ frequency)/sum of frequencies; median via cumulative frequency.

Solution

Step 1: Let n be the unknown frequency for the 20-minute row. Total crossings = $6+8+n+3+2 = 19+n$.

Step 2: Total delay-minutes = $0(6)+10(8)+20n+30(3)+40(2) = 0+80+20n+90+80 = 250+20n$.

Step 3: Set up the mean equation: $(250+20n)/(19+n) = 14.8$.

Step 4: Cross-multiply: $250+20n = 14.8(19+n) = 281.2+14.8n$.

Step 5: Collect the n terms on one side: $20n-14.8n = 281.2-250$, so $5.2n = 31.2$.

Step 6: Solve: $n = 31.2/5.2 = 6$.

Step 7: Total crossings = $19+6 = 25$, which is odd, so the median is the 13th crossing once sorted.

Step 8: Cumulative frequency: 0 min covers positions 1–6; 10 min covers positions 7–14 ($6+8=14$) — the 13th position falls inside the 10-min group.

Step 9: Median delay = 10 minutes.

Answer: $n = 6$; median delay = 10 minutes.

Tip

When a frequency is unknown, let the TOTAL count also be an expression in terms of that unknown (here, $19+n$) — don't assume the total is fixed unless the problem states it directly.

Trap

A rushed student fixes the denominator at 19 (the sum of the known frequencies only) instead of $19+n$, forgetting that the missing frequency also adds to the total number of crossings.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$n \approx 1.56$	Kept the denominator fixed at 19 instead of $19+n$, solving $(250+20n)/19 = 14.8$, which ignores that the unknown frequency also adds to the total crossing count.
$n = 6$, median = 20 minutes	Correctly solved for n but then found the median by picking the value in the middle ROW of the 5-row table (the 3rd of 5 delay categories, 20 min) instead of counting cumulative frequency to the true 13th crossing.
$n \approx -2.26$	Wrote the total delay-minutes as $250+n$ instead of $250+20n$, forgetting to multiply the 20-minute row's delay by its own frequency n before adding it into the sum.

Shortcut

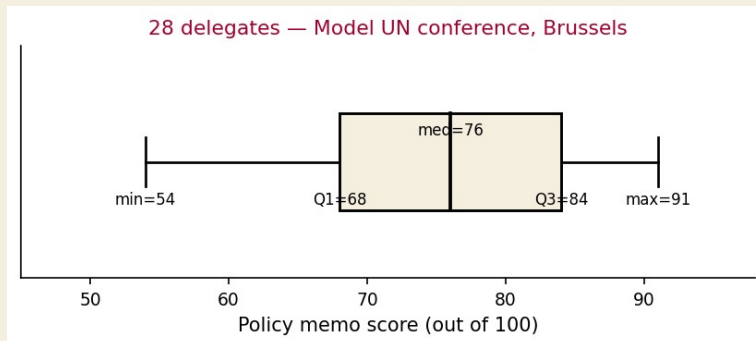
No faster path than setting up the mean equation directly — but if solving ever produces a fractional or negative n , that's a built-in red flag that a setup mistake was made, since a frequency must be a positive whole number.

Desmos Tip

Graph $y=(250+20x)/(19+x)$ and $y=14.8$ in the calculator and read the intersection's x -value, or use the equation solver on $(250+20n)/(19+n)=14.8$, to confirm $n=6$.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 6 (Hard)

At a Model United Nations conference in Brussels, the scores of 28 delegates on a 100-point policy memo (graded on a 40-item rubric) are shown in a box plot: minimum = 54, Q1 = 68, median = 76, Q3 = 84, maximum = 91. Delegates spent an average of 3 hours writing the memo. What is the interquartile range of the scores, and how many delegates scored above 84?



Formula / Rule Used: $IQR = Q3 - Q1$; each quartile of a box plot represents 25% of the data set.

Solution

Step 1: Identify the irrelevant details: the memo's 100-point scale, the 40-item rubric, and the 3-hour average writing time don't affect the box plot's summary statistics — set them aside.

Step 2: Find Q1 and Q3 directly from the box plot: $Q1 = 68$, $Q3 = 84$.

Step 3: Compute the interquartile range: $IQR = Q3 - Q1 = 84 - 68 = 16$.

Step 4: Recall the quartile definition: each quarter of a box plot (below Q1, Q1-to-median, median-to-Q3, above Q3) always contains exactly 25% of the data, regardless of the actual score values.

Step 5: Since 'above 84' means above Q3, that region is exactly the top 25% of delegates.

Step 6: Convert to a count: 25% of 28 delegates = $0.25 \times 28 = 7$ delegates.

Answer: $IQR = 16$; 25% of delegates (7 delegates) scored above 84.

Tip

Before touching any numbers in a box-plot word problem, identify which facts actually belong to the five-number summary and discard details that only describe the scenario.

Trap

The '100-point, 40-item' memo details tempt students into computing a percentage relative to the point scale (like $84/100 = 84\%$) instead of recognizing that 'above Q3' is always exactly 25% of the data by definition, independent of the score scale.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$IQR = 37$	Computed the full range ($\text{max} - \text{min} = 91 - 54 = 37$) instead of the interquartile range ($Q3 - Q1$).
84% of delegates scored above 84	Confused the raw score value at Q3 (84 out of 100 points) with the percentage of delegates above that score, treating the score itself as if it were a percentile.
14 delegates (50%) scored above	Assumed 'above Q3' means the entire upper half of the box plot (above

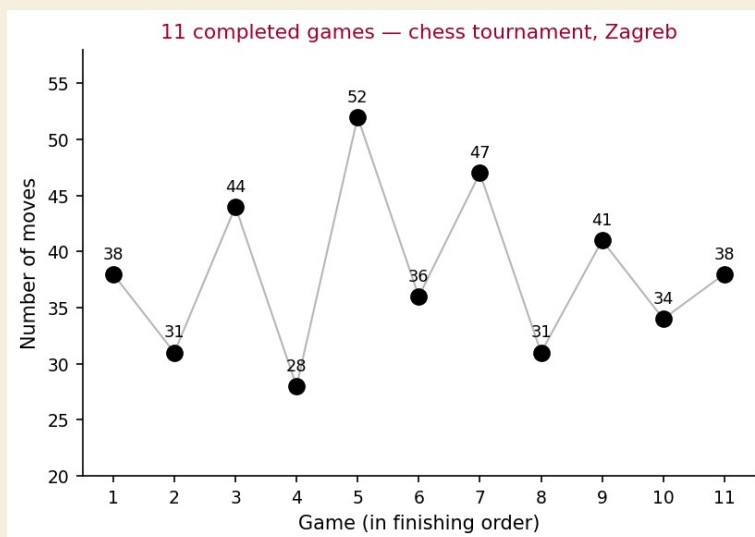
the median) rather than just the top quarter above Q3, doubling the correct percentage.

Shortcut

No calculation is needed for the percentage part — by definition each quartile always holds 25% of the data, so 'above Q3' is always 25% regardless of the actual scores; just multiply 0.25 by the total count.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 7 (Hard)

In a chess tournament in Zagreb, the number of moves in each of 11 completed games was recorded in the order the games finished: 38, 31, 44, 28, 52, 36, 47, 31, 41, 34, 38. What is the interquartile range of the number of moves?



Formula / Rule Used: $IQR = Q3 - Q1$, found by splitting the ordered data at the median (excluding the median itself when n is odd).

Solution

Step 1: Sort the 11 move counts from least to greatest: 28, 31, 31, 34, 36, 38, 38, 41, 44, 47, 52.

Step 2: With $n = 11$ (odd), the overall median is the 6th value: 38.

Step 3: To find Q1, take the lower half of the data, EXCLUDING the median itself: 28, 31, 31, 34, 36 (5 values).

Step 4: Q1 is the median of this lower half, the 3rd value: $Q1 = 31$.

Step 5: To find Q3, take the upper half of the data, again EXCLUDING the median: 38, 41, 44, 47, 52 (5 values).

Step 6: Q3 is the median of this upper half, the 3rd value: $Q3 = 44$.

Step 7: $IQR = Q3 - Q1 = 44 - 31 = 13$.

Answer: $IQR = 13$ ($Q1 = 31$, $Q3 = 44$)



For odd n , always exclude the actual median value from BOTH the lower and upper halves before finding $Q1$ and $Q3$ — including it shifts both quartiles inward and changes the IQR.

Trap

The list is given in the order the games finished, not sorted — using position-in-the-original-list instead of position-in-sorted-order, or accidentally including the median in one or both halves, are the two most common ways students corrupt $Q1$ and $Q3$ here.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
IQR = 10	Included the overall median (38) in both the lower and upper halves when splitting the data, instead of excluding it, which shifts $Q1$ up to 32.5 and $Q3$ down to 42.5.
IQR = 24	Computed the range ($\text{max} - \text{min} = 52 - 28 = 24$) instead of the interquartile range.
IQR = 11.5	Correctly excluded the median from the lower half ($Q1 = 31$) but accidentally included it in the upper half, computing $Q3$ as the average of 41 and 44 (42.5) instead of the correct 44.

Shortcut

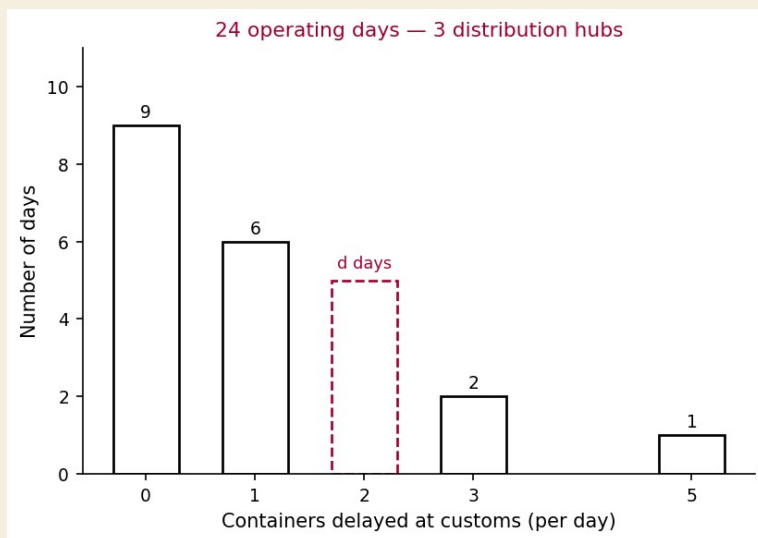
Once sorted, you don't need to write out the two halves as separate lists — just count in 3 positions from each end (since each half has 5 values) to read off $Q1$ and $Q3$ directly.

Desmos Tip

Enter the moves as a list $L1 = \{28, 31, 31, 34, 36, 38, 38, 41, 44, 47, 52\}$ in the calculator and use $\text{quartile}(L1, 1)$ and $\text{quartile}(L1, 3)$ to instantly check $Q1$ and $Q3$ against your hand calculation.

Data Displays — Histograms, Dot Plots & Box Plots — Problem 8 (Very Hard)

A logistics company with distribution hubs in Rotterdam, Hamburg, and Gdańsk tracked the number of containers delayed at customs each day over 24 operating days. The results are shown in this frequency table: 0 delays: 9 days; 1 delay: 6 days; 2 delays: d days; 3 delays: 2 days; 5 delays: 1 day. The company's insurance policy waives its deductible for the month only if the mean number of daily delays over all 24 days is less than 1. First find the value of d , then determine whether the deductible is waived.



Formula / Rule Used: Weighted mean from a frequency table, used to test a real-world threshold condition.

Solution

Step 1: Find the missing frequency d using the total day count: $9+6+d+2+1 = 24$.

Step 2: Solve: $18+d = 24$, so $d = 6$.

Step 3: Compute the total delay-count sum: $0(9)+1(6)+2(6)+3(2)+5(1) = 0+6+12+6+5 = 29$.

Step 4: Compute the mean: $\text{mean} = 29/24 \approx 1.21$ delays per day.

Step 5: Compare to the policy threshold: the deductible is waived only if the mean is LESS THAN 1.

Step 6: Since 1.21 is not less than 1, the deductible is NOT waived.

Answer: $d = 6$; $\text{mean} \approx 1.21$ delays/day, so the deductible is NOT waived.

Tip

Always solve for a missing frequency using the total-count equation FIRST (frequencies must sum to the stated total), then plug that value into the weighted-sum formula for the mean — don't try to solve both at once.

Trap

It's tempting to add d directly into the delay-total sum instead of $2d$, since the '2 delays' row is easy to skim past — this understates the mean just enough to flip the real-world conclusion.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Deductible IS waived (mean ≈ 0.96)	Added d directly into the total delay sum instead of $2d$ (forgot to multiply the '2 delays' row by its frequency d before summing), understating the



	mean just enough to drop it below the threshold of 1.
d = 7	Miscounted the known frequencies as 17 instead of 18 (e.g., dropped the single day with 5 delays), then solved $17+d=24$ to get $d=7$ instead of $d=6$.
Deductible IS waived (mean ≈ 0.40)	Correctly computed the mean as $29/24 \approx 1.21$ but then divided again by the 3 distribution hubs mentioned in the setup, wrongly treating the threshold as a per-hub average instead of the overall daily mean.

Shortcut

Since three known frequencies plus d must sum to 24, find d from one quick equation ($18+d=24$) before ever touching the weighted-mean formula.

Desmos Tip

Once $d=6$ is known, type $(0*9+1*6+2*6+3*2+5*1)/24$ directly into the calculator to get the mean in one line and check it against 1 before answering the waiver question.

12. Scatterplots

Scatterplots — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Scatterplot	Διάγραμμα διασποράς
Line of best fit	Ευθεία βέλτιστης προσαρμογής
Residual	Υπόλοιπο (residual)
Positive/negative association	Θετική/αρνητική συσχέτιση
Nonlinear model	Μη γραμμικό μοντέλο

Scatterplots — Condensed Theory

A scatterplot displays the relationship between two numerical variables, and a line (or curve) of best fit summarizes that relationship for prediction.

Line of Best Fit

$$y = mx + b \text{ (linear case)}$$

The line represents predicted values; the actual data points are the true values. It need not pass through any actual point.

Residuals

$$\text{Residual} = \text{actual value} - \text{predicted value}$$

A positive residual means the actual point lies above the line (the model underestimated); negative means below (overestimated).

Interpreting Slope & Intercept in Context

Slope = predicted change in y per one-unit increase in x. Y-intercept = predicted value of y when x = 0 — sanity-check whether that's realistic in context.

Types of Association

Positive: as x increases, y increases. Negative: as x increases, y decreases. No association: no clear pattern.

Nonlinear Models

$$\text{Quadratic: } y = ax^2 + bx + c. \quad \text{Exponential: } y = a \cdot b^x.$$



Exponential growth/decay curves; a quadratic model is symmetric around a vertex.



Watch Out

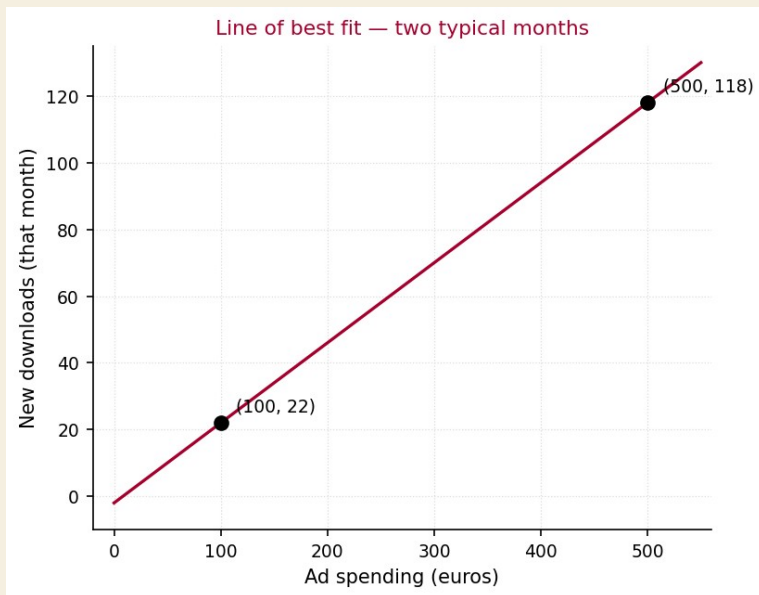
To find two points ON the line of best fit (not the data points themselves), read directly off the line — data points scattered near it are not the same as points on it.

Scatterplots — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Scatterplots — Problem 1 (Very Hard)

A mobile-game studio in Rotterdam ran a monthly analysis linking its social-media advertising spending (in euros) to the number of new app downloads that month. Studying 14 months of data, analysts noted that a month with 100 euros of ad spending averaged 22 new downloads, while a month with 500 euros of ad spending averaged 118 new downloads. Using these two typical months as points on the line of best fit, (a) find the predicted number of new downloads for a month with 350 euros of spending, and (b) determine whether the model's y-intercept is realistic in context.



Formula / Rule Used: Slope-intercept equation from two data points; realism check on the y-intercept.

Solution

Step 1: Translate the two described months into ordered pairs: (100, 22) and (500, 118), where x = euros spent and y = new downloads.

Step 2: Find the slope: $m = (118 - 22)/(500 - 100) = 96/400 = 0.24$ downloads per euro.

Step 3: Use point-slope form with (100, 22): $22 = 0.24(100) + b$, so $22 = 24 + b$.

Step 4: Solve for b : $b = 22 - 24 = -2$.

Step 5: Write the line of best fit: $y = 0.24x - 2$.

Step 6: Predict downloads for $x = 350$: $y = 0.24(350) - 2 = 84 - 2 = 82$ downloads.

Step 7: Check the y-intercept: at $x = 0$ the model predicts $y = -2$ downloads, which is impossible since a download count cannot be negative, so the intercept is not realistic — it is just an artifact of extending the line past the observed data.

Answer: 82 downloads predicted at 350 euros; the y-intercept of -2 is not realistic because download counts cannot be negative.

Tip

When a word problem describes two 'typical' months or cases instead of giving a table, translate them into ordered pairs first, then treat the problem exactly like finding a line through two points.

Trap



A rushed student might treat 22 and 118 as slope values instead of y-coordinates paired with their given x-values, or accept -2 downloads as a valid prediction without questioning whether a negative count makes sense.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
≈1064 downloads predicted	Flipped the slope ratio, computing $(500-100)/(118-22) = 400/96$ instead of $96/400$, giving an inflated slope and a wildly overestimated prediction at $x = 350$.
106 downloads predicted	Set $b = 22$ directly from the point $(100, 22)$ without solving $22 = 24 + b$ for b , effectively treating the y-coordinate itself as the intercept and using $y = 0.24x + 22$.
82 downloads predicted, with the y-intercept of -2 accepted as realistic	Computed the prediction correctly but didn't check whether a negative predicted download count at $x = 0$ makes sense, accepting -2 as plausible instead of recognizing counts can't be negative.

Shortcut

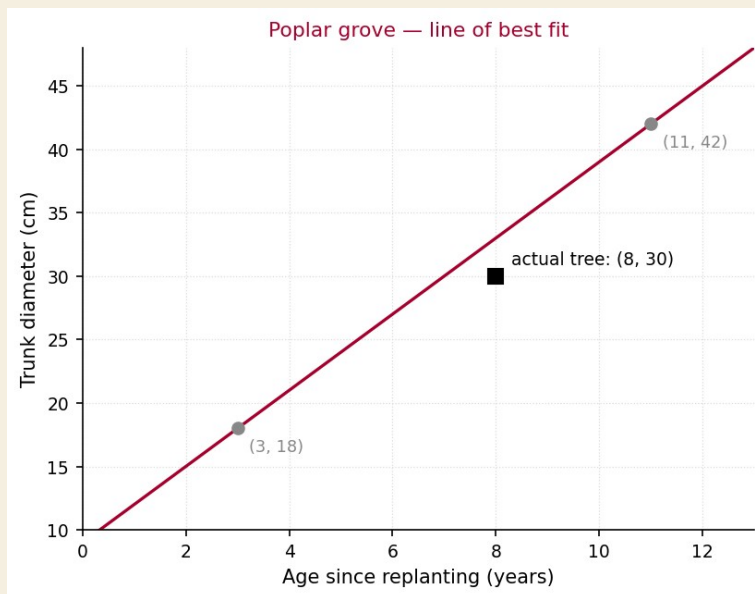
Since the two x-values are 400 apart, the slope $96/400 = 0.24$ can be found in one line without formally writing point-slope form first.

Desmos Tip

Plot $(100, 22)$ and $(500, 118)$ in Desmos and use its built-in linear regression on the two points (or graph $y = 0.24x - 2$) to confirm the line crosses below $y = 0$ near $x = 0$.

Scatterplots — Problem 2 (Hard)

Researchers studying tree growth in a park outside Ljubljana plotted age since replanting (years, x) against trunk diameter (cm, y) for a grove of poplar trees. The line of best fit for the grove passes approximately through the points $(3, 18)$ and $(11, 42)$. One particular tree in the grove is 8 years old and has an actual trunk diameter of 30 cm. Find the residual for this tree, and state whether the line overestimated or underestimated its diameter.



Formula / Rule Used: Line of best fit from two points, then residual = actual - predicted.

Solution

Step 1: Find the slope of the line of best fit: $m = (42 - 18)/(11 - 3) = 24/8 = 3$ cm per year.

Step 2: Use point-slope form with $(3, 18)$: $18 = 3(3) + b$, so $18 = 9 + b$.

Step 3: Solve for b : $b = 9$.

Step 4: Write the equation of the line of best fit: $y = 3x + 9$.

Step 5: Find the predicted diameter for an 8-year-old tree: $y = 3(8) + 9 = 24 + 9 = 33$ cm.

Step 6: Compute the residual: residual = actual - predicted = $30 - 33 = -3$.

Step 7: Since the residual is negative, the actual diameter (30 cm) is below the predicted diameter (33 cm), so the line of best fit overestimated this tree's diameter.

Answer: Residual = -3 cm; the line overestimated the tree's diameter.

Tip

When only two approximate points on the line are given, build the full equation first (slope, then intercept), then substitute the specific x -value asked about before computing any residual.

Trap

A student might compute predicted - actual ($33 - 30 = +3$) instead of actual - predicted, flipping the sign and concluding the tree was underestimated when it was actually overestimated.

Process of Elimination

Tempting Wrong Answer

Exactly What Went Wrong



Residual = 3; line underestimated	Computed predicted - actual ($33 - 30 = 3$) instead of actual - predicted, flipping the sign and reversing the over/under conclusion.
Residual = -21 (using $y = 3x + 27$)	Solved $18 = 9 + b$ by adding instead of subtracting, getting $b = 27$ instead of $b = 9$, which produced a badly inflated predicted diameter of 51 cm.
Residual = 6; line underestimated	Forgot to add the y-intercept when computing the predicted diameter, using predicted = $3(8) = 24$ instead of $3(8) + 9 = 33$.

Shortcut

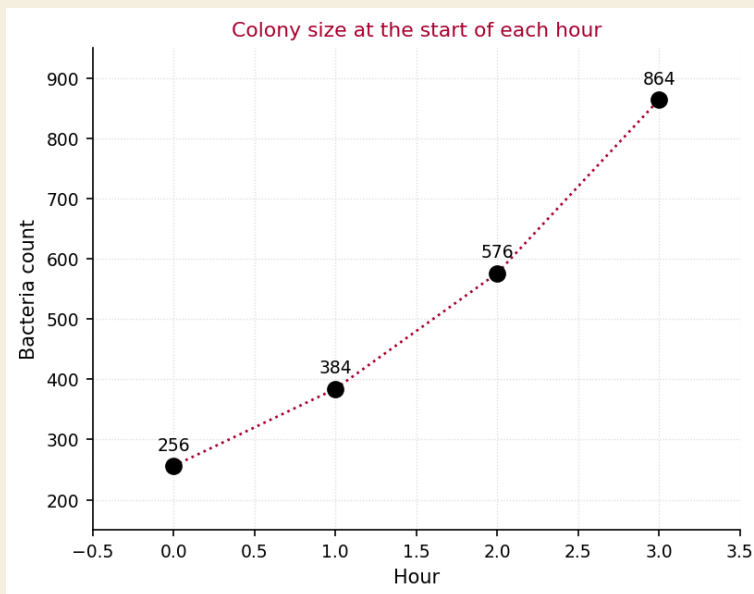
There is no way to skip building the full equation here, since age 8 is not one of the two given points — the two-step process (build the line, then predict) is already the fastest reliable path.

Desmos Tip

Enter $y = 3x + 9$ in Desmos and use the table feature at $x = 8$ to instantly confirm the predicted value of 33 before comparing it to the actual 30.

Scatterplots — Problem 3 (Very Hard)

A microbiology lab in Ghent records the number of bacteria in a colony at the start of each hour: 256 at hour 0, 384 at hour 1, 576 at hour 2, and 864 at hour 3. A lab assistant must decide whether a linear, quadratic, or exponential model best describes this growth before writing a report. (a) Determine which type of model fits the pattern and justify the choice using the data. (b) Using a model of the form $y = a \cdot b^x$, predict the bacteria count at hour 5.



Formula / Rule Used: Constant-difference test vs. constant-ratio test to identify the model; exponential equation $y = a \cdot b^x$.

Solution

Step 1: Test for a constant difference (the signature of a linear model): $384 - 256 = 128$, $576 - 384 = 192$, $864 - 576 = 288$ — these differences are not constant, so the data is not linear.

Step 2: Test for a constant ratio (the signature of an exponential model): $384/256 = 1.5$, $576/384 = 1.5$, $864/576 = 1.5$ — the ratio between consecutive terms is constant at 1.5, which confirms an exponential model rather than quadratic (a quadratic model would show constant second differences, not a constant ratio).

Step 3: Since $y = a \cdot b^x$ and the constant multiplier per hour is 1.5, the growth factor is $b = 1.5$.

Step 4: Find a using hour 0: at $x = 0$, $y = a \cdot b^0 = a$, and the data gives $y = 256$ at hour 0, so $a = 256$.

Step 5: Write the model: $y = 256 \cdot (1.5)^x$.

Step 6: Predict hour 5: $y = 256 \cdot (1.5)^5 = 256 \cdot 7.59375 = 1944$ bacteria.

Answer: Exponential model, $y = 256 \cdot (1.5)^x$; predicted count at hour 5 is 1944 bacteria.

Tip

Test consecutive differences first for a constant amount (linear); if that fails, test the ratio of consecutive y -values for a constant multiplier (exponential) before ever assuming quadratic.

Trap

A student might see the increasing differences (128, 192, 288) and wrongly conclude the data is quadratic because growth is 'speeding up,' without checking that the ratios are constant — which is the actual signature of exponential growth, not just any acceleration.

**Process of Elimination**

Tempting Wrong Answer	Exactly What Went Wrong
≈1269 bacteria at hour 5	Never tested for a constant ratio, instead extrapolated linearly using the average difference between consecutive terms (about 202.67 per hour), badly underestimating the true exponential growth.
1920 bacteria at hour 5	Used the correct model $y = 256 \cdot 1.5^x$ but computed 1.5^5 by multiplying $1.5 \times 5 = 7.5$ instead of repeated multiplication ($1.5^5 = 7.59375$), giving $256 \cdot 7.5 = 1920$.
2916 bacteria at hour 5	Used the hour-1 value (384) as the initial value a instead of the hour-0 value (256), shifting the entire exponential model up by one time step: $384 \cdot 1.5^5 = 2916$.

Shortcut

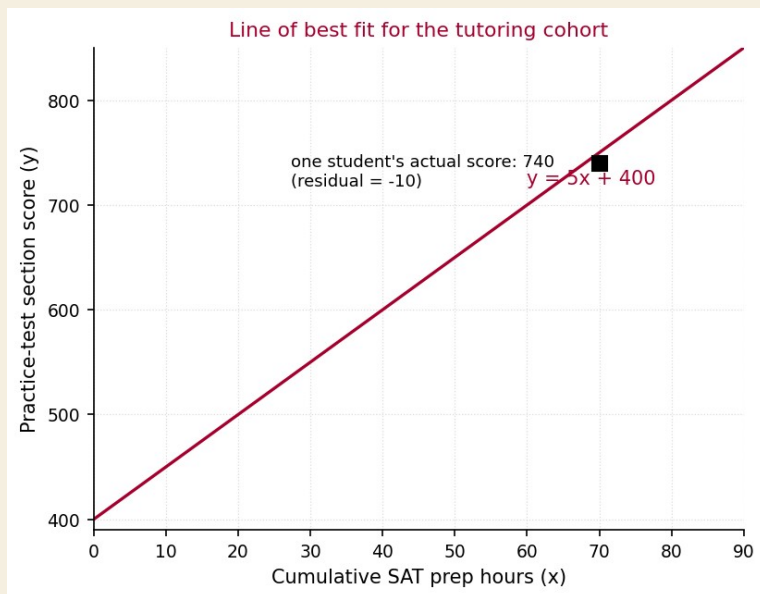
Once a constant ratio is confirmed, there is no need to also test second differences for a quadratic fit — a constant ratio alone guarantees exponential and automatically rules out both linear and quadratic.

Desmos Tip

Graph $y = 256 \cdot 1.5^x$ in Desmos and check it passes through (0,256), (1,384), (2,576), (3,864) exactly, then read the y-value at $x = 5$ directly from the table.

Scatterplots — Problem 4 (Hard)

In a scatterplot relating cumulative SAT prep hours (x) to a student's practice-test section score out of 800 (y) for a tutoring cohort, the line of best fit is $y = 5x + 400$. One student in the cohort had an actual practice-test score of 740, and the residual for that student was -10. How many hours of SAT prep had that student completed?



Formula / Rule Used: Residual = actual - predicted, solved in reverse for the input variable x .

Solution

Step 1: Recall the residual formula: residual = actual - predicted.

Step 2: Rearrange to solve for the predicted value: predicted = actual - residual.

Step 3: Substitute the known values: predicted = $740 - (-10) = 740 + 10 = 750$.

Step 4: Set this predicted score equal to the line of best fit expression: $750 = 5x + 400$.

Step 5: Subtract 400 from both sides: $5x = 750 - 400 = 350$.

Step 6: Divide by 5: $x = 350/5 = 70$.

Step 7: The student had completed 70 hours of SAT prep.

Answer: 70 hours of SAT prep.

Tip

When a residual and one of the two values (actual or predicted) are given, first isolate whichever value is missing using predicted = actual - residual (or actual = predicted + residual), then substitute into the line equation to solve for the remaining unknown.

Trap

A student might compute predicted = actual + residual ($740 + (-10) = 730$) instead of actual - residual, since the direction of the residual formula is easy to misremember, leading to $5x = 330$ and an incorrect $x = 66$.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$x = 66$ hours	Computed predicted = actual + residual ($740 + (-10) = 730$) instead of



	actual - residual, misremembering the direction of the residual formula, then solved $730 = 5x + 400$ to get $x = 66$.
x = 150 hours	Correctly found predicted = 750 but forgot to subtract 400 from both sides before dividing, solving $5x = 750$ directly to get $x = 150$ instead of $x = 70$.
750 hours	Correctly computed the predicted score (750) but stopped there and reported it as the final answer, instead of continuing to substitute into $y = 5x + 400$ and solve for x .

Shortcut

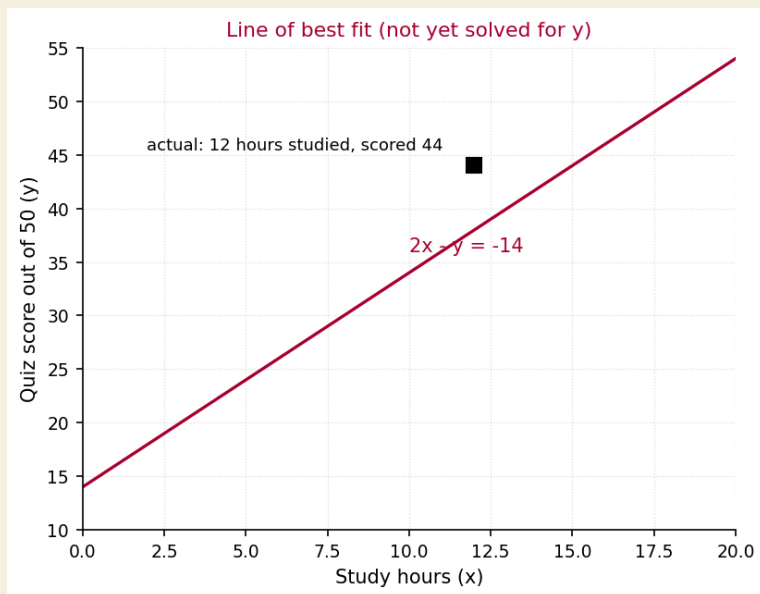
Once predicted = 750 is found, solving $750 = 5x + 400$ is a single two-step linear equation — there is no faster path since x is embedded inside the line's formula.

Desmos Tip

Graph $y = 5x + 400$ in Desmos and trace or use the table to find where $y = 750$, which confirms $x = 70$ without redoing the algebra by hand.

Scatterplots — Problem 5 (Hard)

A researcher studying study hours (x) versus quiz score out of 50 (y) for a class models the relationship with the equation $2x - y = -14$ as the line of best fit, not yet solved for y . A particular student studied 12 hours and scored an actual 44 on the quiz. Find the residual for this student, and state whether the line overestimated or underestimated the score.



Formula / Rule Used: Rearrange standard form to slope-intercept form, then residual = actual - predicted.

Solution

Step 1: Recognize that $2x - y = -14$ is not solved for y , so it cannot be used directly as a prediction rule.

Step 2: Rearrange: $2x - y = -14$, so $-y = -14 - 2x$, and multiplying both sides by -1 gives $y = 2x + 14$.

Step 3: Substitute the student's study hours, $x = 12$, into the corrected equation: predicted $y = 2(12) + 14$.

Step 4: Compute: $2(12) = 24$, so predicted $y = 24 + 14 = 38$.

Step 5: Recall residual = actual - predicted.

Step 6: Compute the residual: $44 - 38 = 6$.

Step 7: Since the residual is positive, the actual score (44) is above the predicted score (38), so the line of best fit underestimated this student's quiz score.

Answer: Residual = 6; the line underestimated the student's score.

Tip

Whenever a line of best fit is given in standard form ($Ax + By = C$) or any form not solved for y , isolate y first — never substitute x into the unsolved equation and call the result the predicted value.

Trap

A student might make a sign error while isolating y (getting $y = 2x - 14$ instead of $y = 2x + 14$), predicting $y = 2(12) - 14 = 10$ and computing a badly wrong residual of $44 - 10 = 34$.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
Residual = 34; line underestimated	Made a sign error isolating y from $2x - y = -14$, getting $y = 2x - 14$ instead of $y = 2x + 14$, which produced a predicted score of only 10 instead of 38.
Residual = -6; line overestimated	Computed predicted - actual ($38 - 44 = -6$) instead of actual - predicted, flipping the sign and reversing the over/underestimate conclusion even though the magnitude was right.
Residual = 8; line underestimated	Made an arithmetic slip computing $2(12)$, getting 22 instead of 24, which shifted the predicted score to 36 and the residual to 8.

Shortcut

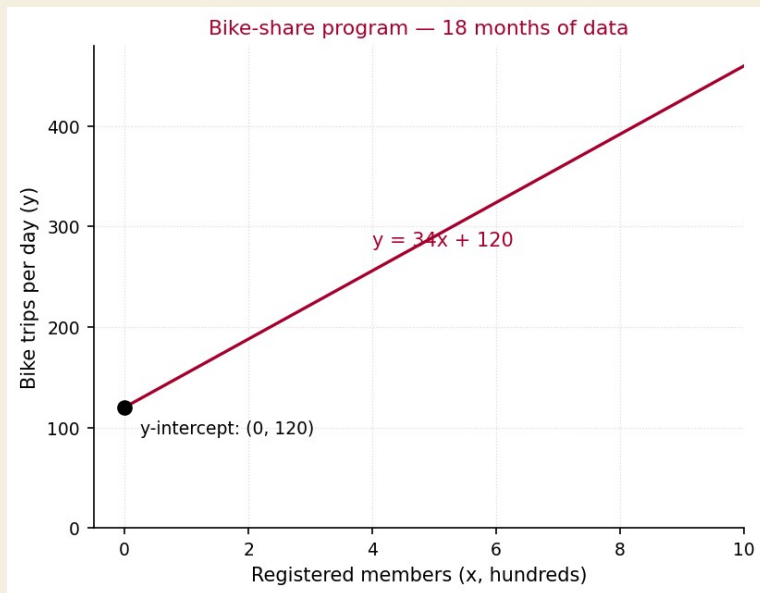
Rearranging once to $y = 2x + 14$ gives a reusable prediction rule for the whole dataset, so it's worth doing before plugging in any x -value rather than manipulating the standard-form equation freshly each time.

Desmos Tip

Type $2x - y = -14$ directly into Desmos, which graphs implicit equations, and trace to $x = 12$ to confirm the predicted $y = 38$ before computing the residual by hand.

Scatterplots — Problem 6 (Hard)

A city bike-share program in Utrecht tracked, over 18 months, the number of registered members (x , in hundreds) against the number of bike trips taken per day (y). Analysts also recorded the average trip duration each month, which turned out to be irrelevant to this particular model. The line of best fit relating members to daily trips is $y = 34x + 120$, and the program currently operates 45 docking stations across the city. Using the model, predict the number of daily bike trips when there are 650 registered members, and state whether the y -intercept is realistic in context.



Formula / Rule Used: Slope/intercept interpretation with unit conversion before substitution.

Solution

Step 1: Note that x is defined in hundreds of members, so 650 registered members corresponds to $x = 650/100 = 6.5$.

Step 2: Substitute $x = 6.5$ into the line of best fit: $y = 34(6.5) + 120$.

Step 3: Compute $34(6.5) = 221$.

Step 4: Add: $y = 221 + 120 = 341$ predicted daily trips.

Step 5: Identify that the average trip duration and the 45 docking stations are not part of this linear model and are not needed for the prediction.

Step 6: Interpret the y -intercept (120): at $x = 0$ (zero registered members), the model predicts 120 trips per day, which is not fully realistic taken literally, since zero registered members should produce close to zero member trips — the intercept would only make sense if it reflects trips by non-member/casual riders.

Answer: Predicted trips are about 341 per day; the y -intercept of 120 is not realistic taken literally, since it implies daily trips even with zero registered members.

Tip

Before substituting, double-check what each variable's units actually are (e.g., ' x in hundreds of members') and ignore any recorded facts that never appear in the equation itself.

Trap



A student might substitute $x = 650$ directly into $y = 34x + 120$ (ignoring that x is measured in hundreds of members), or try to work the 45 docking stations into the slope or intercept interpretation, both of which are irrelevant to the actual linear model given.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
$\approx 22,220$ predicted trips	Substituted $x = 650$ directly into $y = 34x + 120$ without converting to the model's units (hundreds of members), since x should have been $650/100 = 6.5$, not 650.
386 predicted trips	Mistakenly added the 45 docking stations into the calculation ($341 + 45 = 386$), treating the irrelevant docking-station count as part of the linear model instead of recognizing it plays no role in $y = 34x + 120$.
341 predicted trips, with the y-intercept of 120 accepted as realistic	Computed the prediction correctly but didn't examine what the y-intercept means in context, accepting 120 trips at zero members as reasonable instead of questioning it.

Shortcut

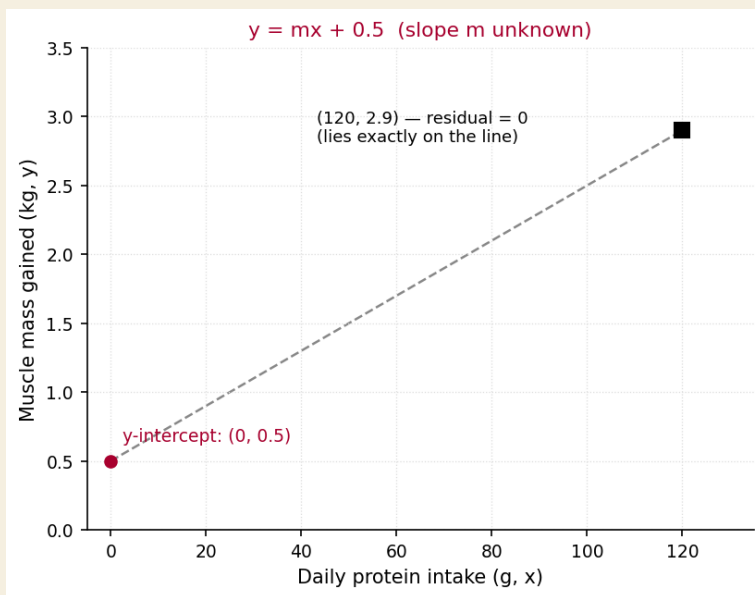
Convert the member count to the model's units ($650/100 = 6.5$) in one step before plugging in, rather than rescaling the whole equation to raw member counts.

Desmos Tip

Graph $y = 34x + 120$ in Desmos and use the table feature to evaluate at $x = 6.5$, instantly confirming $y = 341$ without arithmetic errors.

Scatterplots — Problem 7 (Very Hard)

A sports dietician in Thessaloniki collects data on daily protein intake (grams, x) versus lean muscle mass gained in a month (kg, y) for her clients, and models the relationship with the line of best fit $y = mx + 0.5$, where m has not yet been determined. One client consumed 120 grams of protein per day, and her data point had a residual of exactly 0 — her actual muscle mass gain was 2.9 kg. (a) Find the value of m . (b) Using this equation, predict the muscle mass gain, rounded to the nearest tenth of a kg, for a client who consumes 180 grams of protein per day.



Formula / Rule Used: A residual of 0 means the point lies exactly on the line of best fit; solve for the unknown slope, then predict.

Solution

Step 1: Recognize that a residual of 0 means actual = predicted for this client, so her point (120, 2.9) satisfies the line exactly.

Step 2: Substitute (120, 2.9) into $y = mx + 0.5$: $2.9 = m(120) + 0.5$.

Step 3: Subtract 0.5 from both sides: $120m = 2.9 - 0.5 = 2.4$.

Step 4: Solve for m : $m = 2.4/120 = 0.02$.

Step 5: Write the completed line of best fit: $y = 0.02x + 0.5$.

Step 6: Predict the gain for $x = 180$ grams: $y = 0.02(180) + 0.5$.

Step 7: Compute: $0.02(180) = 3.6$, so $y = 3.6 + 0.5 = 4.1$ kg.

Answer: $m = 0.02$; predicted muscle mass gain at 180 g protein/day is 4.1 kg.

Tip

When a data point is described as having a residual of 0, treat it as if it were just handed to you as a point that satisfies the equation exactly, and substitute it in to solve for the missing constant before doing anything else.

Trap

A student in a hurry sets $mx = 2.9$ directly (ignoring the +0.5 intercept already in the equation) instead of subtracting 0.5 first, since it's tempting to treat the actual value itself as equal to only the mx term.

Process of Elimination



Tempting Wrong Answer	Exactly What Went Wrong
$m \approx 0.024$, predicted gain ≈ 4.85 kg	Set $120m = 2.9$ directly instead of subtracting the 0.5 intercept first, skipping $2.9 - 0.5 = 2.4$ and computing $m = 2.9/120 \approx 0.0242$ instead of $m = 0.02$.
predicted gain = 3.6 kg	Correctly solved for $m = 0.02$ but then computed only $0.02(180) = 3.6$ for the final prediction and forgot to add back the y-intercept of 0.5.
$m \approx 0.028$, predicted gain ≈ 5.6 kg	Made a sign error rearranging $2.9 = 120m + 0.5$, adding 0.5 instead of subtracting it ($120m = 2.9 + 0.5 = 3.4$), giving $m \approx 0.0283$ instead of 0.02.

Shortcut

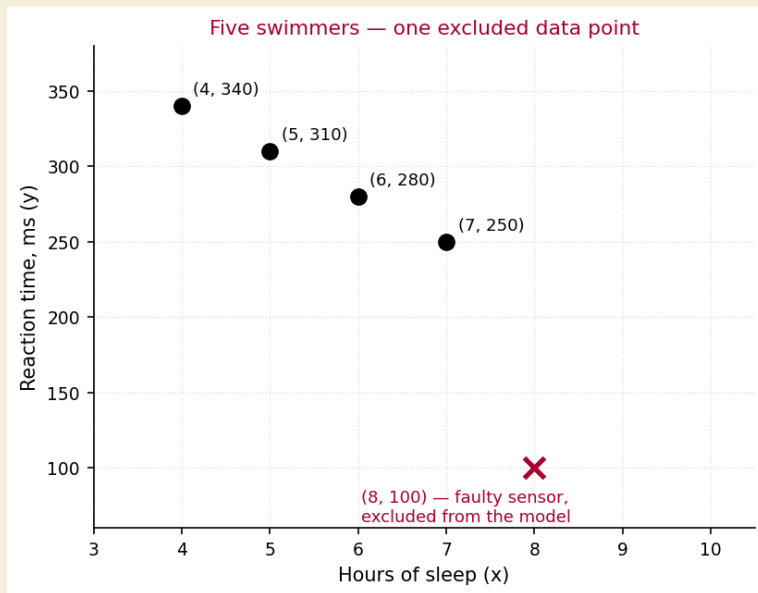
Since the intercept 0.5 is already fixed, the only unknown is m — subtract 0.5 and divide in one line ($120m = 2.4$, $m = 0.02$) without ever writing a full point-slope setup.

Desmos Tip

Once $m = 0.02$ is found, graph $y = 0.02x + 0.5$ in Desmos and use the table feature at $x = 180$ to confirm the predicted 4.1 kg.

Scatterplots — Problem 8 (Very Hard)

A sleep researcher in Krakow records five swimmers' hours of sleep the night before a meet (x) against their reaction time off the blocks in milliseconds (y): (4, 340), (5, 310), (6, 280), (7, 250), and (8, 100). The coach explains that the swimmer at (8, 100) had a faulty starting sensor that night, so that data point should be excluded from the model. (a) Using the line of best fit built from two of the remaining reliable points, predict the reaction time for a swimmer who gets 10 hours of sleep. (b) State whether this prediction is a reliable interpolation or a risky extrapolation, and why.



Formula / Rule Used: Identify and exclude an outlier before building the line of best fit; then judge interpolation vs. extrapolation.

Solution

Step 1: Exclude the flagged outlier (8, 100), leaving the reliable points (4, 340), (5, 310), (6, 280), and (7, 250).

Step 2: Use two of the reliable points, (4, 340) and (7, 250), to find the slope: $m = (250 - 340)/(7 - 4) = -90/3 = -30$ ms per hour of sleep.

Step 3: Use point-slope form with (4, 340): $340 = -30(4) + b$, so $340 = -120 + b$.

Step 4: Solve for b : $b = 340 + 120 = 460$.

Step 5: Write the line of best fit: $y = -30x + 460$.

Step 6: Predict reaction time at $x = 10$: $y = -30(10) + 460 = -300 + 460 = 160$ ms.

Step 7: Compare $x = 10$ to the range of the reliable data (4 to 7 hours): since 10 falls outside that range, this prediction is an extrapolation, not an interpolation, so it is less reliable than a prediction made within the observed range.

Answer: 160 ms predicted at 10 hours of sleep; this is a risky extrapolation, since $x = 10$ falls outside the 4-to-7 hour range of the reliable data used to build the line.

Tip

Before building any line of best fit from a table, scan for a point that clearly breaks the pattern (like a data-entry or equipment error the problem tells you about) and exclude it, then check whether your final x -value actually falls inside or outside the range you used.

**Trap**

A student might build the line using all five points, including the flagged faulty outlier (8, 100), which drags the slope far steeper than the true underlying trend and produces an unrealistic (even negative) prediction.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
-20 ms predicted	Included the excluded outlier (8, 100) when computing the slope, using (4, 340) and (8, 100): $m = (100 - 340)/(8 - 4) = -60$, giving line $y = -60x + 580$ and an impossible negative prediction at $x = 10$.
520 ms predicted	Made a sign error computing the slope, dropping the negative sign from $(250 - 340)/(7 - 4) = -30$ and using $m = 30$ instead, giving line $y = 30x + 220$ and a prediction that rises instead of falls with more sleep.
160 ms, labeled as a reliable interpolation	Correctly computed the predicted reaction time but didn't check that $x = 10$ falls outside the 4-to-7 range used to build the line, incorrectly treating the extrapolated prediction as trustworthy interpolation.

Shortcut

Once the outlier is identified, pick the two reliable points that are farthest apart (4 and 7) to compute the slope — this minimizes rounding and matches what a quick eyeball trend line would give.

Desmos Tip

Plot all five points in Desmos, visually confirm (8, 100) breaks the downward trend, then graph $y = -30x + 460$ over just the reliable points to see that $x = 10$ sits clearly outside their span.

13. Sampling, Bias & Margin of Error

Sampling, Bias & Margin of Error — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

Term (English)	Greek — Ελληνικά
Sample	Δείγμα
Population	Πληθυσμός
Random sampling	Τυχαία δειγματοληψία
Bias	Μεροληψία (σφάλμα)
Margin of error	Περιθώριο σφάλματος

Sampling, Bias & Margin of Error — Condensed Theory

This skill tests whether a study or survey design produces results that can be trusted and generalized — and how confidently.

Population vs. Sample

A reliable sample must be both large AND randomly selected from the population you actually want to draw conclusions about.

Types of Bias

Sampling bias: the population itself wasn't randomly selected. Self-selection bias: participants volunteered. Nonresponse bias: certain groups didn't respond. A larger sample never fixes a biased selection method.

Margin of Error

True population value $\approx$ sample value $\pm$ margin of error

A larger sample size produces a smaller margin of error. The margin of error describes the estimated mean/proportion, not every individual.

Drawing Valid Conclusions

Results generalize ONLY to the exact population that was randomly sampled. Correlation is never proof of causation — only a controlled, randomized experiment supports a cause-and-effect claim.



Watch Out

When comparing two studies, a smaller sample size (all else equal) means a LARGER margin of error —



this trips up students who assume 'smaller number' always means more precise.

Sampling, Bias & Margin of Error — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

Sampling, Bias & Margin of Error — Problem 1 (Hard)

A tech company in Larnaca, Cyprus has 84,000 registered users of its ride-sharing app. To estimate how many users would pay for a premium subscription, the company selects a simple random sample of registered users and asks them. Based on the sample results, the company estimates that 31,500 of all 84,000 registered users would pay for premium. If the sample contained exactly 189 users who said they would pay for premium, and this sample proportion was used directly to produce the population estimate, how many users were in the sample?

Formula / Rule Used: Generalizing a sample to a population: the sample proportion equals the population proportion, so set up and solve a proportion equation.

Solution

Step 1: Find the proportion of the population estimated to want premium: $31500/84000 = 0.375$ (or $3/8$).

Step 2: Since the population estimate was generated by applying the sample proportion, the sample must share this same proportion: $189/n = 0.375$.

Step 3: Solve for n by dividing: $n = 189/0.375$.

Step 4: Compute: $189/0.375 = 504$.

Step 5: So the sample contained 504 users.

Answer: 504 users

Tip

When a problem gives you the final population estimate and asks you to work backward to the sample size, first convert the population estimate into a proportion (population estimate $\div$ total population), then apply that SAME proportion to the sample count.

Trap

It's tempting to treat the population estimate (31,500) as if it were itself the sample size, or to divide the two counts in the wrong order — either move produces a nonsensical or reversed value instead of correctly solving $189/n = 0.375$ for n .

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
71	Set up the proportion inverted, as $189/n = 84000/31500$ instead of $189/n = 31500/84000$, so dividing by the reciprocal ratio gave 71 instead of 504.
31,500	Stopped after finding the population estimate and reported that population count directly instead of continuing on to solve for the



	unknown sample size n .
236	Miscalculated $31500/84000$ as 0.8 instead of 0.375 (flipped which number goes on top), then solved $189/0.8 = 236$ instead of $189/0.375 = 504$.

Shortcut

Recognize $31,500/84,000$ simplifies to $3/8$; since $189 = 3 \times 63$, the sample size must be $8 \times 63 = 504$, avoiding decimal division entirely.

Desmos Tip

Type $189/(31500/84000)$ into the calculator to get 504 directly, skipping manual cross-multiplication.

Sampling, Bias & Margin of Error — Problem 2 (Hard)

A researcher surveys a random sample of 300 students at a school in Vienna about their commute to school. She reports that for every 5 students who take public transit, 3 walk, and 2 bike — no student in the sample uses any other method. If the school has 2,400 students total and the sample is representative, how many of all students at the school would be expected to walk to school?

Formula / Rule Used: Multi-part ratio scaled to a population: convert the ratio sentence into a fraction of the whole, then apply that fraction to the population.

Solution

Step 1: Add the ratio parts: 5 (transit) + 3 (walk) + 2 (bike) = 10 parts total.

Step 2: Find the fraction of students who walk: $3/10$.

Step 3: Check this fraction against the sample as a sanity check: $300 \times 3/10 = 90$ students walked in the sample of 300.

Step 4: Apply the same fraction to the full school population of 2,400 students: $2400 \times 3/10$.

Step 5: Compute: $2400 \times 3/10 = 720$.

Step 6: So 720 students at the school would be expected to walk.

Answer: 720 students

Tip

When a ratio is given as a sentence with three (or more) quantities, first add all the parts to get the total, then form the fraction for the specific category the question asks about before scaling to the population.

Trap

It's easy to grab the wrong category's number from the sentence since transit, walk, and bike all appear together — always re-read which single category the question is actually asking about before you scale up.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
900	Left the bike riders out when adding the ratio parts (used $5+3=8$ instead of



	$5+3+2=10$), then computed $2400 \times \frac{3}{8} = 900$.
1,200	Used the transit fraction ($\frac{5}{10} = \frac{1}{2}$) instead of the walk fraction ($\frac{3}{10}$), computing $2400 \times \frac{1}{2} = 1200$.
480	Used the bike fraction ($\frac{2}{10} = \frac{1}{5}$) instead of the walk fraction, computing $2400 \times \frac{1}{5} = 480$.

Shortcut

No shortcut is needed beyond finding the fraction and multiplying — but you can skip the sample sanity-check (90 students) entirely since it isn't required to answer the population question.

Desmos Tip

Type $2400 \times \frac{3}{10}$ directly into the calculator to confirm 720 without simplifying the fraction by hand first.

Sampling, Bias & Margin of Error — Problem 3 (Very Hard)

The city council of Rotterdam wants to know what fraction of all 620,000 residents support a proposed bike-lane expansion. The council posts a link to an online survey on the city's official social media page and invites anyone who sees the post to click and respond. The survey stays open for two weeks and receives 1,850 responses — more than double the roughly 800 responses the council's mailed paper surveys usually generate. Of the 1,850 online respondents, 74% support the expansion. A council member wants to report that '74% of Rotterdam residents support the bike-lane expansion.' What single flaw in the sampling method makes this specific conclusion invalid, regardless of how many responses were collected?

Formula / Rule Used: Identifying flaws in a study design: check whether participants were randomly selected or self-selected before trusting any generalization to the population.

Solution

Step 1: Note that the sample was NOT drawn randomly from all 620,000 residents — whether someone is included depended entirely on whether they saw the post and chose to click it.

Step 2: Consider whether the sample size (1,850) is the issue — it isn't: 1,850 is actually a fairly large sample, and comparing it to the 800 mailed responses is irrelevant to whether the method itself is unbiased.

Step 3: Consider whether the two-week window caused the problem — timing isn't the flaw either; even a random sample collected over two weeks would still be valid.

Step 4: Recognize the real issue: this is a voluntary response sample. People with strong opinions (often more motivated to click and respond) are systematically over-represented compared to indifferent or opposed residents.

Step 5: Because respondents chose themselves rather than being randomly selected, the 74% reflects motivated social-media users, not a representative cross-section of all 620,000 residents.

Step 6: Conclude that the defining flaw is self-selection (voluntary response) bias, not sample size or survey duration.

Answer: Voluntary response (self-selection) bias: only residents who chose to click the online survey are represented, so the 74% cannot be generalized to all Rotterdam residents.

**Tip**

When a study lets participants decide for themselves whether to respond (a posted link, a call-in poll, a mail-back form), check for self-selection bias first — no sample size, however large, fixes a non-random selection method.

Trap

The '1,850 vs. 800' comparison is designed to tempt you into thinking a bigger sample fixed the problem — sample size only affects margin of error, not whether the sample was randomly selected in the first place.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
The sample size (1,850) is too small to be reliable	Confuses margin-of-error concerns (which shrink as n grows) with selection bias; a bigger sample only makes a biased method's estimate more precisely wrong, not more valid.
The two-week survey window is too short to capture opinions	Focuses on an irrelevant detail (duration) instead of recognizing that no survey window length fixes a fundamentally non-random selection process.
The survey undercounts residents without internet access	Names a real but secondary issue (undercoverage) instead of the defining flaw — even if every resident had internet access, letting people self-select into responding would still bias the result.

Shortcut

As soon as you see 'anyone who wants to can click and respond,' flag voluntary response bias immediately — there's no need to evaluate sample size, timing, or response rate at all.

Sampling, Bias & Margin of Error — Problem 4 (Very Hard)

Two independent research firms use separate random samples to estimate the percentage of a country's residents who support raising the minimum wage. Firm A surveys 400 residents and reports 52% support, with a margin of error of ± 4 percentage points. Firm B surveys 1,600 residents and reports 49% support, with a margin of error of ± 2 percentage points. A news anchor claims, 'Firm A's poll proves support is higher than Firm B's poll shows, since 52% is greater than 49%.' Based on the two margins of error, is the anchor's claim statistically justified, and over what single range of percentages do the two firms' confidence intervals overlap?

Formula / Rule Used: Confidence interval comparison: build estimate $\pm$ margin of error for each result and check whether the intervals overlap before comparing point estimates.

Solution

Step 1: Find Firm A's confidence interval: $52\% \pm 4\%$ gives (48%, 56%).

Step 2: Find Firm B's confidence interval: $49\% \pm 2\%$ gives (47%, 51%).

Step 3: Compare the two intervals for overlap: A runs from 48% to 56%, and B runs from 47% to 51%.

Step 4: The overlapping region is where both intervals hold at once: the higher of the two lower bounds (48%) to the lower of the two upper bounds (51%), so 48% to 51%.

Step 5: Because the intervals overlap, a true population value like 49% or 50% is consistent with BOTH surveys — the surveys are not actually in conflict.

Step 6: Since the intervals overlap, the gap between 52% and 49% is not necessarily statistically meaningful, so the anchor's claim is not justified.

Answer: Not justified — the confidence intervals overlap from 48% to 51%, so the two results are not significantly different.

Tip

To judge whether one poll's estimate is genuinely higher than another's, always build BOTH confidence intervals (estimate $\pm$ margin of error) and check for overlap — never compare the raw point estimates alone.

Trap

The anchor's mistake is comparing only the point estimates (52% vs. 49%) while ignoring that each carries a margin of error; once you build the actual intervals, they clearly overlap, so the claim isn't supported.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Justified — Firm A's estimate (52%) is greater than Firm B's (49%)	Compares only the point estimates and ignores both margins of error, which is exactly the anchor's flawed reasoning restated rather than actually checked.
Not justified — the intervals overlap from 47% to 56%	Reports the full combined span of both intervals (their union, from the lowest low to the highest high) instead of the overlapping region where both intervals hold simultaneously (their intersection).
Justified, because Firm B has a larger sample and smaller margin of error, so its 49% is more trustworthy	Correctly notes Firm B's estimate is more precise but wrongly treats that as disproving Firm A — a smaller margin of error only narrows B's own interval, it doesn't shrink or invalidate A's overlapping interval.

**Shortcut**

Skip building both full intervals and just check: does $|52 - 49| = 3$ exceed the sum of the two margins ($4 + 2 = 6$)? Since $3 < 6$, the intervals must overlap — compute the exact overlap range only if the question specifically asks for it.

Desmos Tip

Plot the two intervals as number-line segments (or type the inequalities $48 \leq x \leq 56$ and $47 \leq x \leq 51$) to visually confirm where they overlap.

Sampling, Bias & Margin of Error — Problem 5 (Hard)

At a three-day music festival in Porto with 30,000 total attendees, organizers randomly sample 900 attendees and ask if they plan to return next year. Based on the sample, they estimate that $64\% \pm 3\%$ of all attendees plan to return. Rounded to the nearest hundred, what is the range of the number of attendees who could be expected to return next year?

Formula / Rule Used: Converting a margin-of-error percentage range into a population count range: find both percentage bounds first, then scale each to the population.

Solution

Step 1: Find the lower bound of the estimated proportion: $64\% - 3\% = 61\%$.

Step 2: Find the upper bound of the estimated proportion: $64\% + 3\% = 67\%$.

Step 3: Apply the lower bound to the total attendance: $0.61 \times 30,000 = 18,300$.

Step 4: Apply the upper bound to the total attendance: $0.67 \times 30,000 = 20,100$.

Step 5: Both results already land on whole hundreds, so no further rounding is needed.

Step 6: The expected number of returning attendees ranges from 18,300 to 20,100.

Answer: 18,300 to 20,100 attendees

Tip

When a margin of error is attached to a percentage, first convert it into a percentage RANGE (estimate minus margin, estimate plus margin), then scale each end of that range to the population separately — never scale just the point estimate if a range is asked for.

Trap

It's tempting to apply the $\pm 3\%$ range to the 900-person sample instead of the 30,000-person population, since the sample size is mentioned right before the percentage — but the question asks about ALL attendees.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
19,200 people	Applied only the point estimate (64%) to the population, ignoring the margin of error entirely and reporting a single value instead of the requested range.
900 to 1,200	Applied the percentage range to the 900-person sample instead of the 30,000-person population, scaling from the wrong base number



	mentioned in the problem.
17,400 to 21,000	Doubled the margin of error to $\pm 6\%$ instead of using $\pm 3\%$, computing bounds of 58% and 70% before scaling to the population.

Shortcut

Since $30,000 \times 1\% = 300$, find both bounds fast: the center is $64 \times 300 = 19,200$, then subtract/add $3 \times 300 = 900$ to get 18,300 and 20,100.

Desmos Tip

Type 0.61×30000 and 0.67×30000 into the calculator directly to get both bounds in two steps without hand-converting percentages.

Sampling, Bias & Margin of Error — Problem 6 (Very Hard)

A high school in Athens has 1,200 students. The student council wants to estimate how many students would support extending the lunch break by 15 minutes. Two methods are proposed: Method 1 — the council president asks the first 60 students who arrive at school on Monday morning. Method 2 — a computer selects 60 students at random from the complete enrollment list. Only Method 2 is carried out, and 42 of the 60 selected students support the extension. Which method would have produced a valid basis for generalizing to the full student body, and, using the results of the valid method, how many of the school's 1,200 students would be expected to support the extension?

Formula / Rule Used: First determine which sampling method is valid (every student must have an equal chance of selection), then scale that method's sample proportion to the full population.

Solution

Step 1: Evaluate Method 1: selecting 'the first 60 students to arrive' is not random — early arrivers may differ systematically (bus routes, class schedules) from the full student body, so this is a convenience sample.

Step 2: Evaluate Method 2: a computer randomly selecting 60 students from the complete enrollment list gives every student an equal chance of selection, so this is a valid simple random sample.

Step 3: Since only Method 2 is valid for generalizing to the population, use its results: 42 of the 60 sampled students supported the extension.

Step 4: Find the sample proportion: $42/60 = 0.7$ (or 70%).

Step 5: Apply this proportion to the full school population: 1200×0.7 .

Step 6: Compute: $1200 \times 0.7 = 840$.

Step 7: So Method 2 is the valid method, and 840 students would be expected to support the extension.

Answer: Method 2; 840 students expected to support the extension.

Tip



Before doing any scaling arithmetic, always check whether the sampling method itself was actually random — 'the first people who show up' is a classic convenience sample and invalidates generalizing no matter how clean the numbers look afterward.

Trap

A student who jumps straight to the arithmetic might try to use or blend in Method 1's results — but Method 1 was never actually carried out, and even if it had been, its non-random selection makes it unusable for the estimate.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
Method 1; 840 students	Computed the population estimate correctly but misidentified Method 1 (arriving early isn't random) as the valid sampling method instead of Method 2.
Method 2; 720 students	Correctly identified Method 2 as valid but mistook the sample size (60) for a percentage, computing 1200×0.6 instead of 1200×0.7 .
Method 2; 42 students	Correctly identified Method 2 but reported the raw sample count (42) instead of scaling it up to the full population of 1,200.

Shortcut

Once Method 2 is confirmed valid, recognize $42/60$ simplifies to $7/10$, so $1200 \times 7/10 = 840$ can be found with simple mental math instead of long division.

Desmos Tip

Type $1200*(42/60)$ into the calculator to confirm 840 in one step.

Sampling, Bias & Margin of Error — Problem 7 (Hard)

A pollster surveys a random sample of 800 registered voters out of a district's 25,000 registered voters to estimate support for candidate Nowak. The pollster reports a 95% confidence interval of (46%, 58%) for the true percentage of all 25,000 registered voters who support Nowak. A newspaper wants to convert this into a range for the actual number of votes Nowak could expect, rounded to the nearest hundred votes. What is that range?

Formula / Rule Used: Applying a confidence interval already stated as a population percentage directly to the population count.

Solution

Step 1: Identify the two ends of the given confidence interval: 46% and 58%.

Step 2: Recognize the interval already describes the full population of registered voters (25,000) — the 800-person sample size was only used to produce the interval and isn't needed again here.

Step 3: Apply the lower bound to the population: $0.46 \times 25,000 = 11,500$.

Step 4: Apply the upper bound to the population: $0.58 \times 25,000 = 14,500$.

Step 5: Both values already land on the nearest hundred, so the range is 11,500 to 14,500 votes.

Answer: 11,500 to 14,500 votes

Tip

When a confidence interval is already stated as a percentage of the full population, apply it directly to the population size — don't get pulled back into re-scaling from the sample size, which was only relevant to how the pollster built the interval in the first place.

Trap

The 800-person sample size is mentioned right before the interval, tempting you to scale the percentages to 800 instead of the actual population of 25,000 that the interval describes.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
368 to 464 votes	Applied the 46%–58% interval to the 800-person sample instead of the 25,000-person population, scaling from the irrelevant number mentioned in the problem.
13,000 votes	Averaged the interval down to a single point estimate ($52\% \times 25,000 = 13,000$) instead of reporting the full range the question asked for.
1,150,000 to 1,450,000 votes	Forgot to convert the percentages to decimals before multiplying (used 46 and 58 directly instead of 0.46 and 0.58), inflating both bounds by a factor of 100.

Shortcut

Since $25,000 \times 1\% = 250$, compute $46 \times 250 = 11,500$ and $58 \times 250 = 14,500$ using mental multiplication instead of decimal multiplication.

Desmos Tip

Type $0.46 * 25000$ and $0.58 * 25000$ into the calculator to get both bounds in two quick steps.

Sampling, Bias & Margin of Error — Problem 8 (Very Hard)

A gym chain in Ljubljana has 18,000 members. A random sample estimates that 10,800 of all 18,000 members (a proportion of 60%) would renew their annual membership, with a reported margin of error of ± 5 percentage points. Reporters convert this into a 95% confidence interval for the actual NUMBER of members (not the percentage) who would renew, and correctly find the lower bound to be 9,900 members. What is the confidence interval's upper bound, in members?

Formula / Rule Used: A margin-of-error confidence interval is symmetric around its point estimate: the two bounds sit equal distances from the center.

Solution

Step 1: Confirm the point estimate in raw numbers: 60% of 18,000 = 10,800 members, matching the given estimate.

Step 2: A confidence interval built from a symmetric margin of error is centered on the point estimate, so the lower and upper bounds sit equal distances from 10,800.

Step 3: Find that distance using the given lower bound: $10,800 - 9,900 = 900$ members.

Step 4: Add the same distance to the point estimate to find the upper bound: $10,800 + 900$.

Step 5: Compute: $10,800 + 900 = 11,700$.

Step 6: Check: this matches $65\% \times 18,000 = 11,700$, confirming the ± 5 percentage-point margin correctly translates to ± 900 members.

Step 7: So the confidence interval's upper bound is 11,700 members.

Answer: 11,700 members

Tip

Recognize that a margin-of-error interval is symmetric around the point estimate — once you know the distance from the center to ONE bound, the other bound is the same distance on the opposite side, so you don't need to recompute percentages from scratch.

Trap

The real trap is treating the interval's full width (1,800 members, lower bound to upper bound) as the amount to add to the point estimate, instead of the 900-member distance from the center to one bound — that mistake produces 12,600 instead of 11,700.

Process of Elimination

Tempting Wrong Answer	Exactly What Went Wrong
9,900 members	Misread the question and reported the given lower bound back as the answer instead of computing the upper bound.
10,800 members	Reported the point estimate (the center of the interval) instead of continuing on to find the actual upper bound.
12,600 members	Added the interval's full width (1,800) to the point estimate instead of just the 900-member distance from the center to one bound, computing $10,800 + 1,800 = 12,600$.

Shortcut



Since the margin is ± 5 percentage points and $18,000 \times 1\% = 180$, the member-count margin is $5 \times 180 = 900$ directly — skip finding the lower bound's distance by subtraction and just add 900 to 10,800.

Desmos Tip

Type $10800+900$ and $10800-900$ into the calculator to instantly confirm both interval bounds (11,700 and 9,900) match.



Quick Reference — Master Summary

Use this page as a final review before test day.

The 13 Skills at a Glance

Skill	First Move	Main Trap
Lines	Find slope first; convert to slope-intercept form.	Perpendicular slope: using m instead of $-1/m$.
Systems	Check if you can combine equations directly for the asked expression.	Solving fully for x and y when only a combination was needed.
Inequalities	Flip the sign only when multiplying/dividing by a negative.	Forgetting to flip — or flipping when adding/subtracting.
Absolute Value	Isolate the bars, then split into two cases.	Not checking both solutions in the ORIGINAL equation.
Exponents	Rewrite with a common base before comparing exponents.	Distributing an exponent over addition — $(a+b)^n \neq a^n + b^n$.
Radicals	Convert to fractional exponents before combining.	Forgetting to renormalize scientific notation coefficients.
Ratios & Proportions	Use the 'x' method — a common multiplier.	Answering with the wrong part after solving for x .
Probability	Identify independent/dependent and mutually exclusive/not first.	Forgetting the denominator changes without replacement.
Unit Conversions	Set up the fraction so units cancel diagonally.	Forgetting to square/cube the WHOLE conversion factor.
Statistics	Use $\text{Sum} = \text{Mean} \times n$ to recover totals fast.	Assuming mean and median move the same way when data changes.
Data Displays	Read the five-number summary directly off a box plot.	Confusing a longer whisker with more data points.
Scatterplots	Read predicted values from the LINE, not the data points.	Using a data point instead of a point actually on the line.
Sampling	Check random selection first — sample size is secondary.	Assuming a larger sample fixes a biased selection method.

Golden Rules

- Write down what the question actually asks for LAST — many Hard problems bury a units or sign switch in the final step.
- If a problem gives one extra fact you haven't used yet, check whether you need it before submitting an answer.
- When a problem can be solved two ways, the faster way is usually the one that solves directly for the combined quantity being asked, not each variable separately.



- Use Desmos to confirm an algebraic answer whenever the relationship can be graphed — but never as a substitute for knowing the rule.
- A 'twist' is not a trick question — it's the same rule applied in a less familiar form. Name the rule before you touch the algebra.